

Cystic Diseases of the Kidney

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CLASSIFICATION OF RENAL CYSTIC DISEASES

Renal cystic diseases encompass a large number of sporadic and genetically determined congenital, developmental, and acquired conditions that have in common the presence of cysts in one or both kidneys. “Renal cysts” are cavities lined by epithelium and filled with fluid or semisolid matter. Cysts are derived primarily from tubules. Whereas cystic kidneys of different etiologies may appear morphologically similar, the same etiologic entity may cause a wide spectrum of renal abnormalities. Classifications of renal cystic diseases are based on morphologic, clinical, and genetic information (Box 45.1) and change as the understanding of the underlying etiologies and pathogenesis continues to expand.

DEVELOPMENT OF RENAL EPITHELIAL CYSTS

Epithelial cysts develop from preexisting renal tubule segments and are composed of a layer of partially dedifferentiated epithelial cells enclosing a cavity filled with either a urine-like liquid or semisolid material. They may develop in any tubular segment between Bowman’s capsule and the tip of the renal papilla, depending on the nature of the underlying disorder. After achieving a size of perhaps a few millimeters, most cysts lose their attachments to their parent tubule segment.

Pathophysiologic processes that contribute to the development of cysts include disruption of programs responsible

for the establishment and maintenance of normal tubular diameter (i.e., convergent extension, or the process of cell intercalation by which cells elongate along an axis perpendicular to the proximal–distal axis of the tubule and actively crawl among one another to produce a narrower, longer tubule; and oriented cell division or alignment of the mitotic spindle axis and cell division with the proximal–distal axis of the tubule),^{1,2} excessive cell proliferation, active solute and fluid transport into the expanding cysts, crosstalk between epithelial cells and interstitial macrophages, and interactions between epithelial cells and extracellular matrix (ECM)^{3,4} (Fig. 45.1).

Renal cysts have been regarded as benign neoplasms that arise from individual cells or restricted segments of the renal tubule. Transgenic insertions of activated proto-oncogenes and growth factor genes into rodents result in the formation of renal cysts. Therefore processes that stimulate renal cell proliferation along with the inability to maintain planar cell polarity have the potential to generate the cystic phenotype.

Conditional knockouts of *Pkd1* or of ciliogenesis genes (*Ift88* and *Kif3a*) at various time points have shown that the timing of their inactivation determines the rate of development of cystic disease.^{5–8} Inactivation in newborn mice leads to rapid cyst development. Inactivation after about 13 days leads to slowly progressive disease evident only in the adult kidneys, but progression can be hastened by maneuvers such as ischemic or reperfusion injury to stimulate cell proliferation.⁹ The underlying rate of epithelial cell proliferation may account for increased susceptibility to cyst development during nephrogenesis as well as for the migration of cysts as the kidney matures in humans and in animal models of polycystic

Box 45.1 Classification of Cystic Kidney Disorders

Autosomal dominant polycystic kidney disease (ADPKD)
 Autosomal recessive polycystic kidney disease (ARPKD)
 Hepatic kidney and/or liver diseases associated with altered maturation of PKD proteins in the endoplasmic reticulum (*PRKCSH*, *SEC63*, *GANAB*, *ALG8*, *SEC61B*, *DNAJB11*)
 Tuberous sclerosis complex
 von Hippel–Lindau syndrome
 Familial renal hamartomas associated with hyperparathyroidism–jaw tumor syndrome
 Hepatocyte nuclear factor-1–associated nephropathy
 Oro-facial-digital syndrome
 Autosomal dominant tubulointerstitial kidney disease
 Hereditary recessive ciliopathies with interstitial nephritis, cysts, or both:
 Nephronophthisis
 Joubert syndrome
 Meckel syndrome
 Bardet–Biedl syndrome
 Alström syndrome
 Nephronophthisis variants associated with skeletal defects (skeletal ciliopathies)
 Renal cystic dysplasias:
 Multicystic kidney dysplasia
 Other cystic kidney disorders:
 Simple cysts
 Localized or unilateral renal cystic disease
 Medullary sponge kidney
 Acquired cystic kidney disease
 Renal cystic neoplasms:
 Cystic renal cell carcinoma
 Multilocular cystic nephroma
 Cystic partially differentiated nephroblastoma
 Mixed epithelial and stromal tumor
 Cysts of nontubular origin:
 Cystic disease of the renal sinus
 Perirenal lymphangiomas
 Subcapsular and perirenal urinomas
 Pyelocalyceal cysts

kidney disease (PKD) from being located predominantly proximal to appearing predominantly distal and in the collecting duct.¹⁰ Immature early tubules (S-shaped bodies) exhibit very high rates of proliferation. Later, when the epithelium differentiates into nephron segments recognizable on light microscopy, proliferative indices become very low in proximal tubules but remain elevated in the distal nephrons and collecting ducts. In pediatric and adult kidneys, proliferative indices are very low in all tubular segments but remain higher in collecting ducts than in proximal tubules.^{11,12}

The finding of fluid secretion in renal epithelial cysts led to a reinvestigation of fluid secretion mechanisms in otherwise normal renal tubules. Beyond the loop of Henle, tubule cells have the capacity to secrete solutes and fluid upon stimulation with cyclic adenosine monophosphate (cAMP).¹³ This secretory flux operates in competition with the more powerful mechanism by which Na^+ is absorbed through apical epithelial Na^+ channels. Under conditions in which Na^+ absorption is diminished, the net secretion of NaCl and fluid can be observed at rates that could have a significant impact on the

net economy of body salt and water content. Thus renal cystic disease has led to a heightened appreciation of an “ancient” solute and water secretory mechanism that has been largely overlooked in modern studies of renal physiology.¹⁴

Three decades ago it was noted that a germ-free environment inhibits cyst development in CFW mice and in a model of PKD induced by nordihydroguaiaretic acid; the administration of endotoxin rescued the cystic phenotype.^{15,16} Chemokines and cytokines were found at high concentrations in cyst fluid and produced by epithelial cells of the cyst lining.¹⁷ In later studies, alternatively activated (M2) macrophages aligned along cyst walls were detected in polycystic kidneys from conditional *Pkd1* knockout and the *Pkd2*^{Wsi8/-} model.^{19,20} Macrophage depletion inhibited epithelial cell proliferation and cyst growth and improved renal function. These observations led to the hypothesis that M2 macrophages contribute to cell proliferation in PKD, as has been described during development, during recovery from acute kidney injury, and in cancer.

Evidence indicates that alterations in focal adhesion complexes, basement membranes, and ECM contribute to the pathogenesis of PKD.²¹ Focal adhesion complexes contain integrin $\alpha\beta$ heterodimer receptors and multiple structural and signaling molecules, including polycystin 1 (PC1). The integrin receptors link the actin cytoskeleton, laminin $\alpha\beta\gamma$ heterotrimers, and collagens in the basement membrane. Integrins β_4 and β_1 may mediate the increased adhesion of cyst-lining epithelial cells to laminin-322 and collagen and are all overexpressed in cystic tissues.^{22,23} Periostin, an ECM protein, and its receptor α_v integrin as well as α_1 and α_2 integrins are also overexpressed.^{24,25} Laminin-322 and periostin stimulate, whereas antibodies to laminin-332 and α_v integrin inhibit cyst formation in three-dimensional gel culture.^{18,24} Renal cystic disease develops in β_1 integrin knockout and laminin α_5 hypomorphic mutant mice, the latter associated with overexpression of laminin-322.^{26,27} Mutations to the gene *COL4A1*, which encodes procollagen type IV, have been found in patients with autosomal dominant (hereditary) angiopathy with nephropathy (consisting of hematuria and bilateral renal cysts), aneurysms, and muscle cramps (HANAC syndrome).

Evidence accumulated during nearly 20 years strongly suggests that the primary cilium is essential to maintain epithelial cell differentiation and that structural and functional defects in the primary cilium of tubular epithelia have a central role in various forms of human and rodent cystic diseases. The “primary cilium” is a single hairlike organelle that projects from the surface of most mammalian cells, including epithelial and endothelial cells, neurons, fibroblasts, chondrocytes, and osteocytes. It is involved in left–right embryonic patterning as well as in mechanosensing (renal tubular and biliary epithelia), photosensing (retinal pigmented epithelia), and chemosensing (olfactory neurons).^{28–32} In renal tubule epithelial cells, the cilium projects into the lumen and is thought to have a sensory role (Fig. 45.2). The cilium arises from the mother centriole in the centrosome. The centrosome comprises a mother centriole and a daughter centriole plus a “cloud” of pericentriolar material.³³ The centrosome serves as the microtubule-organizing center for interphase cells, and the mother and daughter centrioles form the poles of the spindle during cell division.

The first clue connecting PKD and cilia was that PC1 and polycystin 2 (PC2) homologs in *Caenorhabditis elegans* are

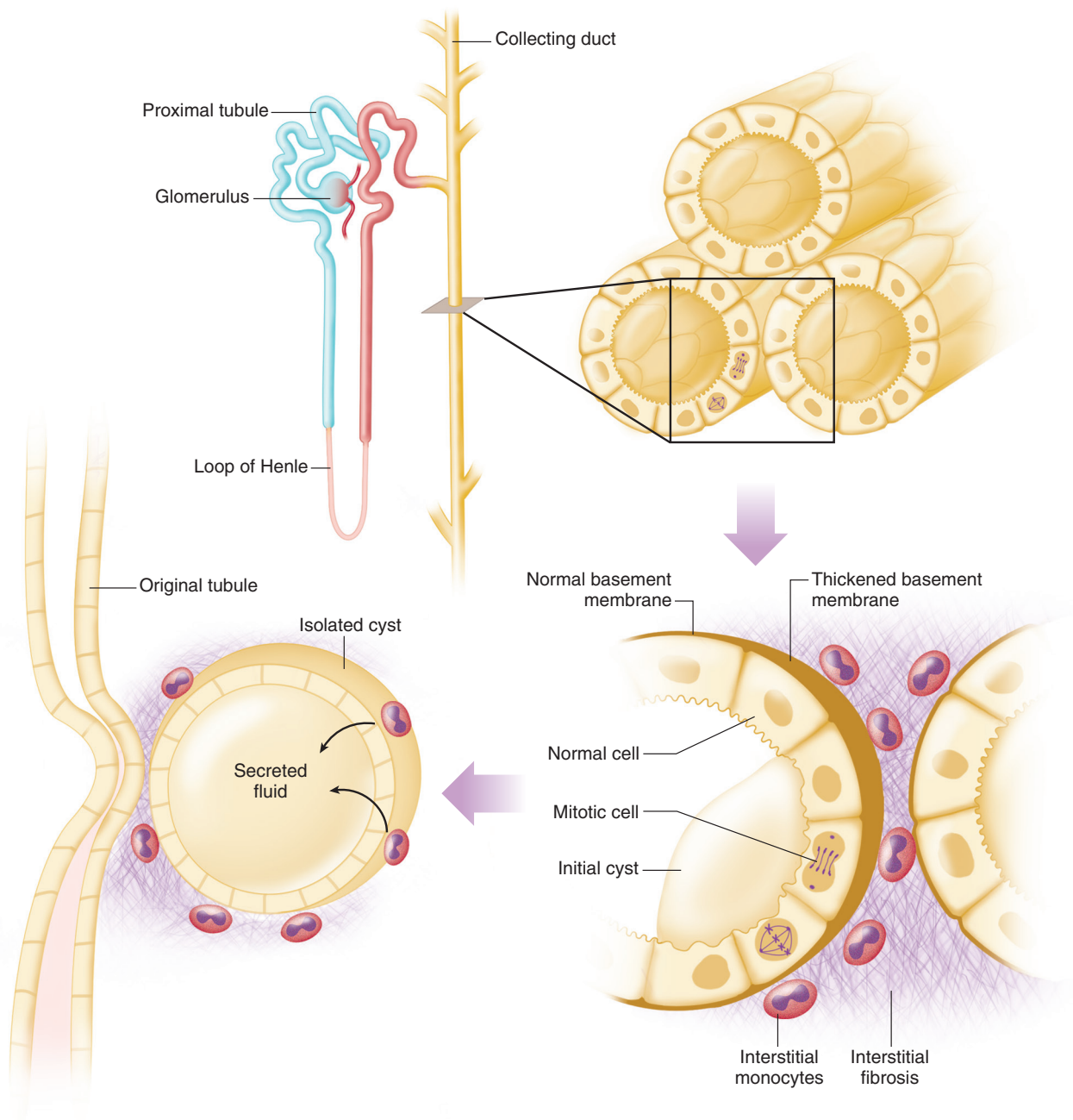


Fig. 45.1 Evolution of cysts from renal tubules. Abnormal proliferation of tubule epithelium begins in a single cell after a “second-hit” process disables the function of the normal allele or if the level of functional polycystin falls below a specific threshold. Repeated cycles of cell proliferation lead to expansion of the tubule wall into a cyst. The cystic epithelium is associated with thickening of the adjacent tubule basement membrane and with an influx of inflammatory cells into the interstitium. The cystic segment eventually separates from the original tubule, and net epithelial fluid secretion contributes to the accumulation of liquid within the cyst cavity.

located in cilia of male sensory neurons; loss of these proteins is associated with mating behavior defects.³⁴ Next, a known intraflagellar transport (IFT) protein, polaris, was found to be defective in a hypomorphic mouse mutant, *orpk*, in which PKD, left-right patterning defects, and a variety of other abnormalities develop, and that has shortened cilia in the kidney.³⁵ Subsequently, the proteins mutated in other rodent models of PKD—such as the *cpk* (centrin) and *inv* (inversin) mice, autosomal dominant PKD (ADPKD; PC1 and PC2), autosomal recessive PKD (ARPKD; fibrocystin), autosomal recessive ciliopathies (see later in the chapter), and possibly

tuberous sclerosis complex (TSC) and von Hippel-Lindau (vHL) disease—have been localized to the ciliary axoneme, the basal body, or centrosomal structures.^{30,36,37} Conditional inactivation of the ciliary motor protein KIF3A (kinesin family member 3A) in collecting duct epithelial cells reproduced all of the clinical and biologic features of PKD.³⁸

The polycystin complex on cilia may detect changes in flow and transduce it into a Ca^{2+} influx through the PC2 channel, thus functioning as a mechanosensor,³⁹ although a chemosensory role or ligand receptor has not been excluded. The Ca^{2+} influx may in turn induce release of Ca^{2+} from

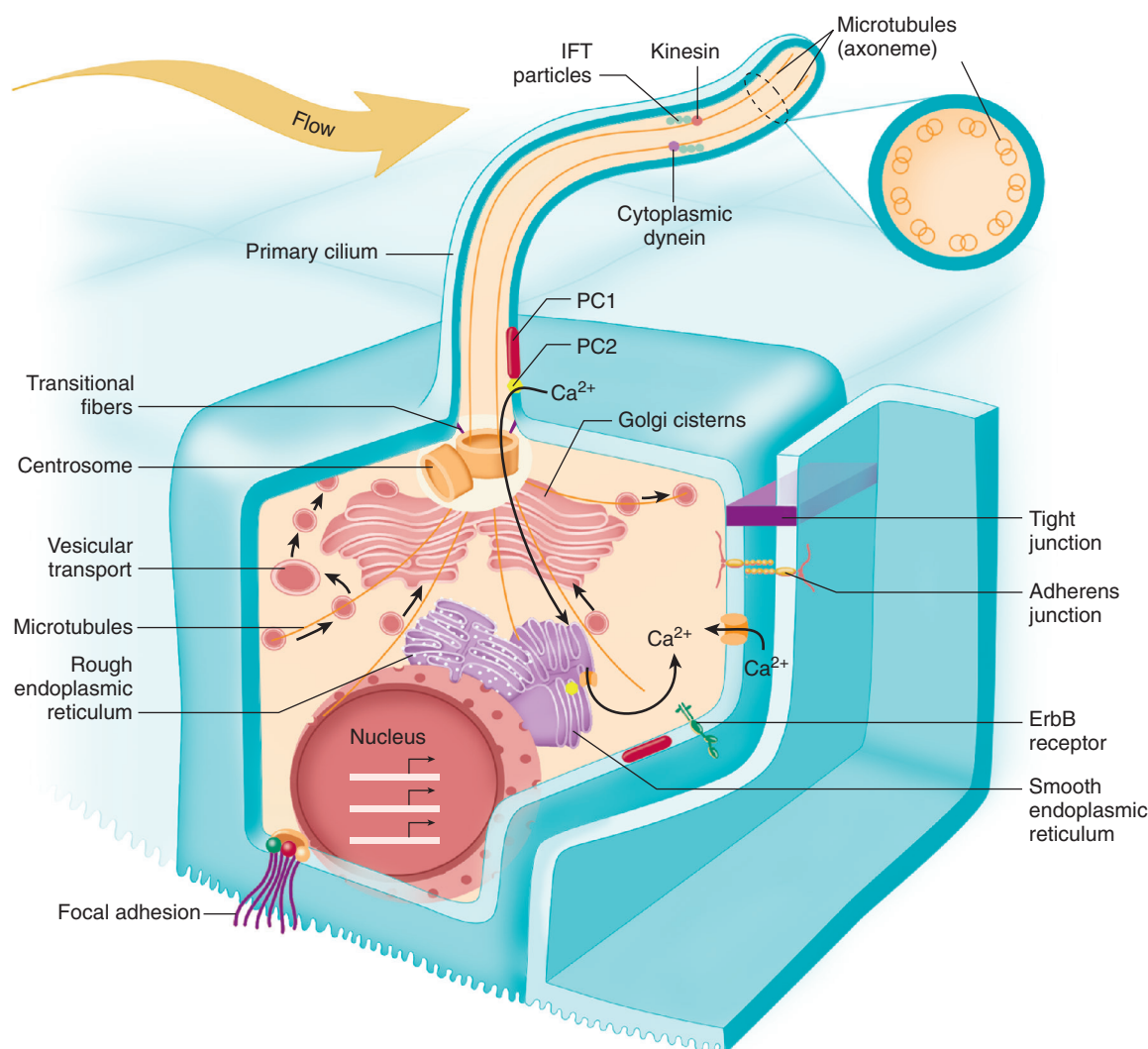


Fig. 45.2 Diagram depicting the primary cilium and hypothetical functions of the polycystins. Polycystin 1 (PC1) and polycystin 2 (PC2) are found on the primary cilium, a single hairlike structure that projects from the apical surface of the cell into the lumen. It consists of a membrane continuous with the cell membrane and a central axoneme composed of nine peripheral microtubule doublets. It arises from the mother centriole in the centrosome, the microtubule-organizing center of the cell. The centrosome comprises a mother centriole and a daughter centriole plus a “cloud” of pericentriolar material. In response to mechanical stimulation of the primary cilium by flow, the PC1 and PC2 complex mediates Ca^{2+} entry into the cell. This triggers Ca^{2+} -induced release of Ca^{2+} from the smooth endoplasmic reticulum (ER) through ryanodine receptors. The function of the polycystins extends beyond the cilium because PC1 is also found in the plasma membrane and PC2 is predominantly expressed in the ER. PC2 is an intracellular Ca^{2+} channel that is required for the normal pattern of $[\text{Ca}^{2+}]_i$ responses involving ryanodine receptors and inositol 1,4,5-trisphosphate (IP3) receptors and may also affect the activity of store-operated Ca^{2+} channels. *ErbB*, Epidermal growth factor receptor; *IFT*, intraflagellar transport. (Reproduced from Torres VE, Harris PC, Pirson Y. Autosomal dominant polycystic kidney disease. *Lancet*. 2007;369:1287–1301.)

intracellular stores. The increased Ca^{2+} concentration in intracellular microenvironments may then modulate specific signaling pathways that regulate cellular differentiation, proliferation, and apoptosis, such as cAMP, receptor-tyrosine kinase, extracellular signal-regulated kinase (ERK), and mammalian target of rapamycin (mTOR) signaling.⁴

The ciliocentric model of cystogenesis is attractive but may be too reductionist. Several cyst-associated proteins have other functions, including participation in cell–cell and cell–matrix interactions at adherens junctions and focal adhesions. Dysfunction of these subcellular domains most likely contributes to the aberrant epithelial growth and tubular architecture that are common to virtually all forms of renal cystic disease. Although ciliary dysfunction may be

the initiating event in cystogenesis, defects in other cellular mechanisms may modulate the final cystic disease phenotype.

HEREDITARY CYSTIC KIDNEY DISORDERS

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

EPIDEMIOLOGY

ADPKD occurs worldwide and in all races, with a prevalence estimated to be between 1 in 400 and 1 in 1000.⁴⁰ The yearly incidence rates for end-stage kidney disease (ESKD) due

to ADPKD are 8.7 and 6.9 per million (1998–2001, United States),⁴¹ 7.8 and 6.0 per million (1998–1999, Europe),⁴² and 5.6 and 4.0 per million (1999–2000, Japan)⁴³ in men and women, respectively. Age-adjusted gender ratios greater than unity (1.2–1.3) suggest a more progressive disease in men than in women. Approximately 30,000 patients have ESKD due to ADPKD in the United States (1:3500 individuals aged 65–69 years).⁴⁴ The proceedings of a 2014 Kidney Disease: Improving Global Outcomes (KDIGO) Controversies Conference to assess the current state of knowledge related to the evaluation, management, and treatment of ADPKD have been published.⁴⁵

GENETICS AND GENETIC MECHANISMS

ADPKD is inherited as an autosomal dominant trait with complete penetrance in terms of cyst development. Therefore each child of an affected parent has a 50% chance of inheriting the abnormal gene. Most patients with ADPKD have an affected parent, but at least 10% of families can be traced to an apparent de novo mutation.^{46,47} ADPKD is genetically heterogeneous with two major genes identified, *PKD1* (chromosome region 16p13.3) and *PKD2* (4q21).^{48–51} Recently, two other genes with an ADPKD-like phenotype have been described, *GANAB* and *DNAJB1*^{52,53} (see the “Polycystic Kidney and/or Liver Diseases Associated With Altered Maturation of PKD Proteins in the Endoplasmic Reticulum” section).

In groups identified through renal clinics, *PKD1* accounts for about 78% of pedigrees and *PKD2* about 15%, and no mutation is detected in about 7%.^{54,55} *PKD2* may account for up to approximately 25% of mutation-characterized cases in population-based studies and those focused on milder patients.^{56,57} In the latest version of the ADPKD Mutation Database (PKDB), 1272 *PKD1* mutations are described, accounting for 1874 families, with 202 mutations to *PKD2* causing disease in 438 pedigrees.⁵⁸ For *PKD1*, about 65% of mutations are predicted to truncate the protein, leaving about 35% nontruncating.^{54,55,59,60} Corresponding levels for *PKD2* are about 87% truncating and 13% nontruncating; about 3% of ADPKD mutations are larger rearrangements involving deletion or duplication of at least one exon.^{54,55,59–61} A next-generation sequencing (NGS) method for ADPKD screening based on sequencing the locus-specific long-range polymerase chain reaction products has been described.⁶² Such methods can identify unusual mutations, such as gene conversions with one of the pseudogenes. Recently, exon capture and NGS approaches for screening the ADPKD have been described to efficiently screen *PKD1*, despite the segmental duplication of this locus.^{63,64}

Inheritance of two *PKD1* or two *PKD2* alleles with fully inactivating mutations is lethal in utero.⁶⁵ Individuals heterozygous for both a *PKD1* and *PKD2* mutation live to adulthood but have more severe renal disease than those heterozygous for only one mutation.⁶⁶ Patients with a *PKD1* mutation have more severe disease than those with a *PKD2* mutation, the average ages at ESKD being 58.1 years and 79.7 years, respectively.⁶⁷ Viable ADPKD cases homozygous/compound heterozygous for *PKD1* pathogenic variants suggest the presence of hypomorphic alleles.⁶⁸ Some nontruncating *PKD1* changes have been suggested to be not fully penetrant alleles, resulting in ESKD at 55 years in patients with truncating *PKD1* mutations and 67 years in those with nontruncating changes.^{67,69} Further bioinformatics analysis of nontruncating mutations has identified two populations,

those that are fully inactivating and those that are incompletely penetrant, associated with milder kidney disease.⁵⁹

A small number (<1%) of patients with ADPKD exhibit early-onset disease, with a diagnosis made in utero or in infancy from the presence of enlarged echogenic kidneys that may resemble those seen in ARPKD.^{70,71} Most early-onset cases have been linked to *PKD1*, but a family with *PKD2* mutation and perinatal death in two severely affected infants has been described.⁷² Some cases of early-onset ADPKD, or cases mimicking ARPKD, are due to an in trans combination of two *PKD1* mutations, at least one of which is hypomorphic.^{68,73,74} Studies of a *Pkd1* mouse model with a missense change, p.R3277C, confirmed the hypomorphic nature of this allele and its role in causing early-onset disease.^{10,68} Uniparental isodisomy involving a hypomorphic *PKD2* allele can also cause early-onset ADPKD.⁷⁵ Mutations in other cystogenes, such as *HNF1B* [associated with the autosomal dominant tubulointerstitial kidney disease (ADTKD); see later discussion] and *PKHD1* (the ARPKD gene), in combination with an ADPKD mutant allele have also been associated with early-onset PKD.⁷⁶ The contiguous deletion of the adjacent *PKD1* and *TSC2* (see later discussion of TSC) genes is characterized by childhood PKD with additional clinical signs of TSC.^{77,78}

Significant intrafamilial variability in the severity of renal and extrarenal manifestations points to genetic and environmental modifying factors. Analysis of the variability in renal function between monozygotic twins and siblings supports a role for genetic modifiers.^{79,80} Parental hypertension, particularly in the nonaffected parent, increases the risk for hypertension and ESKD.⁸¹ Parents are as likely to show more severe disease as children are.⁸² Mosaicism can also modulate disease presentation and result in marked intrafamilial variability.^{61,83}

Cysts in ADPKD kidneys appear to be derived through clonal proliferation of single epithelial cells that occurs in fewer than 1% of the tubules. A two-hit model of cystogenesis has been proposed to explain the focal nature of the cysts. In this model, a mutated *PKD1* (or *PKD2*) gene is inherited from one parent, and a wild-type gene is inherited from the unaffected parent. During the lifetime of the individual, the wild-type gene undergoes a somatic mutation and becomes inactivated. Loss of function owing to somatic mutations of the *PKD1* and *PKD2* genes has been identified in the cells lining the cysts in both the kidney and the liver.^{84,85} Support for this model of cystogenesis is provided by the embryonic lethality and severe PKD of homozygous *Pkd1* or *Pkd2* knockout mice, the late development of cysts in the kidney or liver in heterozygous mutant mice, and the increased severity of the disease in *Pkd2*^{WS25/-} mice carrying a *Pkd2* allele (WS25) prone to genomic rearrangement.⁸⁶

Evidence suggests, however, that other genetic mechanisms may also be involved. Transgenic overexpression of *PKD1* or *PKD2* can induce renal cystic disease.^{87–89} The presence of somatic trans-heterozygous mutations in human polycystic kidneys (somatic mutation of the PKD gene not involved by the germline mutation) and the greater severity of cystic disease in mice with trans-heterozygous mutations of *Pkd1* and *Pkd2* than could be predicted by a simple additive effect suggest that haploinsufficiency may play a role in cyst formation.^{66,90} Comparative genomic hybridization and loss of heterozygosity analysis have shown multiple molecular

cytogenetic aberrations in epithelial cells from individual cysts in polycystic kidneys, suggesting the involvement of additional genes in the initiation and progression of the cystic disease.⁹¹ Mice that are homozygous for *Pkd1* hypomorphic alleles indicate that complete inactivation of both *Pkd1* alleles is not required for cystogenesis in ADPKD.^{10,92,93} *Pkd2* haploinsufficiency has been associated with a higher rate of cell proliferation in noncystic tubules of *Pkd2*^{+/-} mice.⁹⁴ These observations suggest that diminished expression of native polycystins below a certain threshold is sufficient to induce renal cystic disease and may also be relevant for the extrarenal manifestations of the disease. Reduction of PC2 levels to 50% of normal in the vascular smooth muscle of *Pkd2*^{+/-} mice causes significant alterations in $[Ca^{2+}]_i$ (intracellular calcium concentration) and cAMP; moreover, it results in higher rates of cell proliferation and apoptosis, contractility, and vasculature susceptibility to hemodynamic stress.⁹⁵ Reduction of PC1 levels to 50% of normal causes significant alterations in $[Ca^{2+}]_i$ homeostasis and increased vascular reactivity with compensatory changes in the transport proteins involved in calcium signaling in the aorta of *Pkd1*^{+/-} mice.⁹⁶

PATHOGENESIS

PC1 (4303 aa; ≈600 kDa, uncleaved and glycosylated) is a receptorlike protein with a large ectodomain (3074 aa) that comprises a number of domains involved in protein–protein and protein–carbohydrate interactions and 16 PKD repeats with an immunoglobulin (Ig) domain–like fold (Fig. 45.3).^{49,50} PC1 also has 11 transmembrane domains and a cytoplasmic tail. PC2 (968 aa; ≈110 kDa) is a six-transmembrane, Ca^{2+} -responsive cation channel of the transient receptor potential (TRP) family (also known as TRPP2).⁵¹ PC1 and PC2 interact via their C-terminal tails with the resulting polycystin complex thought to play a role in intracellular Ca^{2+} regulation. Data also indicate an interdependence of the proteins for maturation and localization.^{90,97} The structure of a homotetrameric PC2 complex has recently been resolved by cryo-electron microscopy. It shows the presence of a large exoplasmic domain named the tetragonal opening for polycystin (TOP domain) that may regulate gating of the channel.^{98,99} A 3 to 1 PC2/PC1 structure has also recently been resolved by cryo EM.¹⁰⁰

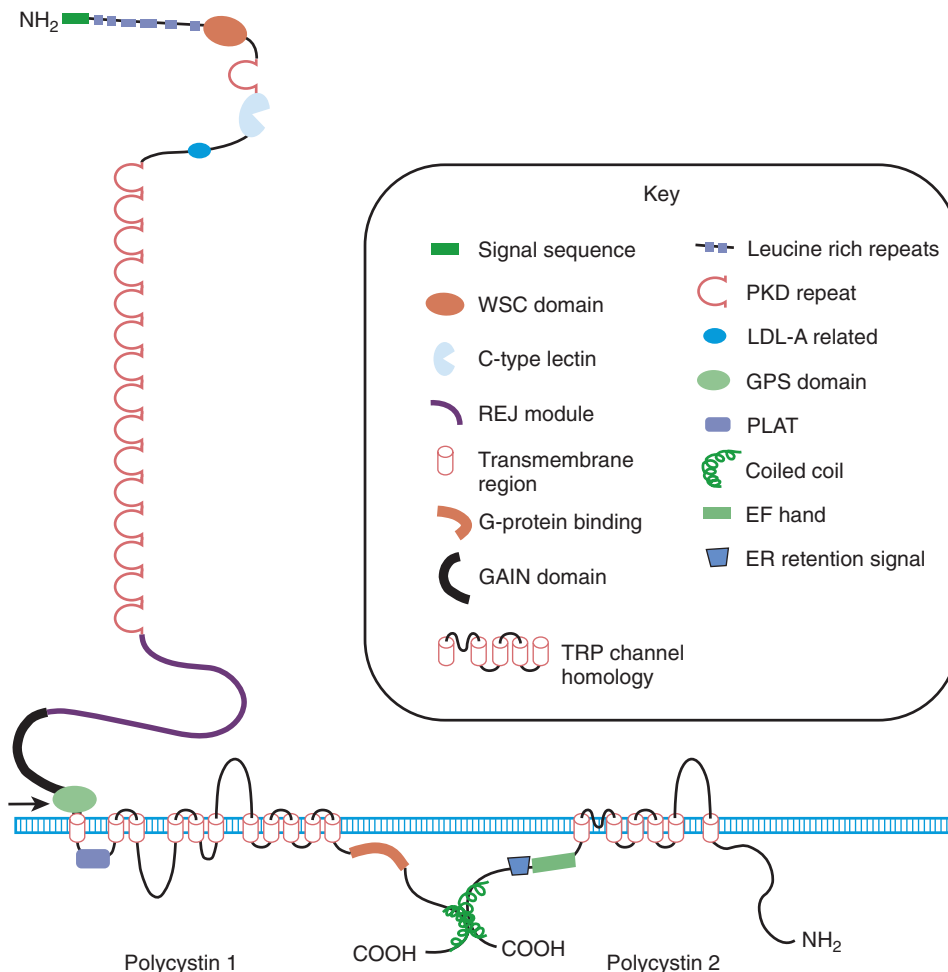


Fig. 45.3 Structure of the polycystin proteins. Polycystin 1 (PC1) is a large protein with an extensive extracellular region, an 11-transmembrane area, and a short cytoplasmic tail. The protein contains a number of recognized domains and motifs (see Key). The protein is cleaved at the GPS (G protein-coupled receptor proteolytic site) domain (arrow). Polycystin 2 (PC2) is a six-transmembrane, transient receptor potential (TRP)-like channel with cytoplasmic N and C termini. The proteins are thought to interact through coiled-coil domains. *ER*, Endoplasmic reticulum; *GAIN*, G protein-coupled receptor-autoproteolysis inducing; *LDL-A*, low-density lipoprotein A module; *PKD*, polycystic kidney disease; *PLAT*, Polycystin 1, lipoxigenase, alpha-toxin; *REJ*, receptor of egg jelly protein.

Like many other proteins implicated in renal cystic diseases, the polycystins are located in the membrane overlying primary cilia.^{101,102} PC2 currents have recently been recorded from patch clamping of individual cilia.¹⁰³ The polycystins may be required for the induction of calcium transients in response to ciliary bending.³⁹ PC1 is also found in plasma membranes at focal adhesion, desmosomes, and adherens junction sites,^{104–107} whereas PC2 is found in the endoplasmic reticulum,^{108–110} and both proteins are abundant in exosomes.^{111,112} The PC1 protein in the plasma membrane may interact with PC2 in the adjacent endoplasmic reticulum. PC2 interacts with the inositol 1,4,5-trisphosphate receptor (IP3R), ryanodine receptor 2 (RyR2), and TRP channels TRPC1, TRPC4, and TRPV4.^{113–115}

Precisely how intracellular calcium homeostasis is altered in ADPKD remains uncertain.¹¹⁶ Cells overexpressing PC2 exhibit an amplified Ca^{2+} release from intracellular stores after agonist stimulation.¹¹⁷ A 50% reduction in PC2 lowers capacitative calcium entry, sarcoplasmic reticulum Ca^{2+} stores, and $[\text{Ca}^{2+}]_i$ in vascular smooth muscle cells (VSMCs).⁹⁵ Increases in $[\text{Ca}^{2+}]_i$ levels evoked by platelet-activating factor are reduced in unciliated B-lymphoblastoid cells from patients with *PKD1* or *PKD2* mutations.¹¹⁸ Loss of PC2 localization to the mitotic spindles by knockdown of the interacting cytoskeletal protein, mDia1, blunts agonist-evoked $[\text{Ca}^{2+}]_i$ increases in dividing cells that lack primary cilia.¹¹⁹ The majority of studies that have measured resting intracellular calcium, endoplasmic reticulum calcium stores, and store-operated calcium entry in primary cell cultures or microdissected samples from human and rodent polycystic tissues have found them to be reduced.^{95,96,113,120–126}

Tissue levels of cAMP are increased in numerous animal models of PKD, not only in the kidney^{10,127–130} but also in cholangiocytes,¹³¹ VSMCs,¹³² and choroid plexus (Fig. 45.4).¹³³ Levels of cAMP are determined by the activities of membrane-bound (under the positive or negative control of G protein-coupled receptors and extracellular ligands) and soluble adenylyl cyclases (ACs), and of cAMP phosphodiesterases (PDEs), themselves subject to complex regulatory mechanisms. The increased levels in cystic tissues may be directly related to changes in $[\text{Ca}^{2+}]_i$ homeostasis. Reduced calcium activates calcium-inhibitable AC-6, directly inhibits calcium/calmodulin-dependent PDE₁ [also increasing the levels of cyclic guanosine monophosphate (cGMP)], and indirectly inhibits cGMP-inhibitable PDE₃^{128,134} (Fig. 45.4). Additional mechanisms include the following:

1. Dysfunction of a ciliary protein complex [comprising A-kinase anchoring protein 150, AC-5/6, PC2, PDE_{4C}, and protein kinase A (PKA)] that normally restrains cAMP signaling via inhibition of AC-5/6 activity by PC2-mediated calcium entry and degradation of cAMP by PDE_{4C} transcriptionally controlled by hepatocyte nuclear factor-1 β (HNF-1 β).¹³⁵
2. Depletion of the endoplasmic reticulum calcium stores, which triggers oligomerization and translocation of stromal interaction molecule 1 (STIM1) to the plasma membrane, where it recruits and activates AC-6.¹²⁵
3. Other contributory factors, such as disruption of PC1 binding to heterotrimeric G proteins, upregulation of the vasopressin V₂ receptor (V₂R), and increased levels of circulating vasopressin or accumulation of forskolin,

lysophosphatidic acid, adenosine triphosphate (ATP), or other AC agonists in the cyst fluid.^{136–139}

The marked amelioration of the cystic disease in collecting duct-specific *Pkd1* knockout mice by a concomitant *Ac6* knockout provides strong support for the central role of calcium-inhibitable AC-6.¹⁴⁰ PDEs are likely important in PKD because maximal rates of degradation by PDEs exceed, by an order of magnitude, those of synthesis by ACs and hence control compartmentalized pools of cAMP, which are likely more crucial than total intracellular cAMP. PDE₁ and PDE₃ may be particularly important. PDE₁ accounts for most PDE activity in renal tubules, it is the only PDE activated by calcium (which is reduced in PKD cells), and its activity is reduced in cystic kidneys.¹³⁴ The knockdown of *pde1a* using morpholinos induces or aggravates the cystic phenotype of wild-type or *pkd2* morphant zebrafish embryos, respectively, whereas PDE_{1a} RNA partially rescues the phenotype of *pkd2* morphants.¹⁴¹ PDE₃ controls a compartmentalized cAMP pool that stimulates mitogenesis in MDCK cells¹⁴² as well as cystic fibrosis transmembrane conductance regulator (CFTR)-driven chloride secretion in pig trachea submucosal and shark rectal glands.^{143,144} A small-molecule, nonselective PDE activator lowers cAMP and inhibits the growth of MDCK cysts.¹⁴⁵

The reduction in $[\text{Ca}^{2+}]_i$ and the increase in cAMP may play a central role in the pathogenesis of PKD (Fig. 45.4). Cyclic AMP stimulates mitogen-activated protein kinase (MAPK)/ERK signaling and cell proliferation in PKD renal epithelial cells in a manner dependent on PKA, the proto-oncogene tyrosine protein kinase Src, and the protein Ras. By contrast, cAMP has an inhibitory effect in wild-type cells.^{146,147} The abnormal proliferative response to cAMP is directly linked to the alterations in $[\text{Ca}^{2+}]_i$ because it can be reproduced in wild-type cells by lowering of $[\text{Ca}^{2+}]_i$.¹⁴⁸ Conversely, calcium ionophores or channel activators can rescue the abnormal response of cyst-derived cells.¹²⁰ Activation of mTOR signaling also occurs downstream from PKA, through ERK-mediated phosphorylation of tuberlin in cystic tissues.^{149,150} Activation of mTOR has in turn been linked to transcriptional activation of aerobic glycolysis and increased levels of ATP, which together with ERK-dependent inhibition of liver kinase B1 (LKB1) and inhibition of AMP kinase (AMPK)^{151–153} further enhance mTOR signaling.¹⁵⁴ Phosphorylation and inhibition of glycogen synthase kinase type 3 β ¹⁵⁵ and direct phosphorylation and stabilization of β -catenin by PKA¹⁵⁶ enhance Wnt/ β -catenin signaling. PKA-dependent upregulation of CREB (cAMP response element-binding transcription factor),¹⁵⁷ Pax-2 (paired box gene 2),^{118,158} and STAT3 (signal transducer and activator of transcription 3)^{159–162} also contribute to the proliferative phenotype of the cystic epithelium. Cyst-derived epithelial cells also exhibit increased expression and apical localization of the epidermal growth factor (EGF) receptors ErbB1 and ErbB2.^{163,164} Activation of these receptors by EGF-related compounds, which are present in cyst fluid, is likely to contribute to the stimulation of MAPK/ERK signaling and cell proliferation.

Upregulation of PKA signaling promotes cystogenesis via phosphorylation of CFTR in the apical membrane, stimulation of chloride-driven fluid secretion,^{163–170} and possibly other mechanisms such as disruption of tubulogenesis¹⁷¹ and effects on cell–ECM and epithelial cell–macrophage interactions.³

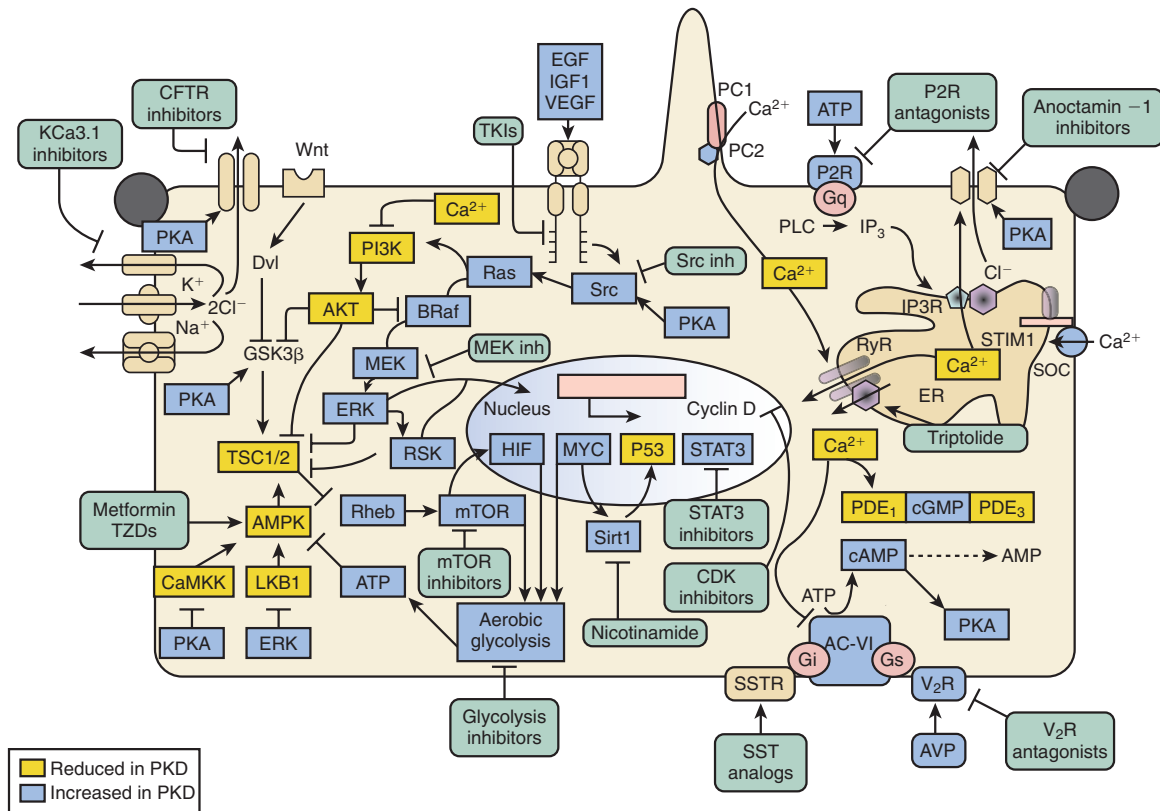


Fig. 45.4 Diagram depicting proposed pathways upregulated or downregulated in polycystic kidney disease and rationale for treatments targeting these pathways (green boxes). Aberrant crosstalk between intracellular calcium (Ca^{2+}) and cyclic adenosine monophosphate (cAMP) signaling may be one of the first consequences of polycystic kidney disease (PKD) mutations. Disrupted calcium may enhance cAMP and protein kinase A (PKA) signaling through activation of calcium-inhibitable adenylyl cyclases and inhibition of calcium-dependent phosphodiesterases [PDEs; PDE₁ and, indirectly, cyclic guanosine monophosphate (cGMP)-inhibited PDE₃]. Enhanced PKA activity may in turn disrupt intracellular calcium homeostasis through phosphorylation of calcium-cycling proteins in the endoplasmic reticulum. PKA-induced phosphorylation of the cystic fibrosis transmembrane conductance regulator (CFTR) allows chloride and fluid secretion into the cysts; anoctamin-1 may synergistically interact with CFTR, further enhancing fluid secretion. PKA activation inhibits cell proliferation in wild-type cells but has a stimulatory effect in PKD cells. Calcium deprivation in wild-type cells and delivery of calcium in PKD cells reverse these effects. A proposed mechanism for the proliferative response in PKD and calcium-deprived wild-type cells is inhibition of phosphoinositide 3-kinase (PI3K) and protein kinase B (AKT), which releases protein BRAF from AKT inhibition. This in turn leads to dysregulation of signaling pathways (BRAF/MEK/ERK; AMPK/mTOR) and transcription factors (HIF1, MYC, P53, STAT3) that control cell cycle progression and energy metabolism. Mislocalization of ErbB (epidermal growth factor receptor) receptors and overexpression of growth factors, cytokines, chemokines, and their receptors further contribute to disease progression. AC-VI, Adenylyl cyclase 6; AMPK, AMP-activated kinase; ATP, adenosine triphosphate; AVP, vasopressin; CaMKK, calcium/calmodulin-dependent protein kinase kinase; CDK, cyclin-dependent kinase; EGF, epidermal growth factor; ER, endoplasmic reticulum; ERK, extracellular signal-regulated kinase; Gi, inhibitory G protein; Gq, a G-protein subunit; Gs, stimulatory G protein; GSK3 β , glycogen synthase kinase 3 β ; HIF, hypoxia-inducible factor; IGF1, insulinlike growth factor 1; inh, inhibition/inhibitor; IP3R, inositol 1,4,5-trisphosphate (IP3) receptor; KCa3.1, a calcium channel; LKB1, liver kinase B1; MEK, mitogen-activated protein kinase kinase; mTOR, mammalian target of rapamycin; P2R, purinergic 2 receptor; PC1, polycystin 1; PC2, polycystin 2; PLC, phospholipase C; Rheb, Ras homolog enriched in brain; RSK, ribosomal s6 kinase; RYR, ryanodine receptor; Sirt1, sirtuin 1; SOC, store-operated channel; SSTR, somatostatin receptor; STAT3, signal transducer and activator of transcription 3; STIM1, stromal interaction molecule 1; TKIs, tyrosine kinase inhibitors; TSC, tuberous sclerosis proteins tuberlin (TSC2) and hamartin (TSC1); TZDs, thiazolidinediones; V₂R, vasopressin V₂ receptor.

Additional ways that extracellular cues detected by the polycystin complex may be transmitted to the nucleus include canonical and noncanonical Wnt, JAK (Janus kinase)/STAT, and NFAT (nuclear factor of activated T cells) pathways.^{169,170} A cleavage event in the G protein-coupled receptor proteolytic site domain, separating the extracellular region from the transmembrane part of the protein, is important for activation of PC1, as is appropriate glycosylation and trafficking.^{90,172,173} It has also been proposed that PC1 may activate transcription directly by cleaving at additional sites and through the translocation of the resulting C-terminal fragments to the nucleus, a process that may be regulated by flow.^{174,175}

PATHOLOGY

Cystic kidneys usually maintain their reniform shape (Fig. 45.5). Their size ranges from minimally or moderately enlarged in early disease to >20 times normal size in advanced disease. Although unusual, striking asymmetry of cyst development may be seen. Both the outer and the cut surfaces show numerous cysts ranging in size from barely visible to several centimeters in diameter. They are distributed evenly throughout both the cortical and medullary parenchyma. The papillae and pyramids are distinguishable in early cases but are difficult or impossible to identify in advanced examples, and the calyces and pelves are often greatly distorted.

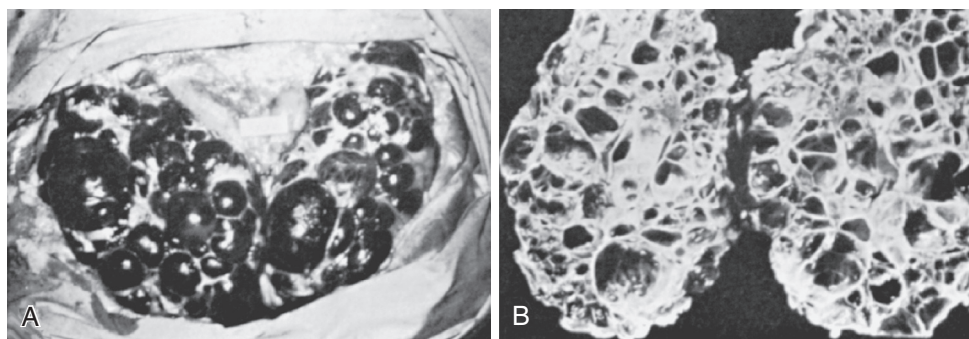


Fig. 45.5 Autosomal dominant polycystic kidney disease (A) in situ and (B) on cut section. Note diffuse, bilateral distribution of cysts. (Courtesy FE Cuppage, Kansas City, KS.)

Nephron reconstruction and microdissection studies revealed that cysts begin as outpouchings from preexisting renal tubules. With enlargement beyond a few millimeters in diameter, most cysts become detached from the tubule of origin. In the early stages of the disease, the noncystic parenchymal elements appear relatively normal because fewer than 1% of the tubules appear to become cystic. The cells in the vast majority of the cysts are not typical of fully differentiated, mature renal tubular epithelium and are thought to be partially dedifferentiated or relatively immature. A minority of the cysts continue to function, as evidenced by their capacity to generate transepithelial electrical gradients and to secrete NaCl and fluid in vitro. The majority of cysts (75%) with Na⁺ levels approximating the level in plasma and relatively leaky apical junctions probably represent cysts with epithelium that is less well differentiated than cysts with low Na concentrations.

Although ADPKD cysts have been thought to arise from all tubular segments, microdissection studies of ADPKD kidneys in the 1960s and 1970s suggested that collecting ducts are diffusely enlarged and that collecting duct cysts are more numerous and larger than those derived from other tubular segments. Most cysts of at least 1 mm in diameter stain positively for collecting duct markers.^{176,177} Studies of *Pkd1* or *Pkd2* rodent models with postnatal development of cystic disease have shown that most cysts originate from the collecting ducts and distal nephron^{86,92,93,178} but that proximal tubule cysts may be common at early stages.¹⁰ Cultured epithelial cells from human ADPKD cysts exhibit a larger cAMP response to 1-deamino-8-D-arginine vasopressin and vasopressin than to the parathyroid hormone, a reaction consistent with a collecting duct origin.¹⁷⁹ These observations indicate that the majority of cysts in adults with ADPKD are derived from the distal nephron and the collecting duct (Fig. 45.6).

At the end stage of the disease, the kidneys are usually several times larger than normal and exhibit innumerable fluid-filled cysts that make up almost all of the total renal mass. In these far-advanced cases, only scant normal-appearing parenchyma may be found in isolated patches. Abundant fibrous tissue is plastered along the surface of the kidney beneath the capsule, and on the cut surfaces of transected kidneys, cysts may be found encapsulated by fibrous bands. Tubulointerstitial fibrosis and arteriolar sclerosis are cardinal features of end-stage polycystic kidney. The disappearance of noncystic parenchyma implicates apoptosis as a primary mechanism in progressive renal dysfunction in ADPKD.

Up to 90% of adults with ADPKD have cysts in the liver.¹⁸⁰ These cysts are lined by a single layer of epithelium resembling that of the biliary tract and contain fluid that resembles the bile salt-independent fraction of the bile. The electrolyte composition and osmolality are similar to those in the serum; the concentrations of phosphorus, cholesterol, and glucose are lower.¹⁸¹ The cysts are derived by progressive proliferation and dilatation of the biliary ductules (biliary microhamartomas or von Meyenburg complexes) and peribiliary glands.^{182,183} Like kidney cysts, liver cysts become detached as they grow, so macroscopic cysts usually do not communicate with the biliary system. Minimal to moderate dilation of the extrahepatic bile ducts is common. In rare kindreds, hepatic changes indistinguishable from those seen in congenital hepatic fibrosis (CHF) can be seen.

DIAGNOSIS

The diagnosis of ADPKD in an individual with a positive family history primarily relies on imaging. Counseling should be performed before testing. Benefits of testing include certainty regarding diagnosis that may influence family planning, early detection and treatment of disease complications, selection of genetically unaffected family members for living-related donor renal transplantation, and now identification of patients for treatment. Potential discrimination in terms of insurability and employment associated with a positive diagnosis should be discussed.

Renal ultrasonography is commonly used because of its low cost and safety (Fig. 45.7). Revised criteria have been proposed to improve the diagnostic performance of ultrasonography in ADPKD (Table 45.1). For at-risk individuals with a 50% chance of having ADPKD (i.e., having a parent or sibling with ADPKD), the presence of at least three (unilateral or bilateral) renal cysts for those aged 15 to 39 years, or of two cysts in each kidney for those aged 40 to 59 years, has a positive predictive value (PPV) of 100% for the diagnosis of ADPKD.¹⁸⁴ For at-risk individuals aged 60 years and older, four or more cysts in each kidney are required. Although the PPVs of these criteria are very high, their sensitivity and negative predictive value (NPV) are low, particularly when applied to 15- to 59-year-old patients with PKD2, and likely hypomorphic *PKD1* alleles. This is a problem in the evaluation of potential kidney donors, in which exclusion of the diagnosis is important. Information on the age at ESKD in other affected family members may be helpful in this setting.¹⁸⁵ A history of at least one affected family member who had ESKD secondary to ADPKD by age 55

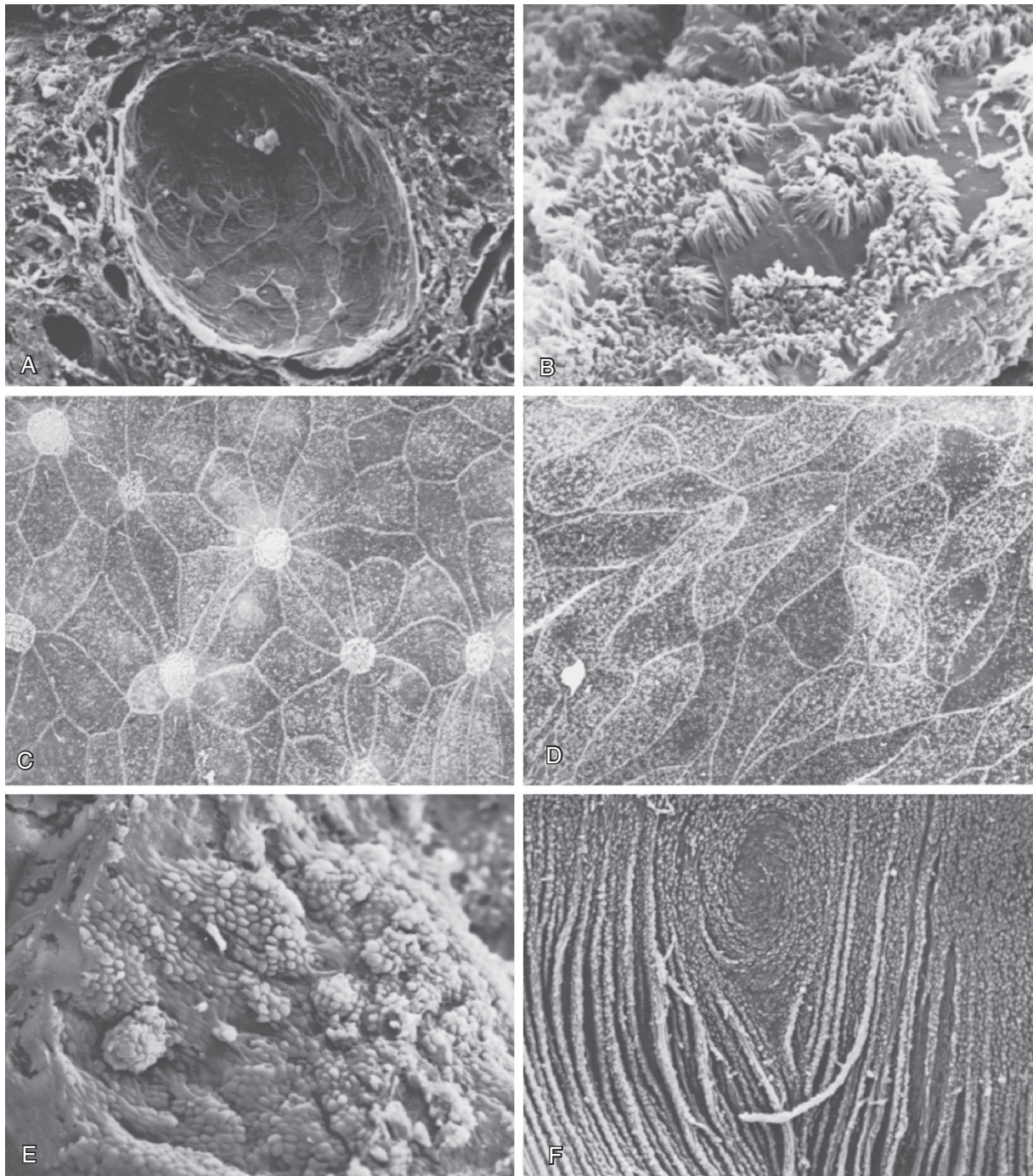


Fig. 45.6 Scanning electron micrographs of epithelium lining a cyst in autosomal dominant polycystic kidney disease. (A) Epithelium typical of glomerular visceral layer ($\times 250$). (B) Epithelium typical of proximal tubule ($\times 3000$). (C) Epithelium typical of cortical collecting duct ($\times 1000$). (D) Epithelium not typical of any normal tubule segment ($\times 1000$). (E) Micropolyps ($\times 250$). (F) Cordlike hyperplasia ($\times 80$). (From Grantham JJ, Geiser JL, Evan AP. Cyst formation and growth in autosomal dominant polycystic kidney disease. *Kidney Int.* 1987;31:1145–1152, with permission.)

years has 100% PPV for PKD1. Conversely, a history of at least one affected family member without ESKD by age 70 or older is predictive of PKD2 or a hypomorphic *PKD1* allele. Different criteria have therefore been proposed to exclude a diagnosis of ADPKD in an individual at risk from a family with an unknown genotype (Table 45.1). An ultrasonographic finding of normal kidneys or one renal cyst in an individual 40 years or older has an NPV of 100%. The absence of any

renal cyst provides near certainty that ADPKD is absent in at-risk individuals aged 30 to 39 years with an NPV of 98.3%. A negative or indeterminate ultrasonography scan result does not exclude ADPKD with certainty in an at-risk individual younger than 30 years. In this setting, the results of magnetic resonance imaging (MRI) or contrast-enhanced computed tomography (CT) provide further assurance, and one study has shown that the finding of a total of

fewer than five renal cysts on MRI is sufficient for disease exclusion.¹⁸⁶

In the absence of a family history of ADPKD, the finding of bilateral renal enlargement and cysts with or without hepatic cysts, as well as absence of other manifestations suggesting a different renal cystic disease, provides presumptive evidence for the diagnosis. Contrast-enhanced CT and MRI provide better anatomic definition than ultrasonography and are more helpful to ascertain the severity and prognosis of the disease (Figs. 45.8 and 45.9).

Genetic testing can be used when the imaging results are equivocal and when a definite diagnosis is required in a younger individual, such as a potential living-related kidney

donor. It may also be helpful in patients with a negative family history, atypical radiologic presentations, and unusually severe or mild disease.¹⁸⁷ Prenatal testing is rarely considered for ADPKD.^{188,189} Preimplantation genetic diagnosis, which is most commonly used in severe genetic diseases with early manifestations, such as cystic fibrosis and ARPKD, is becoming more frequently used in ADPKD, but it is available only in certain countries and the acceptance of this technique is influenced by personal values as well as the severity of the disease.^{188,190}

Genetic testing should be performed by sequence analysis. The large size and complexity of *PKD1* and marked allelic heterogeneity have been barriers to sequence-based molecular testing, but the development of multigene panels and NGS have made this more accessible. Mutation scanning by Sanger sequencing of *PKD1* and *PKD2* or employing a specifically generated NGS panel to screen PKD genes now yields detection rates >90%.^{55,59,63,192} Detection of mutations in the segmentally duplicated *PKD1* gene may be lower by whole-exome sequencing. More than 50% of *PKD1* mutations are unique, with a lower level of unique pathogenic variants to *PKD2*. Approximately one-third of *PKD1* changes are in-frame; careful bioinformatic evaluation, and considering normal and ADPKD population data, can help determine pathogenic from neutral variants. The advances in NGS technologies to screen PKD genes are rendering molecular diagnostics more accessible and less costly in ADPKD and are likely to become more widely used for diagnostics and prognostics.⁵³

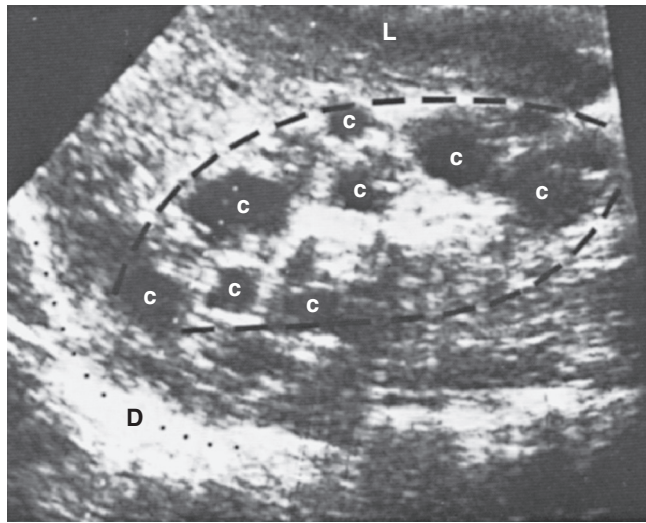


Fig. 45.7 Autosomal dominant polycystic kidney disease seen in a parasagittal or longitudinal sonogram. This view of the right kidney was obtained with the patient in the right anterior oblique position. The approximate outline of the kidney is indicated by the broken line. Some of the larger renal cysts are indicated by Cs. The liver (L) is at the top of the figure. The right dome of the diaphragm (D) is at the lower left.

RENAL MANIFESTATIONS

Cyst Development and Growth

Many manifestations of ADPKD are directly related to renal cyst development and enlargement. A longitudinal study of 241 15–46-year-old patients with creatinine clearance ≤70 mL/min followed prospectively with yearly MRI examinations by the Consortium for Radiologic Imaging Studies of Polycystic Kidney Disease (CRISP) has provided invaluable information to the understanding of how cysts develop and grow.^{193,194}

Table 45.1 Sonographic Criteria for Diagnosis or Exclusion of Autosomal Dominant Polycystic Kidney Disease in At-Risk Individuals^a

Age (years)	Criteria for Positive Diagnosis	Family Genotype					
		Unknown		PKD1		PKD2	
		PPV (%)	Sensitivity (%)	PPV (%)	Sensitivity (%)	PPV (%)	Sensitivity (%)
15–29	≥3 cysts, unilateral or bilateral	100	81.7	100	94.3	100	69.5
30–39	≥3 cysts, unilateral or bilateral	100	95.5	100	96.6	100	94.9
40–59	≥2 cysts in each kidney	100	90.0	100	92.6	100	88.8
≥60	≥4 cysts in each kidney	100	100	100	100	100	100
	Revised Criteria For Diagnosis	NPV (%)	Specificity (%)	NPV (%)	Specificity (%)	NPV (%)	Specificity (%)
	Exclusion						
15–29	≥1 cyst	90.8	97.1	99.1	97.6	83.5	96.6
30–39	≥1 cyst	98.3	94.8	100	96.0	96.8	93.8
40–59	≥2 cysts	100	98.2	100	98.4	100	97.8

^aAt-risk individuals are those known to have a 50% chance of having autosomal dominant polycystic kidney disease (ADPKD) by virtue of having a parent or at least one sibling with ADPKD.
NPV, Negative predictive value; PKD, polycystic kidney disease; PPV, positive predictive value.

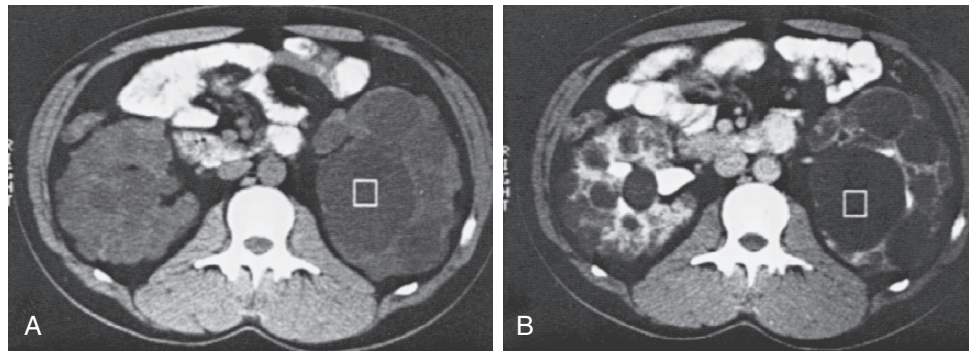


Fig. 45.8 Computed tomography (CT) scans of polycystic kidneys. This male patient has autosomal dominant polycystic kidney disease, and his serum creatinine level is within the normal range. An oral contrast agent was given to highlight the intestine. (A) CT scan without contrast. (B) CT scan at the same level as (A) but after intravenous infusion of iodinated radiocontrast material. The cursor (box) is used to determine the relative density of cyst fluid, which in this case is equal to that of water. Contrast enhancement highlights functioning parenchyma, which here is concentrated primarily in the right kidney. The renal collecting system also is highlighted by contrast material in both kidneys.

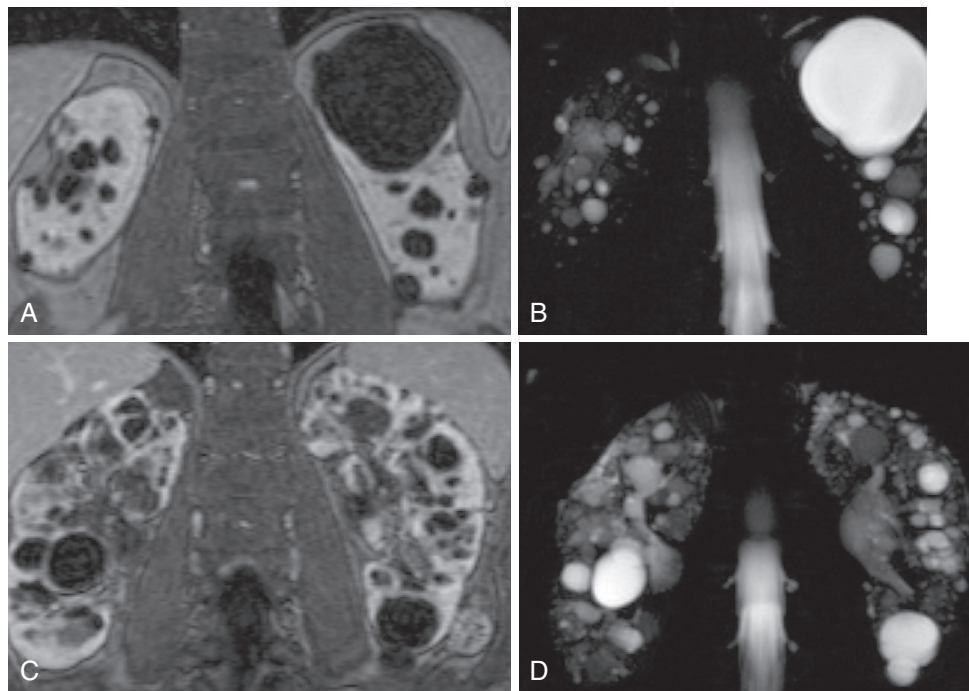


Fig. 45.9 Magnetic resonance imaging studies of two female patients with (A and B) mild and (C and D) moderately severe disease. In neither subject was the serum creatinine value >1.1 mg/dL. For the images in (A) and (C), gadolinium was infused intravenously a few minutes previously. The residual, normal parenchyma between cysts is highlighted by gadolinium. In (B) and (D), heavy T2-weighted images are shown at the same kidney level as in (A) and (C). The cysts are emphasized, illustrating that cysts <3 mm can be detected.

MR images at baseline illustrated the large phenotypic variability in terms of total kidney volume (TKV) in ADPKD. Follow-up images showed the TKV growth to be quasi-exponential, unique to and variable among patients (Fig. 45.10). While TKV changed significantly year after year, glomerular filtration rate (GFR) started declining years later, hence TKV was a good predictor of future estimated GFR (eGFR) decline. A collaborative effort including the PKD Foundation, Food and Drug Administration (FDA), Critical Path Institute, academic centers, and the pharmaceutical industry led to FDA and European Medicines Agency qualification of TKV, together with age and eGFR, as a prognostic biomarker for risk of GFR decline.¹⁹⁵

Measuring TKV, as a prognostic biomarker, does not require high precision. Measurements by the ellipsoid equation and various imaging modalities can be used to inform patients about their prognosis and determine eligibility for clinical trials and therapies as they become available. Its value is limited in atypical cases with markedly asymmetric or coexisting ischemic disease, but the Mayo imaging classification facilitates its use.¹⁹⁶ It is based on criteria to exclude atypical cases and stratify typical cases into five classes (A through E) based on growth rates ($<1.5\%$, $1.5\%–3\%$, $3\%–4.5\%$, $4.5\%–6\%$, or $>6\%$ per year) estimated from patient age and a theoretical starting height-adjusted TKV (150 mL/m). A model based on the classification can predict future GFR

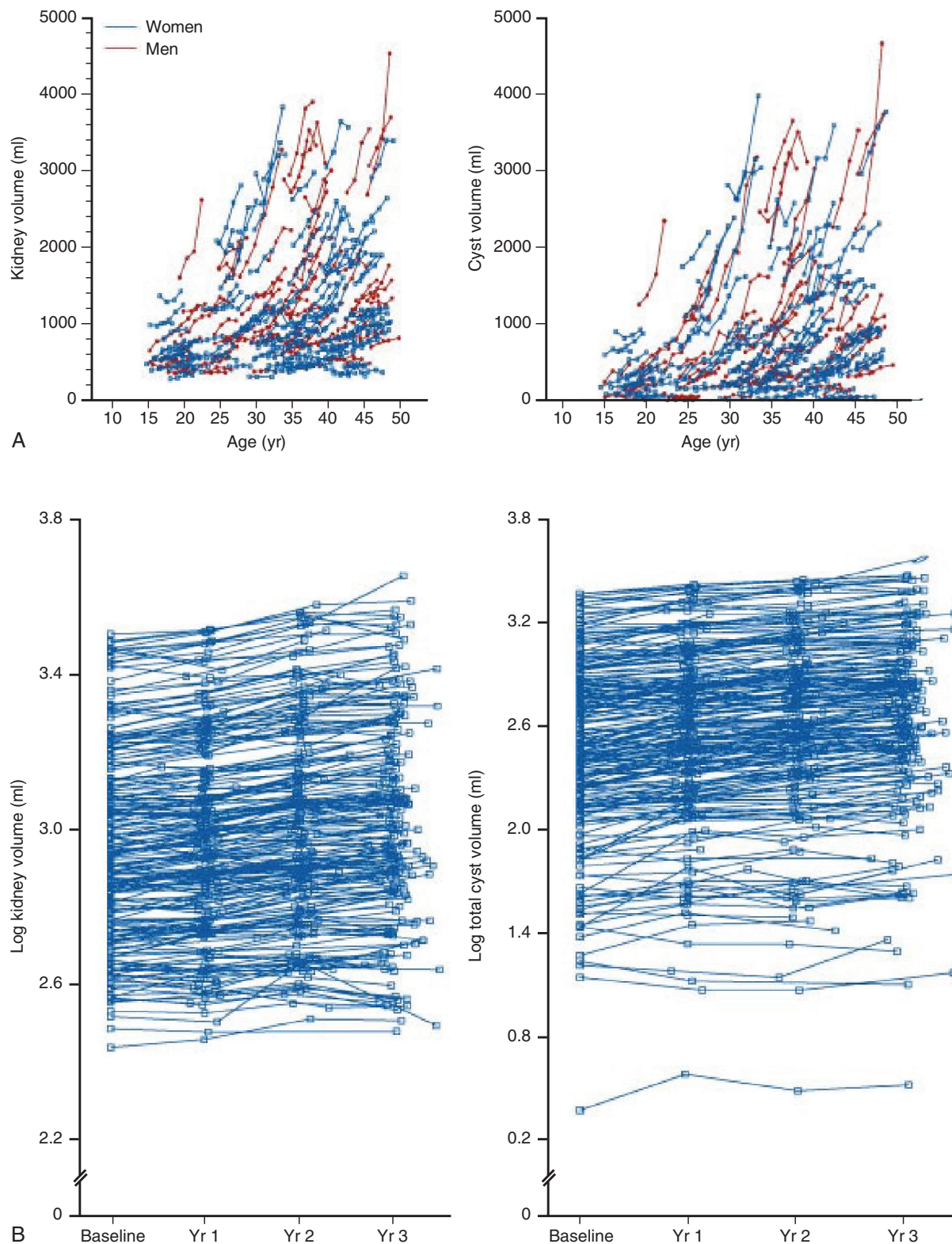


Fig. 45.10 Progression of autosomal dominant polycystic kidney disease. (A) Combined left and right total kidney and cyst volumes in relation to age in women (*blue*) and men (*red*). The lines connecting the four measurements for each patient in the 3 years of follow-up exhibit a concave upward sweep suggestive of an exponential growth process. (B) Log_{10} combined total kidney and cyst volumes in relation to time. The linearity of the four measurements for each patient in the 3 years of follow-up is consistent with an exponential growth process. (Reproduced from Grantham JJ, Torres VE, Chapman AB, et al. Volume progression in polycystic kidney disease. *N Engl J Med*. 2006;354:2122–2130.)

decline with reasonable accuracy (<http://www.mayo.edu/research/documents/pkd-center-adpkd-classification/doc-20094754>). TKV has been used as an endpoint in clinical trials. Measurements for this purpose require high precision, achievable using planimetry, the gold standard, or stereology. Stereology is faster than planimetry but does not segment the kidney, a requirement for advanced image analysis. Fast, automatic segmentations, including a deep learning-based, fully automated approach, capable of replacing humans for the task of segmenting polycystic kidneys, have been developed.^{197,198} Once segmented, advanced MRI processing and analysis such as texture analysis may be superior and/or complementary to TKV in predicting or measuring disease progression.¹⁹⁹

RENAL FUNCTION ABNORMALITIES

Impaired urinary concentrating capacity is common even at early stages of ADPKD. Nearly 60% of children cannot maximally concentrate their urine,²⁰⁰ and plasma vasopressin levels are increased. The vasopressin-resistant concentrating defect is not explained by reduced cAMP or expression of concentration-associated genes, which are consistently increased in animal models. It has not been determined whether the defect is attributable to disruption of the medullary architecture by the cysts or to a cellular defect directly linked to disruption of polycystin function. Newer studies suggest that the urinary concentrating defect and elevated vasopressin values may contribute to cystogenesis. They may also contribute to the glomerular hyperfiltration seen in children and young adults²⁰¹ and to the development of hypertension and the progression of chronic kidney disease (CKD). Defective medullary trapping of ammonia and transfer to the urine caused by the concentrating defect may contribute to the low urine pH values, hypocitruric aciduria, and predisposition to stone formation.

Reduced renal blood flow is another early functional defect.²⁰² It may be caused by changes in intrarenal pressures, neurohumoral or local mediators, and/or intrinsic vascular abnormalities. Mild to moderate persistent proteinuria (150–1500 mg/day) may be found in a significant number of patients in the middle to late stages of the disease and is an indicator of a more progressive disease.²⁰³ Patients with proteinuria may also excrete doubly refractile lipid bodies (oval fat bodies).²⁰⁴

Hypertension

Hypertension (blood pressure ≥ 140 mm Hg systolic/90 mm Hg diastolic), found in approximately 50% of 20–34-year-old patients with ADPKD and normal renal function, is present in nearly 100% of patients with ESKD.²⁰⁵ Development of hypertension is accompanied by a reduction in renal blood flow, an increase in filtration fraction, abnormal renal handling of sodium, and extensive remodeling of the renal vasculature.

The association between renal size and the prevalence of hypertension supports the hypothesis that stretching and compression of the vascular tree by cyst expansion cause ischemia and activation of the renin–angiotensin–aldosterone system (RAAS).²⁰⁶ The expression of PC1 and PC2 in vascular smooth muscle^{207–209} and endothelium,²¹⁰ along with enhanced vascular smooth muscle contractility²¹¹ and impaired endothelium-dependent vasorelaxation,²¹² suggests that a

primary disruption of polycystin function in the vasculature may also play a role in the early development of hypertension and renal vascular remodeling.

Whether circulating angiotensin is instrumental in causing hypertension is controversial.^{213,214} Plasma renin activity and aldosterone values are normal in most studies, but because blood pressures are higher than those of controls, it has been argued that the renin and aldosterone levels are not appropriately suppressed. A 1990 study showed higher levels after short- or long-term administration of an angiotensin-converting enzyme (ACE) inhibitor in normotensive and hypertensive patients with ADPKD and normal renal function than in both normal subjects and patients with essential hypertension.²¹³ Another study found no differences in hormonal or blood pressure responses between patients with ADPKD and patients with essential hypertension matched in terms of renal function and blood pressure under conditions of high and low sodium intake and after the administration of an ACE inhibitor.²¹⁴ Sodium intake was not controlled in the former study, and differences in selection and ethnic composition of the control groups have been offered as possible explanations for the different results.

There is stronger evidence for local activation of intrarenal RAAS. It includes 1. partial reversal of the reduced renal blood flow, increased renal vascular resistance, and increased filtration fraction by short- or long-term administration of an ACE inhibitor^{213,215,216}; 2. shift of immunoreactive renin from the juxtaglomerular apparatus to the walls of the arterioles and small arteries^{217,218}; 3. ectopic synthesis of renin in the epithelium of dilated tubules and cysts^{219,220}; and 4. ACE-independent generation of angiotensin II by a chymase-like enzyme.²²⁰

Nitric oxide–associated endothelium–dependent vasorelaxation has been shown to be impaired in small subcutaneous resistance vessels from patients with normal renal function before the development of hypertension.^{221–223} Other factors proposed to contribute to hypertension in ADPKD include increases in sympathetic nerve activity and plasma endothelin 1 levels as well as insulin resistance.²²⁴

The diagnosis of hypertension in ADPKD is often made late. Twenty-four-hour ambulatory blood pressure monitoring of children or young adults without hypertension often reveals blood pressure elevations, attenuated nocturnal blood pressure dipping, and/or exaggerated blood pressure response during exercise, which may be accompanied by left ventricular hypertrophy and diastolic dysfunction. Early detection and treatment of hypertension are important because cardiovascular disease is the main cause of death in patients with ADPKD.^{40,225} Uncontrolled blood pressure increases the morbidity and mortality from valvular heart disease and aneurysms as well as the risk of proteinuria, hematuria, and a faster decline of renal function. The presence of hypertension also increases the risk of fetal and maternal complications during pregnancy. Normotensive women with ADPKD usually have uncomplicated pregnancies.²²⁶

Pain

Pain is the most frequent symptom (60%) reported by adult patients with ADPKD.^{227,228} Acute pain may be associated with renal hemorrhage, passage of stones, and urinary tract infections. Some patients have chronic flank pain without an identifiable etiology other than the cysts.

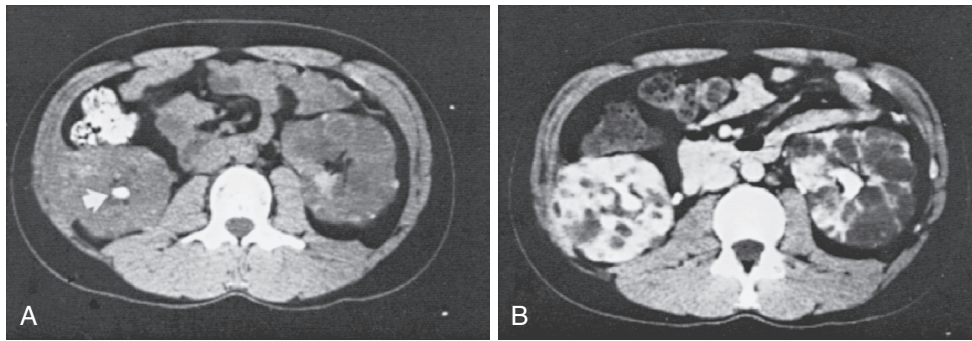


Fig. 45.11 Computed tomography (CT) of polycystic kidneys in a male patient whose serum creatinine level is within the normal range. (A) CT scan without contrast shows a radiopaque stone in the pelvis of the right kidney (*arrow*). (B) CT scan after intravenous administration of an iodinated radiocontrast agent. The stone now is obscured by contrast medium in the renal pelvis.

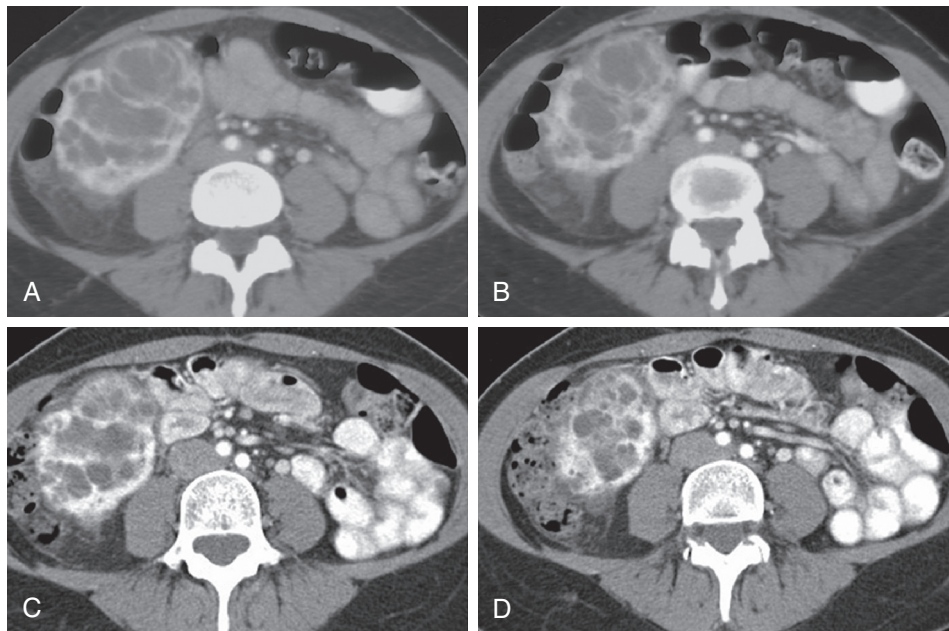


Fig. 45.12 Cyst infection. (A and B) Contrast-enhanced computed tomography (CT) scans demonstrate a 4-cm infected cyst in the anterior portion of the lower pole of the right kidney and inflammatory stranding in the perirenal fat. (C and D) CT scans obtained after 3 weeks of antibiotic therapy show a decrease in the size of the cyst and improved enhancement of the renal parenchyma.

Vascular endothelial growth factor (VEGF) produced by the cystic epithelium²²⁹ may promote angiogenesis, hemorrhage into cysts, and gross hematuria. Symptomatic episodes likely underestimate the frequency of cyst hemorrhage because >90% of patients with ADPKD have hyperdense (CT) or high-signal (MRI) cysts, reflecting blood or high protein content. Most hemorrhages resolve within 2 to 7 days. If symptoms last longer than 1 week or if the initial episode occurs after the age of 50 years, investigation to exclude neoplasm should be undertaken.

Approximately 20% of patients with ADPKD have kidney stones, usually composed of uric acid and calcium oxalate.^{230,231} Metabolic factors include decreased ammonia excretion, low urinary pH, and low urinary citrate concentration. Urinary stasis secondary to the distorted renal anatomy may also play a role. CT of the abdomen before and after contrast enhancement is the best imaging technique to detect small uric acid stones that may be very faint on plain films with tomograms and to differentiate stones from cyst wall and

parenchymal calcifications. Stones may be missed if only a contrast-enhanced CT is obtained (Fig. 45.11). Dual-energy CT can be used to distinguish between calcium and uric acid stones.^{232,233}

As in the general population, urinary tract infections affect women more frequently than men. Most are caused by Enterobacteriaceae.²³⁴ CT and MRI are useful to detect complicated cysts and provide anatomic definition, but the findings are not specific for infection (Fig. 45.12). Nuclear imaging (⁶⁷Ga- or ¹¹¹In-labeled leukocyte scans) may be helpful, but false-negative and false-positive results are possible. Fluorine 18 2-fluoro-2-deoxy-D-glucose (FDG) positron emission tomography (PET) has become a promising agent for detection of infected cysts, but its use to diagnose kidney infections may be difficult because FDG is filtered by the kidneys, is not reabsorbed by the tubules, and appears in the collecting system.^{235–237} Cyst aspiration should be considered when the clinical setting and imaging are suggestive and blood and urine cultures are negative.

Renal cell carcinoma (RCC) is a rare cause of pain in ADPKD. Although it does not occur more frequently than in the patients with other renal diseases,²³⁸ it may manifest at an earlier age in patients with ADPKD, with frequent constitutional symptoms and a higher proportion of sarcomatoid, bilateral, multicentric, and metastatic tumors.²³⁹ A solid mass on ultrasonography, speckled calcifications on CT and contrast enhancement, and tumor thrombus and regional lymphadenopathies on CT or MRI should raise the suspicion of a carcinoma.

Chronic Kidney Disease and Renal Failure

The development of renal failure in ADPKD is highly variable. In most patients, renal function is maintained within the normal range because of compensatory adaptation, despite relentless growth of cysts, until the 4th to 6th decade of life (Fig. 45.13). By the time renal function starts declining, the kidneys usually are markedly enlarged and distorted with little recognizable parenchyma on imaging studies. At this stage, the average rate of decline in GFR is approximately 4.4 to 5.9 mL/min/year.²⁴⁰ The mutated gene (*PKD1* vs. *PKD2*), type of mutation in *PKD1* (truncating versus nontruncating), and modifier genes determine to a significant extent the clinical course of ADPKD (see earlier discussion). Other risk factors are male gender, diagnosis before the age of 30 years, a first episode of hematuria before 30 years of age, onset of hypertension before 35 years of age, hyperlipidemia, low level of high-density lipoprotein cholesterol, and sickle cell trait.^{59,60,241,242} As in other forms of CKD, smoking raises

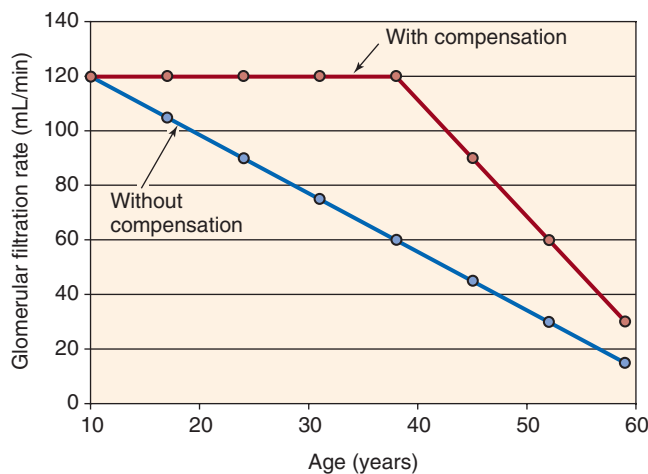


Fig. 45.13 Effects of compensatory maintenance of glomerular filtration rate (GFR) on the pattern of progression in autosomal dominant polycystic kidney disease. It was assumed that, beginning at the age of 10 years, the patient loses an amount of parenchyma each year that normally contributes 2 mL/min of GFR. It was further assumed that each residual normal glomerulus can double the single-nephron GFR by compensatory mechanisms (as seen in normal individuals by the maintenance of total GFR after uninephrectomy for kidney donation). As seen in the model, total GFR is maintained until parenchymal loss precludes complete compensation; at that point, total GFR begins to fall at a rate that appears more “precipitous” than what had actually occurred. This model illustrates that GFR is a poor indicator of autosomal dominant polycystic kidney disease (ADPKD) progression and that more sensitive markers of parenchymal loss are needed to facilitate earlier monitoring.

the risk for ESKD, at least in some patient subsets, such as male smokers with no history of ACE inhibitor treatment.²⁴³

While several factors contribute to renal function decline, a particularly strong relationship with renal enlargement has been noted. CRISP has confirmed this relationship and has shown that kidney and cyst volumes are the strongest predictors of renal functional decline.²⁴⁴ After a mean follow-up of 7.9 years, 30.7% of CRISP enrollees reached stage 3 CKD. Correlations of htTKV at baseline with GFR at different time points increased from -0.22 (GFR at baseline) to -0.65 (GFR at year 8).²⁴⁵ A htTKV of ≥ 600 mL/m at baseline most accurately defined the risk for development of stage 3 CKD within 8 years [area under the curve of 0.84 in a receiver operator characteristic analysis, 95% confidence interval, 0.79 to 0.90]. Kidney volume was a better predictor of GFR decline than baseline age, serum creatinine, blood urea nitrogen (BUN), urinary albumin, or monocyte chemotactic protein-1 (MCP-1) excretion. At the latest analysis, completed after a median follow-up of 13 years, 19% of patients in CRISP have now reached stage 5 CKD or end-stage renal disease (ESRD).²⁴⁶ Baseline htTKV remained a strong independent predictor, with an odds ratio of 1.35 (1.18–1.55) for reaching stage 5 CKD or ESRD for each 100 mL/m increment in htTKV. While PKD genotype was also associated with CKD outcome, it was no longer independently associated with outcomes after adjusting for htTKV, suggesting that the effect of ADPKD genotype on kidney function may largely be mediated by changes in htTKV.

CRISP has also shown that reduced renal blood flow (or increased vascular resistance) is an independent predictor of GFR decline.²⁰² This factor points to the importance of vascular remodeling in the progression of the disease and may account for cases in which the decline of renal function seems to be out of proportion to the severity of the cystic disease. Angiotensin II, transforming growth factor- β (TGF- β), and reactive oxygen species may contribute to the vascular lesions and interstitial inflammation and fibrosis by stimulating the synthesis of chemokines, ECM, and metalloproteinase inhibitors. The expression of MCP-1 and osteopontin are increased in cyst epithelial cells. MCP-1 is found in cyst fluids in high concentrations, and the urinary excretion is increased.²⁴⁷ Other factors such as heavy use of analgesics may contribute to CKD progression in some patients.

Patients with ADPKD and advanced CKD have less anemia than patients with other renal diseases because of enhanced production of erythropoietin by the polycystic kidneys.

EXTRARENAL MANIFESTATIONS

Polycystic Liver Disease

Polycystic liver disease (PLD) is the most common extrarenal manifestation of ADPKD. It is associated with both *PKD1* and non-*PKD1* genotypes.^{52,248,249} Although hepatic cysts are rare in children, their frequency increases with age and may have been underestimated by ultrasonography and CT studies. Their prevalence according to MRI in the CRISP study were 58%, 85%, and 94% in 15- to 24-, 25- to 34-, and 35- to 46-year-old participants, respectively.¹⁸⁰ Hepatic cysts are more prevalent and hepatic cyst volume is larger in women than in men. Women who have multiple pregnancies or who have used oral contraceptive agents or estrogen replacement therapy have more severe disease, suggesting an estrogen

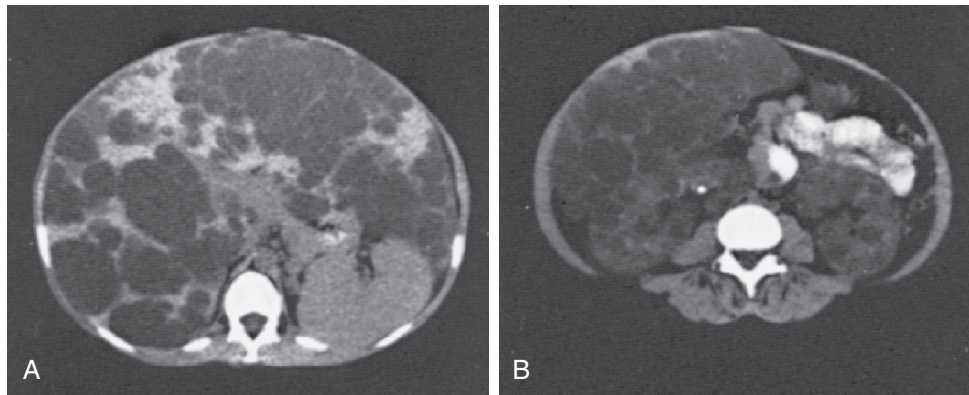


Fig. 45.14 Computed tomography (CT) scan of polycystic liver and kidneys in a female patient with autosomal dominant polycystic kidney disease. The serum creatinine level and liver function test results were within the normal range. An oral contrast agent was given to highlight the intestine, but no intravenous contrast was used. (A) Massive enlargement of the liver caused by intraparenchymal cysts. (B) CT scan at a lower level in the abdomen shows cystic kidneys and the lower portion of the cystic liver.

effect on hepatic cyst growth.^{206,250} Estrogen receptors are expressed in the epithelium lining the hepatic cysts, and estrogens stimulate proliferation of hepatic cyst–derived cells.

Typically, PLD is asymptomatic, but symptoms have become more frequent as the lifespan of patients with ADPKD has lengthened because of dialysis and transplantation. Symptoms may result from the mass effect or from complicating infection and hemorrhage (Fig. 45.14). Symptoms typically caused by massive enlargement of the liver or by mass effect from a single or a limited number of dominant cysts include dyspnea, early satiety, gastroesophageal reflux, and mechanical lower back pain. Other complications caused by mass effect include hepatic venous outflow obstruction, inferior vena cava compression, portal vein compression, and bile duct compression manifesting as obstructive jaundice.²⁵¹

Symptomatic complications of hepatic cysts include cyst hemorrhage, infection, and, rarely, torsion or rupture. The typical manifestation of cyst infection consists of localized pain, fever, leukocytosis, elevated erythrocyte sedimentation rate, and, often, elevated alkaline phosphatase. It is usually monomicrobial and caused by Enterobacteriaceae.^{252–254} MRI sensitively differentiates between complicated and uncomplicated hepatic cyst. On CT, fluid-debris levels within cysts, cyst wall thickening, intracystic gas bubbles, and heterogeneous or increased density have been associated with infection. Radionuclide imaging and FDG-PET scanning have been used for diagnosis.²⁵⁵

Mild dilation of the common bile duct has been observed in 40% of patients with PLD studied by CT and may rarely be associated with episodes of cholangitis.²⁵⁶ Rare associations of PLD include CHF, adenomas of the ampulla of Vater, and cholangiocarcinoma.

Autosomal dominant PLD (ADPLD) occurs as a genetically distinct disease with no or few renal cysts. Similar to ADPKD, ADPLD is genetically heterogeneous, with initially two genes identified: *PRKCSH* (chromosome 19) and *SEC63* (chromosome 6), each accounting for approximately one-third of isolated ADPLD cases.^{257–259} Subsequently, whole-exome sequencing showed that *LRP5* mutations (chromosome 11) are associated with hepatic cystogenesis,²⁶⁰ and recently mutations in *GANAB*, *ALG8*, *SEC61B*, and *PKHD1* have also been associated with ADPLD.^{52,249,261}

Cysts in Other Organs

Cysts are found in the pancreas in approximately 5%, arachnoid in approximately 8%, and seminal vesicles in approximately 40% of patients with ADPKD.^{262–267} Seminal vesicle cysts rarely result in infertility.²⁶⁸ Defective sperm motility is another cause of male infertility in ADPKD.²⁶⁹ Pancreatic cysts are almost always asymptomatic, with very rare occurrences of recurrent pancreatitis. It is uncertain whether the reported association of carcinoma of the pancreas represents more than chance. Arachnoid membrane cysts are usually asymptomatic but may increase the risk for subdural hematomas.^{267,270} Spinal meningeal diverticula may occur with increased frequency and rarely manifest as intracranial hypotension due to a cerebrospinal fluid leak.²⁷¹ Ovarian cysts are not associated with ADPKD.

Vascular Manifestations

Vascular manifestations of ADPKD include IAs and dolichoectasias, thoracic aortic and cervicocephalic artery dissections, and coronary artery aneurysms. They are caused by alterations in the vasculature directly linked to mutations in *PKD1* or *PKD2*. PC1 and PC2 are expressed in VSMCs,^{207–209} *Phd2*^{+/–} VSMCs exhibit increased rates of proliferation and apoptosis, and *Phd2*^{+/–} mice have a greater susceptibility to vascular injury and premature death when hypertension is induced to develop.^{95,132}

IAs occur in approximately 6% of patients with a negative family history and 16% of those with a positive family history of aneurysms²⁷² and are most often asymptomatic. Focal findings such as cranial nerve palsy and seizure result from compression of local structures. The risk of rupture depends on many factors (see later discussion). Rupture carries a 35% to 55% risk of combined severe morbidity or mortality.²⁷³ The mean age at rupture is lower than in the general population (39 years vs. 51 years, respectively). Most patients have normal renal function, and up to 29% have normal blood pressure, at the time of rupture.

Cardiac Manifestations

Mitral valve prolapse observed by echocardiography is the most common valvular abnormality, found in up to 25% of patients with ADPKD.^{274,275} Aortic insufficiency may occur

in association with dilation of the aortic root.²⁷⁶ Although these lesions may progress with time, they rarely require valve replacement. Screening echocardiography is not indicated unless a murmur is detected on physical examination. Nonhemodynamically significant pericardial effusions are frequently observed.²⁷⁷ Cardiomyopathies (dilated, hypertrophic, and left ventricular noncompaction) and atrial fibrillation may occur more frequently in ADPKD patients compared with the general population.²⁷⁸

Diverticular Disease

Colonic diverticulosis and diverticulitis are more common in patients with ADPKD and ESKD than in those with other renal diseases. Whether this increased risk extends to patients before the onset of ESKD is uncertain.²⁷⁹ There have been reports of extracolonic diverticular disease.²⁸⁰ It may become clinically significant in a minority of patients. Subtle alterations in polycystin function may enhance the smooth muscle dysfunction from aging, which is thought to underlie the development of diverticula.

Bronchiectasis

PC1 is expressed in the motile cilia of airway epithelial cells. Bronchiectasis occurs three times more frequently in patients with ADPKD than in control individuals (37% vs. 13%, respectively; $P < .002$), as detected by CT.²⁸¹

TREATMENT

Current therapy of ADPKD has been directed toward limiting the morbidity and mortality from the complications of the disease, but specific therapies are now becoming available.

Hypertension

There is no proven antihypertensive agent of choice in ADPKD. ACE inhibitors or angiotensin receptor blockers (ARBs) increase renal blood flow, have a low side-effect profile, and may have renoprotective properties beyond blood pressure control. Some studies have shown better preservation of renal function or reduction in proteinuria and left ventricular hypertrophy with ACE inhibitors or ARBs than with diuretics or calcium channel blockers,^{282–284} but other studies have been unable to detect the superiority of these drugs.²⁸⁵ A metaanalysis of 142 patients with ADPKD in eight randomized clinical trials showed that ACE inhibitors were more effective in lowering urine protein excretion and slowing kidney disease progression in patients with higher levels of proteinuria, but the overall kidney disease progression was not significantly different (29% in the inhibitor group vs. 41% in the control group).²⁸⁶ Most studies have been limited by inadequate power, short follow-ups, wide ranges of renal function, and the use of doses with inadequate pharmacologic effects.

Equally controversial has been the optimal blood pressure target. In the Modification of Diet in Renal Disease (MDRD) study, patients with ADPKD and a baseline GFR between 13 and 24 mL/min/1.73 m² assigned to a low blood pressure target (≤ 92 mm Hg) had faster declines in GFR than those assigned to a standard blood pressure goal (≤ 107 mm Hg). The reason may be the inability to autoregulate renal blood flow.²⁴⁰ The rate of decline in participants with a baseline GFR between 25 and 55 mL/min/1.73 m² was not affected by the blood pressure target over a mean intervention period of

2.2 years. However, an extended follow-up of these patients showed a delayed onset of kidney failure and a reduced composite outcome of kidney failure and all-cause mortality in the low blood pressure target group (51% of them taking ACE inhibitors) in comparison with those in the standard blood pressure target group (32% of them taking ACE inhibitors).²⁸⁷ The magnitude of this beneficial effect was similar to that observed in patients with other renal diseases.

In study A of the HALT Progression of Polycystic Kidney Disease (HALT-PKD) clinical trial, 558 hypertensive patients with ADPKD (15 to 49 years of age, with eGFRs > 60 mL/min/1.73 m²) were randomly assigned to either a standard blood pressure target (120/70 to 130/80 mm Hg) or a low blood pressure target (95/60 to 110/75 mm Hg) and to either lisinopril plus telmisartan or lisinopril plus placebo.²⁸⁸ In study B, 486 hypertensive patients with ADPKD (18 to 64 years of age, with eGFRs 25 to 60 mL/min/1.73 m²) were randomly assigned to receive lisinopril plus telmisartan or lisinopril plus placebo, with the doses adjusted to achieve a blood pressure of 110/70 to 130/80 mm Hg.²⁸⁹ Both studies showed that an ACE inhibitor alone adequately controlled hypertension in most patients; the addition of an ARB did not show additional benefit, justifying the use of ACE inhibitors for first-line treatment for hypertension in this disease. Study A showed that lowering blood pressure to levels below those recommended by current guidelines in young patients with good kidney function reduced the rate of increase in kidney volume by 14%, the increase in renal vascular resistance, urine albumin excretion (all identified in CRISP as predictors of renal function decline), left ventricular mass (LVM) index, and, marginally (after the first 4 months of treatment), the rate of decline in eGFR.²⁸⁸ The overall effect of low blood pressure on eGFR, however, was not statistically significant, possibly because the reduction of blood pressure to low levels was associated with an acute reduction in eGFR within the first 4 months of treatment. When stratified by disease severity classification by TKV/age, class D and E patients assigned to intensive treatment had slower annual TKV increase (6.4% vs. 7.8%, $P = .033$), and eGFR decline after month 4 (-3.36 vs. -4.44 , $P = .011$) and arguably overall (-3.57 vs. -4.37 , $P = .051$).²⁹⁰ These results underline the importance of early detection and treatment of hypertension in ADPKD.

Several studies suggest that improved blood pressure control in patients with ADPKD over the last 2 decades has been associated with reduced cardiovascular morbidity and mortality. A small prospective study from the University of Colorado showed that rigorous blood pressure control caused a greater decrease in LVM without a detectable effect on renal function.²⁹¹ A population-based study using the UK General Practice Research Database found that increased use of antihypertensive drugs over the period from 1991 to 2008 in patients with ADPKD, particularly of agents blocking the RAAS, was accompanied by reduced mortality.²⁹² The prevalence of left ventricular hypertrophy, assessed by MRI in 543 hypertensive HALT-PKD study A patients (mean age 36 years) with normal renal function at baseline, was 3.9% by nonindexed LVM and 0.9% by LVM index at enrollment, much lower than that observed in earlier studies.²⁹³ This likely reflects the excellent blood pressure control (mean approximately 124/82 mm Hg) and the high utilization of RAAS blockers (61%).²⁹⁴ In Danish patients with ADPKD and ESKD,

cardiovascular and cerebrovascular deaths decreased from 1993 to 2008, possibly owing to a greater effectiveness of antihypertensive treatment.²⁹⁵

There is less evidence that the better control of hypertension over the last 2 decades has delayed the progression to ESKD. Two observational studies suggested that in patients with ADPKD, the average age at start of renal replacement therapy (RRT) has increased considerably during the last 2 decades.^{296,297} A study of European Renal Association–European Dialysis and Transplant Association Registry data on patients starting RRT between 1991 and 2010, spanning 12 European countries with 208 million inhabitants, also showed that mean age at onset of RRT among patients with ADPKD ($n = 20,596$) has risen, albeit considerably less than in the two aforementioned studies, from 56.6 to 58.0 years.²⁹⁸ Although the RRT incidence did not change among patients with ADPKD younger than 50 years, it increased among older patients (>70 years of age). These data suggest that the greater age of patients with ADPKD at the start of RRT may be explained by increased access of the elderly to RRT or by the lower competing risk of mortality prior to the start of RRT, rather than the consequence of effective renoprotective therapies.^{299,300}

Pain

Causes of pain in ADPKD that may require intervention, such as infections, stones, and tumors, should be excluded. Long-term administration of nephrotoxic agents such as nonsteroidal antiinflammatory drugs should be avoided, and narcotic analgesics should be reserved for acute episodes. Psychological evaluation and an understanding and supportive attitude on the part of the physician are essential to minimize the risk for narcotic and analgesic dependence in patients with chronic pain. Reassurance, lifestyle modification, avoidance of aggravating activities, tricyclic antidepressants, and pain clinic interventions such as splanchnic nerve blockade with local anesthesia or steroids may be helpful.^{227,228}

When conservative measures fail, surgical interventions can be considered. Aspiration of large cysts under ultrasonographic or CT guidance is a simple procedure and may help identify the cause of the pain. Sclerosing agents may be used to prevent the reaccumulation of fluid. When multiple cysts contribute to pain, laparoscopic or surgical cyst fenestration through lumbotomy or flank incisions may be of benefit.³⁰¹ Laparoscopy is as effective as open surgical fenestration for patients with limited disease and has a shorter, less complicated recovery period.^{302,303} Surgical interventions do not accelerate the decline in renal function as was once thought, but they do not preserve declining renal function either. Laparoscopic renal denervation or thoracoscopic symptho-splanchnicectomy can be considered, particularly in polycystic kidneys without large cysts.^{304,305} Percutaneous transluminal renal denervation has been proposed as a potential treatment option for PKD-related pain but has not yet been adequately tested.^{306,307} Laparoscopic or retroperitoneoscopic nephrectomy is indicated for symptomatic patients with ESKD. Arterial embolization is an alternative when the surgical risk is high, but its role has not been fully defined.

Cyst Hemorrhage

Cyst hemorrhages are usually self-limiting and respond to conservative management with bed rest, analgesics, and

hydration. When a subcapsular or retroperitoneal hematoma is causing significant decrease in hematocrit and hemodynamic instability, hospitalization, transfusion, and investigation by CT or angiography become necessary. In cases of unusually severe or persistent hemorrhage, segmental arterial embolization can be successful. If not, surgery may be required to control bleeding. Segmental arterial embolization or surgery may be required in some cases. The antifibrinolytic agent tranexamic acid has been successfully used in some cases, but no controlled studies of its use have been performed,³⁰⁸ and the dose needs to be reduced in the presence of renal insufficiency. Potential adverse effects of this treatment include glomerular thrombosis and ureteral obstructions due to clots.

Cyst Infection

Cyst infections are often difficult to treat.²³⁴ Treatment failure may occur because of poor antibiotic penetration into the cysts, but lipophilic agents such as quinolones, as well as trimethoprim–sulfamethoxazole and chloramphenicol, penetrate the cysts consistently. If fever persists after 1 to 2 weeks of appropriate antimicrobial therapy, percutaneous or surgical drainage of infected cysts or, in the case of end-stage polycystic kidneys, nephrectomy should be undertaken. If fever recurs after antibiotic therapy is stopped, a complicating feature such as obstruction, perinephric abscess, or a stone should be excluded. If none is identified, several months of antibiotic therapy may be required to eradicate the infection.

Nephrolithiasis

Treatment is similar to that in patients without ADPKD. Potassium citrate is indicated for the three causes of stones associated with ADPKD: uric acid lithiasis, hypocitraturic calcium oxalate nephrolithiasis, and distal acidification defects. Extracorporeal shock wave lithotripsy and percutaneous nephrostolithotomy have been performed successfully without undue complications.

End-Stage Kidney Disease

Patients with ADPKD do better on dialysis than patients with other causes of ESKD, perhaps because of higher levels of erythropoietin and hemoglobin or lower comorbidity.³⁰⁹ Despite renal size and increased risk for hernias, peritoneal dialysis is usually possible.

Transplantation is the treatment of choice for ESKD in ADPKD. There is no difference in patient or graft survival between patients with ADPKD and other ESKD populations. Graft survival after living-donor transplants is also no different in patients with and without ADPKD. However, data are more limited for those with ADPKD, in whom living-related-donor transplantation was not widely practiced in the past. In 1999, for instance, 30% of kidney transplants for ADPKD patients were from living donors, compared with 12% in 1990.

Complications after transplantation are no greater in the ADPKD population than in the general population, and specific complications directly related to ADPKD are rare. Cyst infection is not increased after transplantation, and there is no significant increase in the incidence of symptomatic mitral valve prolapse or hepatic cyst infection. One study showed a higher rate of diverticulosis and bowel perforation in ADPKD. Whether ADPKD increases the risk for development of new-onset diabetes mellitus after transplantation is controversial.

Pretransplantation nephrectomy, commonly used in the past, has fallen out of favor. By 1 and 3 years following renal transplantation, kidney volumes decrease by 37.7% and 40.6%, and liver volumes increase by 8.6% and 21.4%, respectively.³¹⁰ Indications for nephrectomy include a history of infected cysts, frequent bleeding, severe hypertension, and massive renal enlargement with extension into the pelvis. There is no evidence for an increased risk for development of RCC in native ADPKD kidneys after transplantation. When nephrectomy is indicated, hand-assisted laparoscopic nephrectomy is associated with less intraoperative blood loss, less postoperative pain, and faster recovery than open nephrectomy and is increasingly being used.³⁰³

Polycystic Liver Disease

PLD is usually asymptomatic and requires no treatment. When it is symptomatic, therapy is directed toward reducing cyst volume and hepatic size. Noninvasive measures include avoiding ethanol, other hepatotoxins, and possibly cAMP agonists (e.g., caffeine), which have been shown to stimulate cyst fluid secretion *in vitro*. Estrogens are likely to contribute to cyst growth, but the use of oral contraceptive agents and postmenopausal estrogen replacement therapy are contraindicated only if the liver is significantly enlarged and the risk for further hepatic cyst growth outweighs the benefits of estrogen therapy. Rarely, symptomatic PLD may require invasive measures to reduce cyst volume and hepatic size. Options include percutaneous cyst aspiration and sclerosis, laparoscopic fenestration, open surgical hepatic resection/cyst fenestration, selective hepatic artery embolization, and liver transplantation.^{311,312} Cyst aspiration is the procedure of choice if symptoms are caused by one or a few dominant cysts or by cysts that are easily accessible to percutaneous intervention. To prevent the reaccumulation of cyst fluid, sclerosis with minocycline or 95% ethanol is often successful. Laparoscopic fenestration can be considered for large cysts that are more likely to recur after ethanol sclerosis or if several cysts are present that would require multiple percutaneous passes to be treated adequately. Partial hepatectomy with cyst fenestration is an option because PLD often spares a part of the liver with adequate preservation of hepatic parenchyma and liver function.³¹³ In cases in which no segments are spared, liver transplantation may be necessary.

When a hepatic cyst infection is suspected, any cyst with an unusual appearance by imaging should be aspirated for diagnostic purposes. The best management is percutaneous cyst drainage in combination with antibiotic therapy. Long-term oral antibiotic suppression or prophylaxis should be reserved for relapsing or recurrent cases. Antibiotics of choice are trimethoprim-sulfamethoxazole and the fluoroquinolones, which are effective against the typical infecting organisms and concentrate in the biliary tree and cysts.

Intracranial Aneurysm

Widespread presymptomatic screening is not indicated because it yields mostly small aneurysms in the anterior circulation with a low risk of rupture. Indications for screening in patients with good life expectancy include a family history of aneurysm or subarachnoid hemorrhage, previous aneurysm rupture, preparation for major elective surgery, high-risk occupations (e.g., airline pilots), and patient anxiety despite

adequate information.²⁷² MR angiography does not require intravenous contrast material. CT angiography is a satisfactory alternative when there is no contraindication to intravenous contrast agents.

When an asymptomatic aneurysm is found, the recommendation of whether to intervene depends on size, site, and morphology; prior history of subarachnoid hemorrhage from another aneurysm; patient age and general health; and whether the aneurysm is coilable or clippable. The prospective arm of the International Study of Unruptured Intracranial Aneurysms (ISUIA) provided invaluable information to assist in the decision.³¹⁴ The 5-year cumulative rupture rates for patients without a previous history of subarachnoid hemorrhage with aneurysms located in the internal carotid artery, anterior communicating or anterior cerebral artery, or middle cerebral artery were 0%, 2.6%, 14.5%, and 44.0% for aneurysms <7, 7 to 12, 13 to 24, and ≥25 mm, respectively, compared with rates of 2.5%, 14.5%, 18.4%, and 50%, respectively, for the same size categories involving the posterior circulation and posterior communicating artery. Among unruptured noncavernous segment aneurysms <7 mm in diameter, the rupture risks were higher among patients who had a previous subarachnoid hemorrhage from another aneurysm. As reported by the ISUIA, these risks also need to be balanced with those associated with surgical or endovascular surgery. The 1-year mortality and combined morbidity (Rankin score 3 to 5 or impaired cognitive status) and mortality rates were 2.7% and 12.6%, respectively, for open surgery and 3.4% and 9.8%, respectively, for endovascular repair.

The risk for development of new aneurysms or enlargement of an existing one in patients with ADPKD is very low in those with small (<7 mm) aneurysms detected by presymptomatic screening and moderate in those with a previous rupture from a different site.³¹⁵⁻³¹⁷ On the basis of these and the ISUIA data, conservative management is usually recommended for patients with ADPKD who have small (<7 mm) aneurysms detected by presymptomatic screening, particularly in the anterior circulation. Semiannual or annual imaging studies of an aneurysm are appropriate initially, but reevaluation at less frequent intervals may be sufficient after the stability of the aneurysm has been documented. Elimination of tobacco use and aggressive treatment of hypertension and hyperlipidemia should be recommended.

The risk for development of a new aneurysm after an initial negative result is small, about 3% at 10 years in patients with a family history of IAs.³¹⁸ Therefore rescreening of patients with a family history of IAs after an interval of 5 to 10 years seems reasonable.

Novel Therapies

A better understanding of the pathophysiology and the availability of animal models has facilitated the development of preclinical trials and identification of promising candidate drugs for clinical trials (<http://www.clinicaltrials.gov>).

Vasopressin V₂ Receptor Antagonists. The effect of vasopressin, via V₂ receptors, on cAMP levels in the collecting duct, the major site of cyst development in ADPKD, and the role of cAMP in cystogenesis provided the rationale for preclinical trials that showed effectiveness of V₂R antagonists in animal models of ADPKD, ADPKD, and NPHP.^{128,178,319} The initial

studies with tolvaptan, conducted in small numbers of patients with ADPKD treated for 1–3 weeks, showed that split twice-daily doses were necessary for effective V_2R inhibition (urine osmolality to <300 mOsm/kg continuously for 24 hours).¹⁹ The aquaresis induced by daily split doses in patients with CKD stages 1–4 was accompanied by small reductions in GFR without significant changes in renal plasma or blood flow, likely due to activation of tubuloglomerular feedback.^{320,321} These changes, which were rapidly reversible after discontinuation of tolvaptan, underlined the importance of using eGFR values either off treatment (baseline and after washout) or on treatment when comparing the effects of tolvaptan and placebo on the rate of eGFR decline.

The double-blind Tolvaptan Efficacy and Safety in Management of Autosomal Dominant Polycystic Kidney Disease and Its Outcomes (TEMPO) 3:4 trial randomly assigned 1445 patients to tolvaptan at the highest tolerated dose (45/15, 60/30, or 90/30 mg) or placebo.³²² Entry criteria were designed to enrich for patients with relatively preserved renal function and rapidly progressive disease (18–50 years, estimated creatinine clearance ≥ 60 mL/min, TKV ≥ 750 mL). TKV increased 2.8% per year in the tolvaptan group and 5.5% per year in the placebo group ($P < .001$). Tolvaptan significantly reduced the decline in eGFR from 10.1 to 6.8 mL/min/1.73 m² over 3 years, as well as the frequency of kidney pain and urine excretion of albumin and monocyte chemoattractant protein-1. The eGFR benefit at the end of TEMPO 3:4 was maintained for two additional years when all patients received tolvaptan in an open label extension (TEMPO 4:4).³²³ With monitoring every 4 or 3 months in TEMPO 3:4 or TEMPO 4:4, transaminase elevations more than three times upper limit of normal (ULN) occurred in 4.4% of tolvaptan and 1% of placebo patients. A total of 3 of 1271 tolvaptan-treated patients met Hy's law criteria of hepatocellular toxicity. On the basis of the TEMPO 3:4 results, tolvaptan was approved for rapidly progressive ADPKD in Japan, Canada, the European Union, Switzerland, Norway, South Korea, and Australia. In the United States, the FDA did not approve tolvaptan and asked for additional data.

The double-blind Replicating Evidence of Preserved Renal Function: An Investigation of Tolvaptan Safety and Efficacy in later stage ADPKD (REPRISE) was designed as a randomized withdrawal trial to limit early discontinuations due to aquaretic side effects.³²⁴ After an 8-week prerandomization, sequential placebo, and tolvaptan run-in phases, 1370 patients aged 18–55 years with eGFRs of 25–65 mL/min/1.73 m² or 56–65 years with eGFRs of 25–44 mL/min/1.73 m² able to tolerate 60/30-mg tolvaptan, were randomly assigned to tolvaptan or placebo for 12 months with monthly safety and serum creatinine measurements. Three additional measures of serum creatinine were obtained 7–40 days after discontinuation of tolvaptan or placebo. The eGFR change from baseline was -2.34 mL/min/1.73 m² in the tolvaptan and -3.61 mL/min/1.73 m² in the placebo group ($P < .001$). Transaminase elevations over three times ULN occurred in 5.6% of tolvaptan and 1.2% of placebo patients. No cases met Hy's law criteria, likely due to more frequent monitoring and earlier discontinuation of tolvaptan. Transaminase elevations were reversible after stopping tolvaptan. Following the results of this study, tolvaptan has now been approved for use in the US in adult ADPKD patients that are at risk of rapidly progressing disease.

Somatostatin Analogs. Binding of somatostatin to its receptors (somatostatin receptor 1; SSTR1 through SSTR5) inhibits AC and MAPK, cell proliferation, and secretion of several hormones (growth hormone, insulin, glucagon, gastrin, cholecystokinin, vasoactive intestinal peptide and secretin, thyroid-stimulating hormone, and adrenocorticotrophic hormone) and growth factors (insulinlike growth factor 1 and VEGF).^{325,326} All five SSTRs are expressed in renal tubular epithelial cells and cholangiocytes. Because somatostatin has a half-life of approximately 3 minutes, more stable synthetic peptides (octreotide, lanreotide, and pasireotide) have been developed for clinical use. They differ in stability and in affinity for the different SSTRs. Half-lives in the circulation are 2 hours for octreotide and lanreotide and 12 hours for pasireotide. Octreotide and lanreotide bind with high affinity to SSTR2 and SSTR3, with moderate affinity to SSTR5, and have no affinity to SSTR1 and SSTR4. Pasireotide binds with high affinity to SSTR1, SSTR2, SSTR3, and SSTR5. In preclinical studies, octreotide and pasireotide reduced cAMP levels and proliferation of cholangiocytes in vitro, expansion of liver cysts in three-dimensional collagen culture, and development of kidney and liver cysts and fibrosis in PCK rats and *Pkd2*^{WS18/-} mice.^{131,327} In agreement with the longer half-life and higher affinity to a broader range of SSTRs, the effects of pasireotide are more potent than those of octreotide. Pasireotide and tolvaptan have been shown to have an additive beneficial effect in the *Pkd1*^{RC/RC} mouse model of ADPKD.³¹⁹

Several small randomized, placebo-controlled studies of octreotide or lanreotide have been completed.^{328–331} Most of these studies have been of short duration, but two of them have been extended as open-label, uncontrolled studies.^{332,333} The ALADIN study randomly assigned 79 patients with ADPKD and eGFR values of ≥ 40 mL/min/1.73 m² to receive intramuscular injections of a long-acting form of octreotide or placebo and monitored them for 3 years.³³⁴ The primary outcome variable, a mean increase in TKV at 3 years of follow-up, showed numerically smaller growth in the octreotide group than in the placebo group (220 vs. 454 mL). The difference, however, was not statistically significant. A favorable effect was noted on the secondary outcome of kidney function, but this endpoint also did not reach statistical significance. These findings provide support for larger randomized controlled trials to test the protective effect of somatostatin analogs, and an ongoing clinical trial involves 300 patients with ADPKD and CKD stages 3a and 3b.³³⁵ Until the results of larger trials become available, somatostatin analogs should not be prescribed for renoprotection outside of a research study.

The somatostatin analogs have also shown a potential beneficial effect on the progression of PLD. Liver volume decreased by 4% to 6% during the first year of treatment, and this reduction was sustained during the second year. Young female patients appear to have the greatest benefit from this treatment.³³⁶ The addition of everolimus to treatment with octreotide does not provide added benefit.

Octreotide and lanreotide are overall well tolerated. Self-resolving abdominal cramps and loose stools are common in the first few days following the injections. Other adverse effects include injection site granuloma and pain, cholelithiasis, steatorrhea, weight loss, and, rarely, hair loss. The adverse event profile of pasireotide in patients with ADPKD may include hyperglycemia because pasireotide inhibits insulin

more potently than glucagon secretion, whereas the contrary is true for octreotide and pasireotide.³³⁷

Rapalogs. There is overwhelming evidence for enhanced mTOR complex 1 (mTORC1) signaling in PKD cystic tissues, and results of preclinical trials of mTOR-inhibiting rapalogs (sirolimus and everolimus) in rodent models were mostly encouraging. At doses and blood levels achievable in humans, sirolimus and everolimus were effective in a rat model of PKD affecting proximal tubules^{338,339} but not in an ARPKD model affecting the distal nephron and collecting duct.³⁴⁰ Mice tolerate much higher doses and blood levels than do rats and humans, and at these high levels rapalogs were consistently effective in orthologous and nonorthologous mouse models.^{341,342} However, the results of clinical trials have been mostly discouraging,^{343–345} likely because blood levels capable of inhibiting mTOR in peripheral blood mononuclear cells do not inhibit mTOR in the kidney.³⁴⁶

Several strategies may overcome the systemic toxicity and limited renal bioavailability of rapalogs. One is to target the drug specifically to the kidney by conjugating it to folate, and treatment of *bph* mice was effective in reducing renal cyst growth and preserving kidney function without toxicity.³⁴⁷ Another approach takes advantage of the mechanism of action of sirolimus, which forms a complex with the binding protein FKBP12 that competes with phosphatidic acid for binding to mTOR. Phosphatidic acid, a phospholipase D product generated by the hydrolysis of phosphatidylcholine, is required for the association of mTOR with component Raptor in mTORC1 and with component Rictor in mTORC2. One study showed that PKD cells have a slightly higher phospholipase D activity and that blockade of its activity by either specific inhibitors or “alcohol trap” treatment retrains mTORC1 and decreases cell viability and proliferation.³⁴⁸ A third approach is the use of mTOR-catalytic inhibitors that cause a more potent and durable inhibition of mTORC1 than rapalogs, which are currently being tested in rodent PKD models.³⁴⁹

Other Agents. Many other drugs have been shown to be effective in preclinical trials and of potential value for the treatment of human PKD. Some act on tyrosine kinase receptors, downstream signaling pathways such as MAPK and Wnt/ β -catenin signaling, effectors and inhibitors of cell cycling, or energy metabolism to inhibit cell proliferation.^{163,164,350,351} Other potential therapies target fluid secretion (CFTR and KCa3.1 inhibitors) or important pathogenic interactions between epithelial cells and ECM and the interstitial inflammatory microenvironment. Of particular interest are drugs or interventions that have shown effectiveness in animal models and that can be used with relatively little toxicity, such as metformin, the thiazolidinediones, nicotinamide, and caloric restriction.^{154,352–358}

Clinical Trials and Renal Function

In planning for clinical trials for ADPKD, the use of renal function as the primary outcome is an issue. This is because decades of normal renal function remain despite progressive enlargement and cystic transformation of the kidneys. By the time the GFR starts declining, the kidneys are markedly enlarged, distorted, and unlikely to benefit from treatment. Therefore early interventional trials would require unrealistic

periods of follow-up if renal function were to be used as the primary outcome. The results of CRISP have shown that the rate of renal growth is a good predictor of functional decline and justify the use of TKV as a marker of disease progression in clinical trials for ADPKD.^{194,245}

AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE

EPIDEMIOLOGY

ARPKD is generally characterized by relatively rapid, symmetric, bilateral renal enlargement in infants due to collecting duct cysts in association with CHF.^{359–361} Nonobstructive intrahepatic bile duct dilatation (Caroli disease) is variably seen. A minority of cases may occur in older children, teenagers, or young adults, usually with manifestations of portal hypertension or cholangitis. In rare cases, the presentation may be in older adults, mostly with complications of the liver disease but sometimes with renal manifestations such as proteinuria, nephrolithiasis, and renal insufficiency.^{362,363} Prevalence and carrier frequencies are thought to be 1 in 20,000 and 1 in 70, respectively.³⁶⁴ Molecular data indicate that ARPKD is likely to be found in all racial groups.^{365,366}

GENETICS

ARPKD is inherited as an autosomal recessive trait and therefore may occur in siblings but not in the parents. The disease is observed in one-fourth of the offspring of carrier parents. All cases of typical ARPKD are caused by mutations in a gene on chromosome 6p21.1-p12 (*PKHD1*).^{367–369} Studies have shown that *PKHD1* mutations are also responsible for nonsyndromic CHF and Caroli disease.³⁶² Biallelic mutations to the ciliopathy gene *DZIP1L* have recently been associated with an ARPKD-like phenotype in humans.³⁷⁰ Likewise, specific mutations to *PMM2* can result in an ARPKD-like phenotype (see the “Polycystic Kidney and/or Liver Diseases Associated With Altered Maturation of PKD Proteins in the Endoplasmic Reticulum” section).³⁷¹

PKHD1 is among the largest genes in the human genome, extending approximately 470 kb and including 67 exons.^{372,373} Mutations are scattered throughout the gene without “hot spots.”^{365,366} Most have been described in just a single family, but some are more frequent in particular populations (e.g., p.R496X in Finland and c.9689delA in Spain).^{374,375} One missense substitution, p.T36M, has been found on approximately 16% of mutant alleles and seems to be an ancestral mutation that arose in Europe more than a 1000 years ago.³⁷⁶ There is evidence of genotype–phenotype correlations. The presence of two truncating mutations results in a lethal phenotype by the neonatal period. Surviving patients with severe or milder renal disease have at least one nontruncation mutation, indicating that many nontruncating mutations are hypomorphic.^{374,375,377} Despite the importance of the germline mutations, affected sibling pairs may exhibit phenotypes of markedly discordant severity, most likely because of the effect of modifier genes.

PATHOGENESIS

The ARPKD protein, fibrocystin (460 kDa), has a single-transmembrane pass, a large extracellular region containing IPT/TIG (immunoglobulin-like fold shared by plexins and transcription factors) and PbH1 (parallel beta-helix 1) repeats,

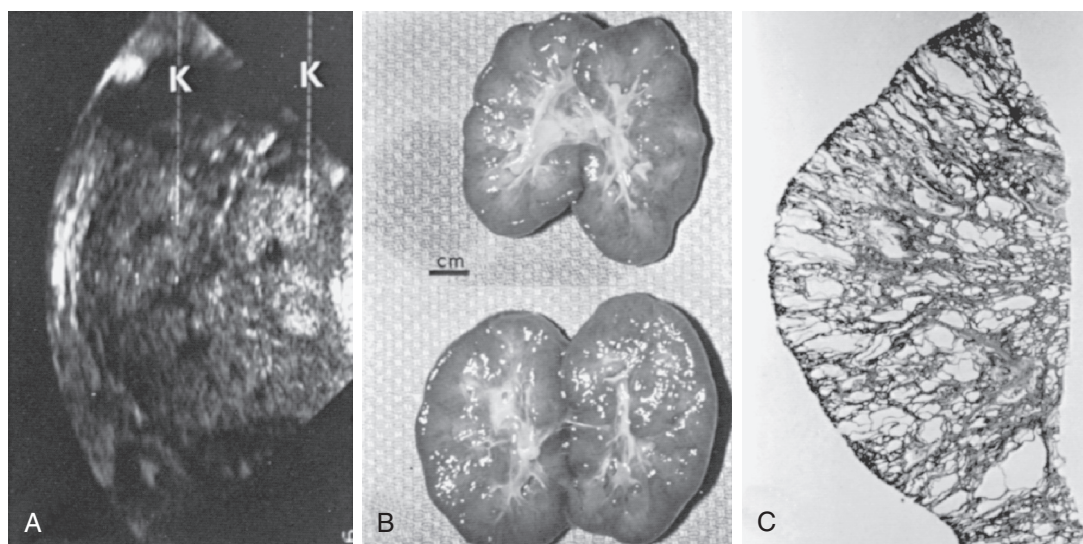


Fig. 45.15 Autosomal recessive polycystic kidney disease in a 32-week-old fetus. (A) A sonogram showing cystic kidneys (K) of a fetus in utero. (B) Gross and (C) microscopic sections show radially oriented cysts of collecting ducts.

and an intracellular carboxyl tail.^{372,373} Cleavage of the protein in the extracellular region at the proprotein convertase site is likely important to form a functional protein.^{112,378} This possibility suggests that fibrocystin may be a cell surface receptor implicated in protein–protein interactions. Like the polycystins, fibrocystin is localized in primary cilia. Fibrocystin, PC2, and the kinesin-2 motor subunit KIF3B (kinesin family member 3B) have been suggested to form a protein complex in which KIF3B acts as a linker between fibrocystin and PC2.³⁷⁹ However, more recent data do not place fibrocystin and PC2 in the same complex.³⁸⁰ Normal kidney and liver in an animal model with the C-tail of the protein deleted also question whether signaling occurs through this part of the protein,³⁸⁰ and highlights the secreted extracellular region that is found in urine alone and on exosomes. *PKHD1* has limited expression to the kidney, pancreas, and liver and the expression of fibrocystin in ureteric bud branches, intrahepatic and extrahepatic biliary ducts, and pancreatic ducts during embryogenesis is consistent with the histologic features of ARPKD.³⁸¹

PATHOLOGY

ARPKD affects both the kidneys and the liver in approximately inverse proportions. That is, the disease may be viewed as a spectrum ranging from severe renal damage and mild liver damage at one end to mild renal damage and severe liver damage at the other. The form with severe renal damage is the more common and is the form that manifests at or near the time of birth. The form with less severe renal damage and more severe liver damage is less common and usually manifests in infancy, childhood, or later.

Kidneys in the perinatal and neonatal forms of ARPKD are symmetrically and bilaterally enlarged up to more than 20 times normal (Fig. 45.15) and may be the cause of dystocia because of their size. Average combined weight of the kidneys in one series was about 300 g (range, 240 to 563 g), a normal combined weight being about 25 g. The renal enlargement is caused by fusiform dilation of collecting ducts to 1 to 2 mm in the cortex and medulla. Almost 100% of collecting

ducts are affected in the most severe cases. Dilation of the collecting ducts occurs in the fetal period, and the glomeruli and more proximal tubular elements of the nephron appear normal. However, there is evidence in early human fetuses (14–24 weeks) that proximal tubule cysts occur, as in some rodent models of recessive PKD,^{382–384} but these are no longer evident after 34 weeks of gestation.³⁸⁵ The dilated collecting ducts are lined by typical cuboidal cells.^{386,387} In many cases of neonatal ARPKD, an overall reduction in size may occur as the children age, and macroscopic cysts may develop. Renal calcifications are common in children with the disease.

In later presentations of ARPKD, mainly with complications of portal hypertension caused by CHF or episodes of cholangitis caused by Caroli disease, the renal involvement may be much less prominent, consisting of medullary ductal ectasia with minimal or no renal enlargement. The picture resembles and may be confused with medullary sponge kidney (MSK), a distinct disease with a far different prognosis (see later discussion).

The hepatic lesion is diffuse but limited to the portal areas. CHF is characterized by enlarged and fibrotic portal areas with apparent proliferation of bile ducts, absence of central bile ducts, hypoplasia of the portal vein branches, and sometimes prominent fibrosis around the central veins. Bulbar protrusions from the walls of dilated ducts also occur, and bridges sometimes form. This malformation has been found to occur occasionally as an isolated event (Caroli disease), but most often it is associated with ARPKD.

DIAGNOSIS

The diagnosis is often made by sonography in utero or shortly after birth. The typical sonogram (Fig. 45.15) shows enlarged kidneys with increased echogenicity in the cortex and medulla, with poor definition of the collecting system and fuzzy delineation of the kidneys from surrounding tissues. Although the appearance of the kidneys on sonography, CT, and MRI may be very suggestive of ARPKD, a definite diagnosis based on renal imaging alone is not possible, particularly in utero and in the neonatal period, when the appearance of the

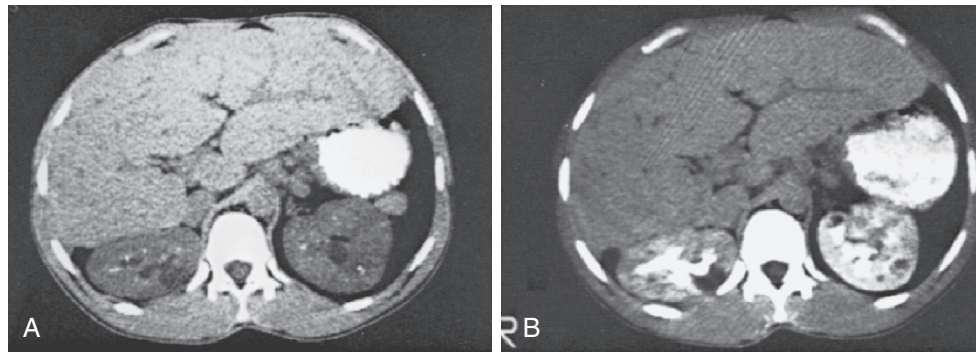


Fig. 45.16 Computed tomography (CT) scan of autosomal recessive polycystic kidney disease in an 18-year-old man whose serum creatinine and liver function test results were within normal ranges. The patient had clinical evidence of portal hypertension (gastric varices and enlarged spleen). An oral contrast agent was given to highlight the intestines. (A) CT scan without contrast enhancement. The liver is enlarged but not cystic. The kidneys are slightly enlarged and contain focal radiodense areas (nephrolithiasis). (B) CT scan after intravenous administration of iodinated radiocontrast agent showing cystic areas in both kidneys. The renal calcifications are now obscured by contrast medium in the collecting systems.

kidney may be indistinguishable from ADPKD and other recessive renal cystic diseases. The family history, ultrasonographic or histologic evaluation of the liver for the presence of hepatic fibrosis, absence of extrarenal malformations associated with multiple malformation syndromes and renal dysplasia, and molecular analysis help in the diagnosis. Family history evaluation is complicated by the finding of increased kidney echogenicity and/or multiple small liver cysts in approximately 10% of *PKHD1* mutation carriers.^{261,388}

Older children and adolescents may present with symptoms and signs related to the hepatic fibrosis and portal hypertension, including gastrointestinal bleeding from varices, hepatosplenomegaly, and hypersplenism, with or without associated renal manifestations such as a urinary concentrating defect, nephrolithiasis, hypertension, and renal insufficiency. Collecting duct ectasia and macrocystic changes may be observed in the kidneys of these patients (Fig. 45.16). Combined use of conventional and high-resolution ultrasonography with MR cholangiography in patients with ARPKD and CHF allows detailed definition of the extent of kidney and hepatobiliary manifestations without requiring ionizing radiation and contrast agents.³⁸⁹

Owing to the severity of disease in ARPKD, there is significant demand for prenatal and preimplantation diagnostics. This interest largely comes from couples with a previously affected pregnancy, detected either in utero (which may have resulted in termination) or with the birth of an affected child (who may have died in the neonatal period).³⁹⁰ Preimplantation genetic diagnosis, which avoids the trauma of a termination of pregnancy in the case of an affected fetus, has been performed in only a few cases.³⁹¹ There is also demand for molecular diagnostics in older patients with less severe disease to differentiate ARPKD from other causes of childhood PKD. Molecular diagnostics is offered using mutation-based diagnostics by Sanger sequencing of the single gene, or now more commonly as a panel of PKD genes including also ciliopathy genes using an NGS approach.^{249,392} Molecular diagnostics have the advantage that DNA from a previously affected family member is not required and that patients with an uncertain diagnosis can be tested. However, it is complicated by the marked allelic heterogeneity of *PKHD1* and prevalence of novel missense variants of uncertain significance

(VUS). If two clearly pathogenic mutations are identified, the diagnosis is highly reliable. However, if only novel VUS are identified, careful bioinformatic and population evaluation is required, although simultaneous evaluation of other PKD/ciliopathy genes can improve the reliability of the molecular results.²⁴⁹ In one study, a definitive prenatal diagnosis (presence or absence of two identified mutations) was feasible in 72% of the cases, and an improved risk assignment (presence or absence of one identified mutation) was possible in an additional 25% of the studied families.³⁶⁵

The phenotype of greatly enlarged and echogenic kidneys in neonates is not pathognomonic for ARPKD, and other cystic disorders should be considered. Rarely (<1% cases), ADPKD manifests in utero or in the neonatal period as clinical symptoms very similar to those of ARPKD.^{70,249,393} In approximately 50% of these cases, an affected parent is recognized only after the diagnosis of a severely affected child.³⁹⁴ In addition, rare de novo, early-onset ADPKD may occur. Most early-onset cases have been linked to *PKD1*, although two *PKD2* families with early-onset disease have been described.^{72,395} CHF is not normally part of the ADPKD phenotype, but rare reports of an association have been described.

The high risk of recurrence of early-onset ADPKD in affected families suggested a common familial modifying background for early and severe disease expression (e.g., mutations or variants in genes encoding other cystoproteins).⁷⁰ Emerging family data suggested that in trans inheritance of a null and an incompletely penetrant *PKD1* allele could result in early-onset ADPKD with an ARPKD-like phenotype.^{68,73,75,249,396} This hypothesis has been confirmed by a knockin mouse model mimicking the naturally occurring *PKD1* variant p.R3277C highlighted by later family studies.^{68,73} Mirroring observation in the human studies, gradual cystic disease developed in *Pkd1*^{RC/RC} animals over 1 year, whereas early-onset, rapidly progressive disease occurred in *Pkd1*^{RC/null} mice.¹⁰ Early-onset severe ADPKD has also been linked to contiguous deletion of *PKD1* and *TSC2*^{297,398} as well as coinheritance of an ADPKD and an *HNF1B* or *PKHD1* allele.⁷⁶

Glomerulocystic kidney disease can rarely manifest in the neonatal period with an ARPKD-like phenotype.³⁹⁹ Rare families with ARPKD-like disease and skeletal and facial

anomalies,^{400,401} or recessively inherited renal and hepatic cystic diseases with hypoglycemia,^{376,402} now identified as hyperinsulinemic hypoglycemia with PKD (HIPKD) due a specific *PMM2* promoter mutation, have been described.³⁷¹ Infantile NPHP and related disorders may also be confused with ARPKD. A number of syndromic congenital hepatorenal disorders (ciliopathies) including ones lethal in infancy can be associated with renal abnormalities resembling those in ARPKD and with CHF. These include Joubert syndrome (JBTS), Meckel syndrome (MKS), Elejalde (acrocephalopoly-syndactyly), and Ivemark (renal–hepatic–pancreatic dysplasia) syndromes and glutaric aciduria type II. Mutations to *ANKS6* have also been shown to sometimes result in an ARPKD-like phenotype but usually also with cardiovascular abnormalities.⁴⁰³ Biallelic mutations to *DZIP1L* result in an ARPKD-like phenotype.³⁷⁰

MANIFESTATIONS

Children with ARPKD are typically identified in utero from the finding of enlarged, echogenic kidneys. In the most severe cases, poor urine output may result in oligohydramnios and the Potter sequence, characterized by typical facies, wrinkled skin, compression deformities of the limbs, and pulmonary hypoplasia. The presentation of ARPKD at birth may be dominated by respiratory difficulties from pulmonary hypoplasia or from restrictive disease caused by massive kidney enlargement. The need for neonatal ventilation predicts the development of CKD and death. Approximately 30% of affected neonates die shortly after birth.^{404–407}

Most patients who survive the neonatal period live to adulthood. Hypertension, electrolyte abnormalities, and renal insufficiency are the major disease complications in surviving infants, with liver disease becoming more important in older patients. Hypertension developed in between 55% and 86% of patients reported in two studies, with blood pressure elevations often seen at birth or at diagnosis.^{404,405} The ectopic expression of components of the RAAS in cystic-dilated tubules suggests that increased intrarenal angiotensin II production contributes to hypertension development.⁴⁰⁸ However, the circulating plasma renin level is usually low,⁴⁰⁶ and the intravascular volume expanded, particularly in patients with concomitant hyponatremia.⁴⁰⁵ Increased sodium reabsorption in the ectatic collecting ducts may contribute to the hypertension,^{409,410} but conflicting data have been reported.⁴¹¹ The inability to concentrate and dilute urine can cause major electrolyte abnormalities. During the first year or two of life, renal function can improve, and renal size relative to body mass often decreases.^{386,412,413} Renal function may remain stable for many years or may slowly progress to renal failure. The consequences of chronic renal insufficiency, growth failure, anemia, and osteodystrophy can become apparent during childhood. Because complications of liver disease become more important as ARPKD children age, careful examination for splenomegaly and blood counts for cytopenias should be regularly performed.

Adolescents and adults present most often with complications of portal hypertension (variceal esophageal bleeding, splenomegaly and hypersplenism with leukopenia, thrombocytopenia, or anemia).³⁶² Up to 50% of patients with CHF may exhibit segmental dilation of intrahepatic bile ducts (Caroli disease), sometimes with episodes of cholangitis or sepsis and

complications of biliary sludge or lithiasis. Hepatocellular function is rarely deranged, and enzyme values are only occasionally mildly elevated. Increased bilirubin or enzyme values suggest the possibility of cholangitis. The kidneys in these patients may be normal or may exhibit various degrees of medullary collecting duct ectasia or macrocystic disease without marked renal enlargement.

Three reports have described cases of ARPKD with IAs.^{414–416} It is not clear whether the prevalence of IAs is increased in ARPKD or whether these cases are coincidental findings.

Heterozygote carriers of *PKHD1* mutations have usually been considered to be entirely normal. However, an ultrasonographic study of 110 obligate carriers of ARPKD showed increased medullary echogenicity in 6 (5.5%), multiple small liver cysts in 10 (9.1%), a moderately increased liver echo pattern suggestive of CHF in 4 (3.6%), and mild splenomegaly in 9 (8.2%) patients.⁴¹⁷ A second study recently identified liver cysts in approximately 10% of *PKHD1* mutation carriers.²⁶¹

TREATMENT

Later studies suggest that the prognosis of ARPKD for children who survive the first month of life is far less bleak than was initially thought.^{359,404–406} In patients with respiratory insufficiency, the cause (pulmonary hypoplasia, abdominal mass, pneumothorax, pneumomediastinum, atelectasis, pneumonia, heart failure) should be assessed fully, and artificial ventilation and aggressive resuscitative measures are indicated. Severely affected neonates may require unilateral or bilateral nephrectomies because of respiratory and nutritional compromise. An aggressive nutritional program and correction of acidosis and other electrolyte disorders are needed to optimize linear growth. The hypertension generally responds to salt restriction and antihypertensive drugs. Like patients with other renal cystic disorders, patients with ARPKD are susceptible to urinary tract infections, so urinary tract instrumentation is best avoided.

For infants with ESKD, peritoneal dialysis is preferable, but hemodialysis is also an option for children with renal failure. Kidney transplantation is limited by body size, but in experienced centers, it can be performed even in small children with a minimal weight of 7 kg.⁴¹⁸ Pretransplantation splenectomy may be indicated for patients with marked leukopenia or thrombocytopenia due to hypersplenism. These patients should receive pneumococcal vaccinations. Rejection rates and survival beyond 3 years for such patients are not different from those in patients with other renal diseases who undergo transplant surgery. Biliary sepsis is a frequent contributor to the mortality in patients with ARPKD who undergo transplant surgery.⁴¹⁹

Surviving patients and those whose disease manifests during adolescence are likely to require portosystemic shunting to prevent life-threatening hemorrhages from esophageal varices. The renal disease may progress to renal failure years later even after successful shunting. Patients with associated nonobstructive intrahepatic biliary dilation (Caroli disease) may have recurrent episodes of cholangitis and may require antimicrobial therapy or segmental hepatic resection. Combined kidney and liver transplantation has been advocated for selected patients with ESKD who have significant bile duct dilation and episodes of cholangitis.^{420–422}

POLYCYSTIC KIDNEY AND/OR LIVER DISEASES ASSOCIATED WITH ALTERED MATURATION OF PKD PROTEINS IN THE ENDOPLASMIC RETICULUM

In the last few years, mutations in genes encoding proteins responsible for the glycosylation, quality control, or translocation across the endoplasmic reticulum membrane of membrane/secreted proteins have been associated with development of cysts in the kidneys and or liver, likely due to inefficient maturation and trafficking of the polycystins

(Fig. 45.17). The disease associated with *PRKCSH* (encoding glucosidase II beta) and *SEC63* (a translocon complex protein) mutations majorly consist of PLD without renal cysts (ADPLD), although a few kidney cysts have been described in 28%–35% of cases, and renal involvement may have been underestimated.^{423–425} Renal inactivation of *PrkcsH* or *Sec63* in mice induces a cystic phenotype that can be rescued by *Pkd1* overexpression.⁴²⁶ More recently, mutations in *GANAB* (glucosidase II alpha), *ALG8* (alpha 1,3-glucosyltransferase), and *SEC61B* (a translocon complex protein) have been identified in patients with phenotypes compatible with ADPKD or ADPLD (Fig. 45.18).^{52,261} Biallelic mutations, including at

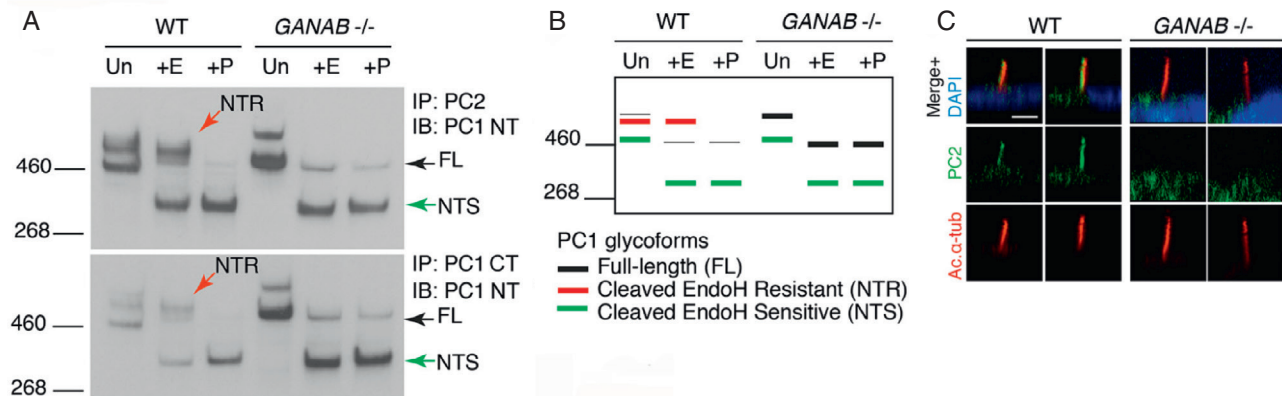


Fig. 45.17 Mutation of the glucosidase II alpha gene, *GANAB*, leads to inefficient maturation and trafficking of polycystins. (A) Deglycosylation analysis of wild-type (WT) and *GANAB*^{-/-} renal cortical tubular epithelial cell membrane protein treated with endoglycosidase H (endoH, +E), PNGase F (+P), or untreated (Un). Immunoprecipitation was used to enrich for the PC1 complex, using C-terminal PC1 (PC1 CT) or PC2 (YCE2) antibodies and immunodetected with the N-terminal PC1 (PC1 NT) antibody (7e12). Complete loss of the mature, N-terminal endoH-resistant PC1 glycoform (NTR; red arrow) was observed in *GANAB*^{-/-} cells, with full-length PC1 (FL) and the immature, N-terminal endoH-sensitive PC1 glycoform (NTS; green arrow) becoming more abundant. (B) Schematic representation of the observed PC1 banding pattern in WT and *GANAB*^{-/-} cells (A). (C) Confocal Z-stack rendering of primary cilia in confluent WT and *GANAB*^{-/-} cells immunostained for acetylated α -tubulin (Ac. α -tub) and PC2, showing no cilia PC2 signal in *GANAB*^{-/-} cells. PC1, Polycystin 1; PC2, polycystin 2. (Modified with permission from Porath B, Gainullin VG, Cornec-Le Gall E, et al. Mutations in *GANAB*, encoding the glucosidase α subunit, cause autosomal-dominant polycystic kidney and liver disease. *Am J Hum Genet.* 2016;98:1193–1207.)

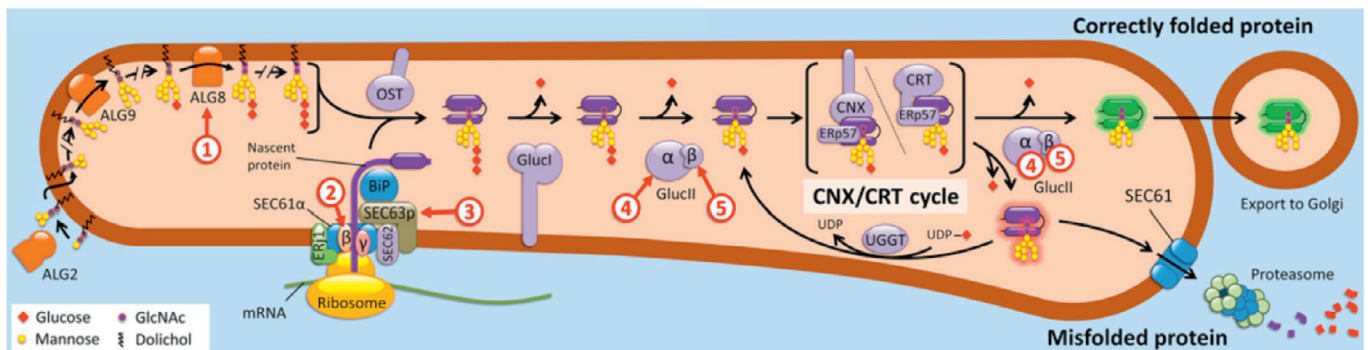


Fig. 45.18 Schematic of the function of autosomal dominant polycystic kidney disease (ADPKD) and ADPLD genes in the endoplasmic reticulum (ER) protein biogenesis pathway. The genes are numbered 1–5. Lipid-linked oligosaccharide precursors of N-linked glycans are initially assembled on dolichol on the cytoplasmic aspect of the ER membrane. These are flipped into the ER lumen, where ALG8 (1) catalyzes the addition of the second glucose residue. Nascent polypeptides undergo cotranslational translocation via the SEC61 translocation pore that is composed of α -, β -, and γ -subunits and is associated with SEC62. SEC63 (3) and ERJ1 act in concert with the major ER HSP70 chaperone BiP to facilitate this translocation process. Oligosaccharyltransferase (OST) catalyzes the attachment of the glycan moiety to asparagine residues. Glucosidase I removes the outermost glucose before glucosidase II, composed of GlcI α (4) and GlcI β (5) subunits, removes the second glucose. This step is necessary for the nascent peptide to enter the calnexin (CNX)/calreticulin (CRT) protein folding and quality control cycle. Glucosidase II (4, 5) subsequently removes the innermost glucose from the N-linked glycan, allowing for exit from the CNX/CRT cycle. If the protein has attained its properly folded conformation, it proceeds along the secretory pathway. Misfolded proteins are recognized and reglycosylated by UGGT, allowing for more time in the folding environment of the CNX/CRT cycle. Eventually, proteins that fail to fold properly undergo ER-associated degradation by retrotranslocation through the SEC61 translocon complex into the cytoplasmic compartment, where they are degraded by the proteasome. (Modified with permission from Besse W, Dong K, Choi J, et al. Isolated polycystic liver disease genes define effectors of polycystin-1 function. *J Clin Invest.* 2017;127(9):3558.)

least one specific in the promoter region of *PMM2* (encoding phosphomannomutase 2, a key enzyme in N-linked glycosylation), have been associated with polycystic kidneys, occasional liver cysts, and hyperinsulinic hypoglycemia³⁷¹ and HIPKD. Mutations in *DNAJB11* (a BiP cofactor, a key chaperone in the endoplasmic reticulum) have been identified in patients with nonenlarged polycystic kidneys with small renal cysts plus liver cysts in some cases leading to ESKD at advanced age.⁵³ Mutations in *SEC61A1* (a translocon complex protein) have been described in two families with ADTKD with anemia.⁴²⁷

TUBEROUS SCLEROSIS COMPLEX

EPIDEMIOLOGY

TSC is an autosomal dominant disease that affects up to 1 in 6000 individuals.

GENETICS

It is caused by mutations in either *TSC1* or *TSC2*. *TSC1* is located on chromosome 9q34 and encodes for hamartin. *TSC2* is located on chromosome 16p13 and encodes for tuberlin. The disease tends to be less severe in patients with *TSC1* mutations than in those with *TSC2* mutations.^{428–430}

PATHOGENESIS

Hamartin and tuberlin physically interact, and this interaction is important for their function. The hamartin–tuberlin complex antagonizes an insulin-signaling pathway that plays an important role in the regulation of cell size, cell number, and organ size.^{431,432} In the absence of growth factor stimulation, tuberlin–hamartin complexes maintain Rheb (Ras homolog enriched in the brain) in an inactive guanosine diphosphate-bound state by stimulating its intrinsic guanosine triphosphatase activity and inhibit downstream signaling from Rheb via mTOR. Growth factor stimulation of phosphoinositide 3-kinase signaling leads to Akt-dependent phosphorylation of tuberlin, dissociation of the tuberlin–hamartin complex, and activation of Rheb and mTOR. Tuberlin or hamartin mutations prevent the formation of tuberlin–hamartin complexes and lead to constitutive activation of mTOR.

DIAGNOSIS

A definite clinical diagnosis of TSC requires one of the following conditions:

- Two major features from the following list: renal angiomyolipoma (AML), facial angiofibromas or forehead plaques, nontraumatic ungual or periungual fibroma, three or more hypomelanotic macules, shagreen patch, multiple retinal nodular hamartomas, cortical tuber, subependymal nodule, subependymal giant cell astrocytoma, cardiac rhabdomyoma, lymphangioleiomyomatosis
- One major feature from the previous list plus two minor features from the following list: multiple renal cysts, nonrenal hamartoma, hamartomatous rectal polyps, retinal achromic patch, cerebral white matter radial migration tracts, bone cysts, gingival fibromas, “confetti” skin lesions, multiple enamel pits

A diagnosis can also be made by the detection of a pathogenic variant in the *TSC1* or *TSC2* gene.

MANIFESTATIONS

Renal involvement is second only to the involvement of the central nervous system (CNS) as a cause of death in patients with TSC. The main renal manifestations are AMLs (including rare epithelioid AMLs), cysts, oncocytomas, clear cell RCC, lymphangiomatous cysts, and (rarely) focal segmental glomerulosclerosis.^{433–436}

AMLs are benign tumors consisting of abnormal blood vessels, smooth muscle, and fat cells. They are derived from perivascular epithelioid cells (PEComas) and exhibit immunoreactivity for both melanocytic markers (as detected by the antibodies HMB-45 and melan-A) and smooth muscle markers (actin and desmin). The diagnosis of renal AML relies on the demonstration of fat in the tumor by imaging studies. The fat appears hyperechogenic on ultrasonography, has a low attenuation value on CT, and on MRI appears bright on T1-weighted images, dark on T2-weighted images with fat saturation, and intermediate on T2-weighted images.^{7,38,42,437} Approximately 5% of renal AMLs contain minimal amounts of fat and are called minimal-fat or fat-poor lesions. The differential diagnosis can be challenging and includes classic AMLs (predominantly composed of smooth muscle cells), epithelioid AMLs, RCC, and oncocytoma. When the distinction between minimal-fat renal AMLs and RCC cannot be reliably established by imaging techniques, imaging-guided percutaneous needle biopsy and staining for melanocyte markers should be considered as an alternative to surgical exploration.⁴³⁸ The risk of bleeding after needle biopsy of a minimal-fat AML does not appear to be higher than that of a biopsy for other renal tumors, particularly when fine needles are used.

In patients with TSC, AMLs are extremely common, are usually multiple and bilateral, and affect both genders, in contrast to the general population, in which they are uncommon, usually single, and mainly found in middle-aged women. They develop after the first year of age, and by the third decade, 60% of patients with TSC have renal AMLs. The lesions express receptors for estrogen and progesterone, and women have more and larger AMLs than men. The early small lesions have a characteristic radial, striated, or wedge-shaped pattern with the base of the wedge facing the surface of the kidney. As the lesions increase in size, they penetrate deeper into the renal parenchyma or become exophytic, extending into the perirenal fat. The main manifestations relate to their potential for hemorrhage (hematuria, intratumoral, or retroperitoneal) and mass effect (abdominal or flank mass and tenderness, hypertension, or renal insufficiency). Because of the potential for renal AML development and growth, it is recommended that renal surveillance in patients with TSC be performed with ultrasonography at diagnosis and thereafter every 1 to 2 years (3 years in those in whom no AMLs are identified and at least yearly in patients with known AMLs).⁴³⁹ In patients with known renal lesions, the serum creatinine concentration should be measured at least once a year.

The epithelioid AML variant is differentiated from the classic variant by the presence of an epithelioid cell component with abundant eosinophilic and granular cytoplasm.⁴⁴⁰ There is no consensus on the percentage of epithelioid cells that is required to make a diagnosis of the epithelioid variant, with values ranging from 10% to 100% in published studies.^{441,442}

In contrast to the uniformly benign prognosis of classic renal AMLs, as described in the preceding section, epithelioid variants infrequently undergo malignant transformation that manifests as local recurrence and/or distal metastases.

Unlike AMLs, renal cysts may be present in the first year of life, and cystic disease may be the presenting manifestation of TSC. *TSC2* and *PKD1* lie adjacent to each other in a tail-to-tail orientation at chromosome 16p13.3. Deletions inactivating both genes are associated with polycystic kidneys diagnosed during the first year of life or early childhood (*TSC2/PKD1* contiguous gene syndrome).^{397,398} Therefore a contiguous gene syndrome should be considered in children with renal cysts and no family history of PKD. Rarely, *TSC2/PKD1* contiguous gene syndrome can be diagnosed in adults.⁴⁴³ Patients with the contiguous gene syndrome usually reach ESKD at an earlier age than patients with ADPKD alone, but the disease severity in patients who have mosaicism is more variable.³⁹⁸ Patients with *TSC1* or *TSC2* mutations without the contiguous gene syndrome also can have renal cysts. The renal cysts in TSC are often lined by a very distinct, perhaps unique, epithelium of markedly hypertrophic and hyperplastic cells with prominent eosinophilic cytoplasm. The combination of cystic kidneys and AMLs has been said to be virtually pathognomonic for TSC.

Oncocytomas—benign tumors derived from intercalated cells in the collecting ducts—and clear cell RCCs occur with increased frequency in TSC. RCCs in TSC have a female predominance and an earlier age of presentation as well as increased bilaterality. Early detection is essential. They should be suspected in cases of enlarging lesions without demonstrable fat and in the presence of intratumoral calcifications.

TREATMENT

AMLs are benign lesions and often require no treatment. Because of their increased frequency and size in women and reports of hemorrhagic complications during pregnancy, patients with multiple AMLs should be cautioned about the potential risks of pregnancy and estrogen administration. Annual reevaluations with ultrasonography or CT are necessary to assess for growth and development of complications. Renal-sparing surgery is indicated for symptoms such as pain and hemorrhage, growth with compromise of functioning renal parenchyma, and inability to exclude an associated RCC. Because AMLs >4 cm are more likely to grow, develop microaneurysms and macroaneurysms, and cause symptoms, some writers suggest that prophylactic intervention should be considered in these cases. Some lesions, because of their size or central location, may be more amenable to selective arterial embolization. Radiofrequency ablation and cryoablation have also been successful in the treatment of renal AMLs <4 cm in diameter without bleeding complications.^{444–446}

Elucidation of the TSC protein function at a molecular level and preclinical studies have identified mTOR as a target for intervention in TSC.^{447,448} The efficacy of mTOR inhibitors has also been demonstrated in patients with TSC who have subependymal giant cell astrocytomas or pulmonary lymphangioleiomyomatosis. Multiple open-label, nonrandomized studies have shown that sirolimus produces a reduction in the volume of renal AMLs, mostly achieved in the first few months of treatment,^{449–452} but the effect is not maintained after discontinuation of treatment. The best data on the efficacy of mTOR inhibitors come from the double-blind EXIST-2 study

(Efficacy and safety of everolimus for subependymal giant cell astrocytomas associated with tuberous sclerosis complex or sporadic lymphangioleiomyomatosis), which included 118 patients, 18 years or older, who had a definite diagnosis of TSC ($n = 113$) or sporadic lymphangioleiomyomatosis ($n = 5$) and at least one renal AML ≥ 3 cm in its largest diameter. The patients were randomly assigned in a 2/1 ratio to everolimus 10 mg/day or placebo, and treated for a median duration of 38 or 34 weeks, respectively.⁴⁵³ At the data cutoff, therapy had been discontinued in 20 patients, because of disease progression in 9 receiving placebo and because of adverse effects in 1 receiving everolimus and 4 receiving placebo. The primary endpoint, at least a 50% reduction in the total volume of all target AMLs identified at baseline, was achieved in 42% of the patients treated with everolimus and in none of the patients treated with placebo ($P < .001$). The median time to response to everolimus was 2.9 months. Progression of AMLs was significantly more common in the placebo group (21% vs. 4%). The response rate of skin lesions was significantly more common with everolimus (26% vs. 0%). The patients treated with everolimus had significantly higher rates of stomatitis (48% vs. 8%) and acnelike skin lesions (22% vs. 5%). On the basis of these findings, the FDA approved everolimus tablets for the treatment of adults with renal AMLs and TSC who do not require immediate surgery. At present, the place of everolimus or other mTOR inhibitors for the treatment of renal AMLs needs to be further defined. Benefits and risks need to be balanced, because the administration of mTOR inhibitors can be associated with significant adverse events, the reduction in renal AML volume is reversible after discontinuation of treatment, and long-term outcome results are not available.

Case reports have suggested efficacy of mTOR inhibitors in patients with unresectable or metastatic malignant epithelioid AMLs.

The main clinical problems associated with cystic disease in TSC are hypertension and renal failure. The treatment consists of strict control of hypertension. Bilateral nephrectomy should be considered before transplantation surgery because of the risk of life-threatening hemorrhage and the development of RCC.

VON HIPPEL-LINDAU SYNDROME

EPIDEMIOLOGY

VHL syndrome is a rare autosomal dominant disease with a prevalence of 1 in 36,000. It is characterized by retinal hemangiomas, clear cell RCCs, cerebellar and spinal heman-gioblastomas, pheochromocytomas, and, less frequently, pancreatic cysts and neuroendocrine tumors, endolymphatic sac tumors of the inner ear, and epididymal cystadenomas.^{454,455} Approximately 20% of patients with VHL syndrome have a de novo mutation: they do not have a family history of VHL syndrome.

GENETICS

The VHL protein (pVHL) is encoded by a highly conserved tumor suppressor gene in chromosome 3p25-p26. Genotype-phenotype correlations in VHL syndrome have classified four VHL subtypes.³⁰⁷ Patients with VHL type 1 have no pheochromocytomas. Mutations in patients with VHL type 1 are mostly loss-of-function type, leading to a truncated pVHL

or no pVHL at all. Patients with complete germline deletions have a lower rate of RCC than patients with partial deletions (22.6% vs. 49%, respectively).⁴⁵⁶ Mutations in patients with VHL type 2 are mostly missense mutations with some residual function. Whereas type 2 families have pheochromocytomas and are divided into subtypes with a low (type 2A) or high (type 2B) risk of RCC, type 2C families present with pheochromocytoma only. Dysregulation of VHL-dependent degradation of the α -subunit of hypoxia-inducible factor (HIF- α) is observed in subtypes 1, 2A, and 2B. By contrast, VHL-dependent HIF- α degradation is not observed in type 2C VHL mutants.

PATHOGENESIS

VHL encodes two protein isoforms with relative molecular masses of approximately 30 and 19 kDa. Although these isoforms may have different functions, both are capable of suppressing RCC growth in vivo. However, the best-characterized function of pVHL is to act as an essential component in the degradation of HIF- α subunits. pVHL is the substrate-binding subunit of an E3 ubiquitin ligase that ubiquitinates HIF- α .⁴⁵⁷ The HIF transcription factors consist of an oxygen-sensitive α -subunit and a constitutively expressed β -subunit, also known as the aryl hydrocarbon receptor nuclear translocator. There are three isoforms of the α -subunit (HIF-1 α , HIF-2 α , and HIF-3 α) and one β -subunit (HIF-1 β). In normoxic conditions, HIF-1 α and HIF-2 α are hydroxylated by prolyl hydroxylases, recognized by pVHL, and targeted for proteasomal degradation. In conditions of low oxygen tension, HIF-1 α and HIF-2 α are stable and bind to HIF-1 β . HIF- α /HIF- β dimers are translocated to the nucleus, bind to R-C-G-T-G DNA sequences called “hypoxia response elements,” and induce expression of numerous proteins, including proteins controlling angiogenesis (e.g., VEGF), erythropoiesis (e.g., erythropoietin), glucose uptake and metabolism (e.g., the Glut1 glucose transporter and various glycolytic enzymes), extracellular pH (e.g., carbonic anhydrases IX and XII), and mitogenesis [e.g., TGF- α and platelet-derived growth factor (PDGF)].⁴⁵⁸

Upregulation of HIF is not sufficient but is necessary to induce the RCC and CNS tumors associated with *VHL* mutations. A homozygous *VHL* missense mutation (p.R200W) causes Chuvash polycythemia. This condition is endemic to the Chuvash population of Russia, but it occurs worldwide.⁴⁵⁹ *VHL* p.R200W homozygosity causes elevations of HIF, VEGF, erythropoietin, and hemoglobin; vertebral hemangiomas; varicose veins; low blood pressure; and premature mortality (42 years; range, 26 to 70 years) related to cerebral vascular events and peripheral thrombosis. The absence of RCCs, spinocerebellar hemangioblastomas, and pheochromocytomas typical of classic VHL syndrome, however, suggest that overexpression of HIF and VEGF is not sufficient for tumorigenesis. By contrast, HIF upregulation is required for the development of VHL-associated RCC, and short hairpin HIF RNAs suppress tumor formation by VHL-defective renal carcinoma cells.

The demonstration that HIF upregulation is not sufficient to induce RCC suggests that pVHL has other cellular functions in addition to controlling HIF levels.^{460,461} These include its ability to regulate apoptosis and senescence as well as its role in the maintenance of primary cilia and orchestration of the deposition of ECM.

Loss of heterozygosity at the *VHL* locus in microscopic renal cysts from patients with inherited VHL syndrome established cyst formation as an early step in the pathogenesis of RCC.⁴⁶² Renal cysts and RCCs from patients with VHL syndrome show increased concentrations of both HIF-1 α and HIF-2 α . Renal cysts in patients with VHL syndrome and in renal clear cell carcinoma cell lines lacking pVHL have either no cilia or sparse, rudimentary cilia.^{36,37,463} Importantly, ectopic expression of the *VHL* gene in renal clear cell carcinoma cell lines restores cilia formation, suggesting that pVHL directly supports ciliogenesis.

DIAGNOSIS

The diagnosis of VHL syndrome should be made in a person with multiple CNS or retinal hemangioblastomas or a single hemangioblastoma plus one of the other characteristic physical abnormalities, or in families with a history of VHL syndrome. In some cases, the diagnosis may be warranted in a patient with a positive family history but without CNS or retinal lesions who has one or more of the less specific findings, with the exception of epididymal cysts, which are too nonspecific.

The molecular genetic diagnosis of VHL syndrome has greatly facilitated the evaluation and management of families whose members have the disease. Using a variety of techniques, the current detection rate of mutations is nearly 100%. Candidates for mutation analysis are patients with classic VHL syndrome (meeting clinical diagnostic criteria) and their first-degree family members and members of a family in which a germline *VHL* gene mutation has been identified (presymptomatic test). Genetic testing should also be considered for patients with findings suggestive but not diagnostic for VHL syndrome (i.e., multicentric tumors in one organ, bilateral tumors, two organ systems affected, one hemangioblastoma or pheochromocytoma in a patient younger than 50 years, or one RCC in a patient younger than 30 years) and for family members who have hemangioblastomas, RCCs, or pheochromocytomas only.

MANIFESTATIONS

Renal cysts usually, but not always, precede the development of renal tumors. VHL syndrome–associated RCC manifests early in life, with a mean age at diagnosis of 35 years. The histology is uniformly clear cell. The cumulative probability of development of RCCs rises progressively beginning at age 20 years, reaching 70% by age 60 years. In contrast to RCCs in the general population, RCCs in individuals with VHL syndrome are more often multicentric and bilateral. Metastatic RCC is the leading cause of death from the syndrome.

TREATMENT

Patients with VHL syndrome need annual physical and ophthalmologic examinations; annual measurements of blood or urinary catecholamines and metanephrines; yearly ultrasonography, MRI, or CT of the abdomen; and yearly or biannual MRI or CT of the head and upper spine.⁴⁶⁴ PET or iodine 131 (¹³¹I)-metaiodobenzylguanidine scanning may be indicated for further evaluation of biochemical or imaging abnormalities. Early detection of complications, especially RCC and CNS lesions, followed by appropriate treatment is essential to reduce mortality from VHL syndrome. Because it tends to be recurrent, bilateral, and multifocal, strategies have been developed to preserve renal parenchyma and

minimize the number of invasive procedures. The National Cancer Institute developed the 3-cm rule for surgical intervention on the basis of absence of documented metastasis from tumors <3 cm. Renal-sparing surgery provides effective initial treatment, with 5- and 10-year cancer-specific survival rates similar to those obtained with radical nephrectomy. Minimally invasive techniques that include percutaneous or laparoscopically guided cryotherapy or radiofrequency ablation represent suitable treatment options for selected patients with VHL syndrome that have high technical success rates and cause minor changes in renal function.^{465,466}

The demonstration that HIF is required for VHL-associated carcinogenesis provides a rationale for therapies targeting this transcription factor.^{467,468} HIF can be downregulated by mTOR inhibitors, heat shock protein 90 inhibitors such as geldanamycin and 17-(allylamino)-17-demethoxygeldanamycin, histone deacetylase inhibitors, topoisomerase I inhibitors, thioredoxin-1 inhibitors, and microtubule disrupters. Two mTOR inhibitors, temsirolimus and everolimus, have shown efficacy in the treatment of sporadic RCC.⁴⁶⁹ Treatments can be directed against HIF-responsive gene products, such as VEGF, or against receptors for VEGF, PDGF, or TGF- β . Sunitinib, a VEGF and PDGF tyrosine kinase inhibitor, is currently used in patients with VHL-associated advanced RCC, pancreatic neuroendocrine tumors, and malignant pheochromocytomas.⁴⁷⁰

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE

ADTKD, previously called medullary cystic kidney disease (MCKD), is characterized by the pathologic appearance of the kidneys (small to normal size with cysts at the corticomedullary junction, irregular thickening of the tubular basement membrane, and marked tubular atrophy and interstitial fibrosis; see Fig. 45.19) and clinical manifestations (polydipsia and polyuria followed by development of renal insufficiency with low-grade proteinuria and a benign urine sediment) similar to those seen in NPHP (see later). The distinguishing features are the pattern of inheritance, distinct pathogenesis, later age at diagnosis and ESKD, and absence of extrarenal organ involvement except for gout.^{471–473}

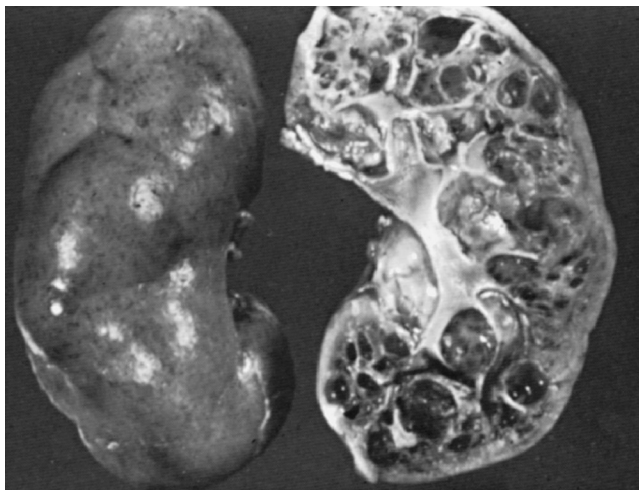


Fig. 45.19 Outer and cut surface of kidney with severe medullary cystic disease.

ADTKD is caused by mutations in at least five genes^{474,475}: *MUC1* encoding the mucoprotein mucin-1 (MUC-1; chromosomal location 1q22)⁴⁷⁶; *UMOD* encoding uromodulin, also known as Tamm–Horsfall protein (chromosome 16p12.3)^{477,478}; *HNF1B* encoding HNF-1 β (chromosome 17q12; see earlier); and *REN* encoding renin (chromosome 1q32.1). Recently mutations in two additional genes, *SEC61A* and *DNAJB11*, have been associated with a renal phenotype overlapping with other forms of ADTKD.^{53,427} Intracellular accumulation of renin or uromodulin in renal tubules of renal biopsies of patients with *SEC61A* or *DNAJB11* mutations, respectively, suggests a role in the folding and trafficking of these proteins.

ADTKD-*MUC1* (or MCKD type 1) is caused by a heterozygous mutation in the variable-number tandem repeat (VNTR) region of *MUC1*.⁴⁷⁶ The common mutation adds one cytosine to a tract of seven cytosine nucleotides, resulting in a frameshift mutation causing truncation of the VNTR and the creation of a new amino acid sequence on the terminal end of the MUC-1 protein. Other rare ADTKD-*MUC1*-associated frame-shifting mutations also result in the same novel amino acid region.⁴⁷⁹ This new protein appears to be improperly processed in the cytoplasm, leading to apoptosis of tubular cells and slowly progressive tubular cell death and nephron dropout, resulting in CKD.⁴⁷⁶ ADTKD-*MUC1* mutations are not readily detected by sequencing methods, including NGS panel analysis, with mass spectrometry of the abnormal VNTR most reliable for diagnostics.⁴⁸⁰

ADTKD-*UMOD* (or MCKD type 2) is caused by a heterozygous mutation, usually a missense mutation in exon 3, 4, 5, 6, 7, or 8, that adds or removes a cysteine residue.^{481,482} The correct protein folding is disrupted, leading to intracellular accumulation of mutant uromodulin, retention in the endoplasmic reticulum, and increased apoptosis.^{483,484} This gain-of-function process explains the fact that *Umod*^{-/-} mice do not have an MCKD phenotype, although they are more susceptible to urinary tract infections and stone formation.^{485–487} A consanguineous family with multiple heterozygous cases and three more severely affected but viable homozygous cases for the same mutation has been described.⁴⁸⁸ The association of a *UMOD* mutation and glomerulocystic kidney disease has been described in one family.⁴⁸⁹ Furthermore, single-nucleotide polymorphisms in *UMOD* have been associated with an increased risk of CKD.⁴⁹⁰ Polydipsia, polyuria, and a tendency to waste sodium are common manifestations of the disease. ADTKD-*UMOD* is frequently but not always associated with hyperuricemia and gout at an early age. It is thought that the accumulation of abnormal uromodulin in the epithelial cells of the thick ascending limb of the loop of Henle prevents the proper function of ion channels, results in mild natriuresis, and increases proximal tubular reabsorption of urate. Hyperuricemia and gout also occur in ADTKD-*MUC1*, but usually at advanced stages of the disease. Patients with ADTKD-*MUC1* and ADTKD-*UMOD* reach ESKD at median ages of 62 and 32 years, respectively, with large patient-to-patient variability.

A diagnosis of ADTKD should be considered in patients with a family history of CKD, bland urinary sediment, and segregation suggesting autosomal dominant inheritance.⁴⁹¹ A history of gout at an early age or a strong family history of gout suggests *UMOD* mutations,⁴⁷¹ whereas a history of anemia in childhood and mildly elevated serum potassium concentrations implicates *REN* mutations.⁴⁹² NGS panel

analysis including all the ADTKD genes is recommended for molecular diagnostic testing, with specific analysis for *MUC1* VNTR variants employed in genetically unresolved cases.^{472,480}

The treatment of choice for ESKD secondary to ADTKD is kidney transplantation.⁴⁹³ If a kidney from a living related donor is considered for transplantation, precautions should be taken to obtain it only from an older relative, who should be subjected to meticulous diagnostic evaluation.

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE-HEPATOCYTE NUCLEAR FACTOR-1B-(ADTKD-*HNF1B*)

HNF-1 β is a transcription factor encoded by the gene *HNF1B* (*TCF2*), which is expressed in polarized epithelia of the pancreas, liver, renal, and genital tracts. HNF-1 β is expressed in the Wolffian duct, the developing mesonephros and metanephros, and the müllerian ducts from the earliest stages of differentiation. In adults, it is expressed in kidney tubules and collecting ducts, oviducts, uterus, epididymis, vas deferens, seminal vesicles, prostate, and testes. HNF-1 β regulates the transcription of a number of genes involved in tubulogenesis, nephron maturation, and tubular transport, including genes mutated in PKD (*PKHD1* and *PKD2*) and another ADTKD gene (*UMOD*) and in autosomal dominant renal hypomagnesemia with hypocalciuria (*FXRD2*).⁴⁹⁴

Heterozygous mutations of *HNF1B* are responsible for a dominantly inherited disease with renal and extrarenal phenotypes, ADTKD-*HNF1B*.^{495–497} As much as 30% to 50% of patients have de novo mutations, and whole-gene deletions occur in approximately 40% of the patients diagnosed in adulthood and in a higher percentage of the patients diagnosed in utero or infancy.^{498–500} The disease is characterized by high phenotypic heterogeneity even within the same family.

The renal manifestations include structural and tubular transport abnormalities.^{495,496,498} The structural abnormalities, often detected in utero, include bilateral hyperechogenic kidneys,⁵⁰¹ unilateral or bilateral agenesis or hypoplasia, multicystic dysplasia, abnormal calyces and papillae, and renal cysts. Histologic evaluation may reveal glomerular cysts and oligomeganephronia. Approximately 80% of congenital solitary kidneys are due to *HNF1B* mutations.

During childhood, the disease is characterized by defective kidney growth and impairment of renal function in approximately 50% of cases with evolution to ESKD in approximately 5% to 10%.⁵⁰²

The renal phenotype in adults is that of a chronic tubulointerstitial nephropathy with bland urinalysis, absence of hematuria, absence of or low-grade proteinuria, low prevalence of hypertension, and slowly progressive kidney failure (yearly eGFR decline of 2 to 2.5 mL/min/1.73 m²). Renal cysts are present in 60% to 80% of the patients but usually are few in number, and unlike in ADPKD, the course of the cystic disease is not characterized by a progressive increase in the number of cysts or in kidney size.

Hypomagnesemia is present in approximately 50% of patients, likely because of the control of the expression of *FXRD2* by HNF-1 β .⁵⁰³ Hypokalemia is also present in 40% of adult patients but has not been observed in children; the underlying mechanism is uncertain. Hyperuricemia and gout have also been associated with the disease with variable frequency. Rarely a generalized dysfunction of proximal

tubular function results in renal Fanconi syndrome. Chromophobe RCCs have been described in a few patients.

The extrarenal manifestations include maturity-onset diabetes of the young type 5 (MODY5), exocrine pancreatic failure, fluctuating liver test abnormalities, and genital tract abnormalities.^{495,496,498} Pancreatic atrophy is found in approximately one-third and diabetes mellitus in approximately half of the patients. For reasons not understood, diabetic nephropathy is extremely rare in these patients. Liver function abnormalities, mainly fluctuating levels of serum alkaline phosphatase and γ -glutamyl transpeptidase, may occur in 40% to 80% of the patients, but liver biopsy findings are usually normal. The genital tract malformations may include absence of fallopian tubes or uterus, vaginal atresia, fusion abnormality such as bicornuate uterus or biseptate vagina, and male genital tract abnormalities. *HNF1B* mutations are found in approximately 18% of women with both renal and uterine malformations but not in patients with isolated uterine malformations. Cognitive defects, autistic features, and epilepsy have also been associated with *HNF1B* mutations.

Patients with ADTKD-*HNF1B* are good candidates for kidney transplantation, but they are at risk for early development of new-onset diabetes after transplantation. The immunosuppressive regimen should avoid tacrolimus and reduce corticosteroid dosage to minimize the risk of development of diabetes mellitus in nondiabetic transplant recipients.⁵⁰⁴ In diabetic patients with ADTKD-*HNF1B* and ESKD, simultaneous pancreatic and kidney transplantation should be considered.

FAMILIAL RENAL HAMARTOMAS ASSOCIATED WITH HYPERPARATHYROIDISM-JAW TUMOR SYNDROME

The autosomal dominant disease familial renal hamartomas associated with hyperparathyroidism-jaw tumor syndrome is characterized by primary hyperparathyroidism (parathyroid adenoma or carcinoma) and ossifying fibroma of the jaw. Kidney lesions may also occur as bilateral cysts, renal hamartomas, or Wilms tumors.⁵⁰⁵ Renal cysts are a common finding, and in some cases they have been clinically diagnosed as PKD. The gene mutated in this disease (*HRPT2*) is ubiquitously expressed and evolutionarily conserved and encodes a protein of 531 amino acids (parafibromin) with moderate identity and similarity to a protein of *Saccharomyces cerevisiae* (cdc73p) that is important in transcriptional initiation and elongation.⁵⁰⁶

AUTOSOMAL RECESSIVE CILIOPATHIES WITH INTERSTITIAL NEPHRITIS AND RENAL CYSTIC DISEASE

Autosomal recessive ciliopathies with interstitial nephritis and renal cystic disease are a large group of diseases caused by mutations in genes encoding ciliary or basal body proteins.^{507,508} The primary cilium is a single hairlike organelle that projects from the surface of most mammalian cells, including epithelial and endothelial cells, neurons, fibroblasts, chondrocytes, and osteocytes.⁵⁰⁹ It arises from the mother centriole, or basal body, in the centrosome. In addition to the basal body, the cilium comprises a transition zone, which regulates protein traffic in and out of the cilium, and the axoneme, which contains a ring of microtubule bundles

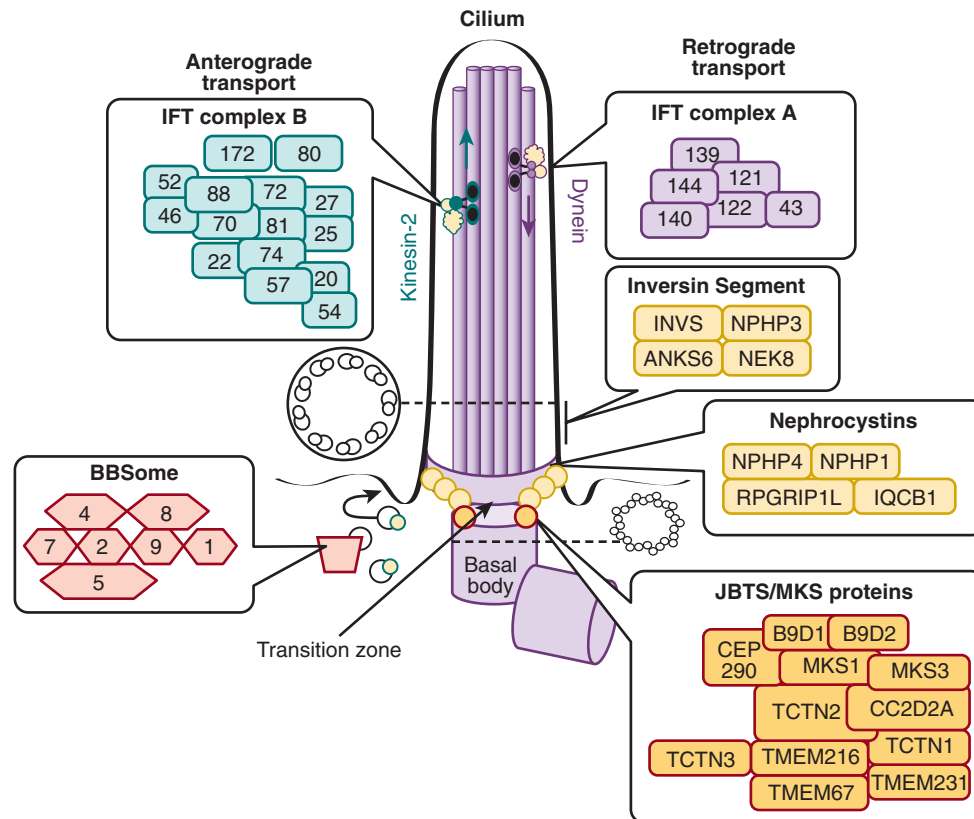


Fig. 45.20 Structure of the primary cilium showing localization of groups of cilioproteins. Primary cilia are sensory organelles that have been implicated in the pathogenesis of polycystic kidney disease, and a large group of the proteins that cause ciliopathies have now been identified. Intraflagellar transport (IFT) moves proteins along the cilium and is involved in cilia formation. Two specific complexes of IFT proteins are involved in anterograde (IFT complex B; kinesin 2 motor) and retrograde (IFT complex A, dynein motor) movements within the cilium. (Numbers inside each complex signify specific IFT proteins). Many of the proteins involved in Meckel syndrome (MKS) and Joubert syndrome (JBTS), and some of the nephronophthisis (NPHP) proteins form a complex at the transition zone and are thought to regulate the protein composition of the cilium. Another group of proteins, including inversin (INVS), are localized to the proximal region of the ciliary axoneme (inversin segment), but their function is less clear. Many of the Bardet-Biedl syndrome (BBS) proteins form the BBSome and regulate trafficking of membrane proteins to the cilium. (Numbers inside BBSome signify various BBS proteins.) Mutations in these ciliopathy proteins alter the normal composition, sensory functions, and signaling of the cilium and are associated with the pleiotropic phenotypes typical of these diseases. *ANKS6*, Ankyrin repeat and SAM (sterile alpha motif) domain-containing protein 6; *B9D1/2*, B9 domain containing proteins 1/2; *CC2D2A*, coiled-coil and C2-domain containing 2A protein; *CEP290*, centrosomal protein of 290 kDa; *IQCB1*, IQ motif containing B₁; *NEK8*, serine/threonine-protein kinase 8; *RPGRIP1L*, RPGR (retinitis pigmentosa guanosine triphosphatase regulator)-interacting protein 1-like protein; *TCTN*, tectonic family member; *TMEM*, transmembrane protein.

connecting the ciliary base to the tip (Fig. 45.20). There is a distinctive proximal segment of the ciliary axoneme known as the inversin compartment, but its function is uncertain. A specific process of IFT involving anterograde transport of cargo toward the tip of the cilium [using the kinesin II motor in association with the IFT complex B (IFT-B) proteins] and retrograde movement [employing the cytoplasmic dynein motor 2 in association with the IFT complex A (IFT-A) proteins] is required for ciliary formation and function.

Primary cilia are important for cell migration or fate determination and tissue patterning during development and for maintenance of cell differentiation thereafter.⁵⁰⁹ The ciliopathies exhibit marked genetic heterogeneity and multiple organ involvement (pleiotropy). NPHP, JBTS, and MKS are autosomal recessive ciliopathies manifesting with cystic kidneys/interstitial nephritis, retinal degeneration, cerebellar/neural tube malformation, and hepatic fibrosis. The impact of the gene involved, the combination of mutations, and likely other genetic and nongenetic factors determine

whether the resulting phenotype is mainly of a developmental (e.g., vermis aplasia, encephalocele, severe renal cystic disease) or degenerative (e.g., retinitis pigmentosa, interstitial nephritis, hepatic fibrosis) nature.⁵⁰⁸ Mutations in the same gene can result in different phenotypes. For example, two *TMEM67* truncating mutations can cause MKS, but also JBTS and NPHP with liver fibrosis [COACH (cerebellar vermis hypoplasia/aplasia, oligophrenia, ataxia, coloboma, and hepatic fibrosis) syndrome]. In the case of the latter disorders, although there is a higher frequency of missense changes, a complete genotype-phenotype correlation has not been defined. Therefore NPHP, JBTS, and MKS can be considered to be allelic disorders.⁵¹⁰ There is also substantial phenotypic and some genetic overlap between these diseases and other disorders such as Bardet-Biedl syndrome (BBS) and Alström syndrome (ALMS). A remarkable feature of these ciliopathies is their large and overlapping genetic heterogeneity, with a total of 68 genes now implicated: NPHP (20), JBTS (34), MKS (13), BBS (21), and ALMS (1; see Table 45.2 for details).⁵¹¹

Table 45.2 Genes Mutated in Autosomal Recessive Ciliopathies With Interstitial Nephritis, Cysts, or Both

Official Gene Name	NPHP	Disease Designations						
		LCA	SLS	JBTS ^a	MKS	BBS	ALMS	OFDS
<i>NPHP1</i>	NPHP1		SLS1	JBTS4				
<i>INVS</i>	NPHP2							
<i>NPHP3</i>	NPHP3				MKS7			
<i>NPHP4</i>	NPHP4		SLS4					
<i>IQCB1</i>	NPHP5		SLS5					
<i>CEP290</i>	NPHP6	LCA10 ^b	SLS6	JBTS5	MKS4	BBS14		
<i>GLIS2</i>	NPHP7							
<i>RPGRIP1L</i>	NPHP8			JBTS7	MKS5			
<i>NEK8</i>	NPHP9							
<i>SDCCAG8</i>	NPHP10		SLS7			BBS16		
<i>TMEM67</i>	NPHP11			JBTS6	MKS3			
<i>TTC21B</i>	NPHP12							
<i>WDR19</i>	NPHP13		SLS8					
<i>ZNF423</i>	NPHP14			JBTS19				
<i>CEP164</i>	NEPH15							
<i>ANKS6</i>	NPHP16							
<i>IFT172</i>	NPHP17							
<i>CEP83</i>	NPHP18							
<i>DCDC2</i>	NPHP19							
<i>MAPKBP1</i>	NPHP20							
<i>INPP5E</i>				JBTS1				
<i>TMEM216</i>				JBTS2	MKS2			
<i>AHI1</i>				JBTS3				
<i>ARL13B</i>				JBTS8				
<i>CC2D2A</i>				JBTS9	MKS6			
<i>OFD1</i>				JBTS10				OFDSI
<i>KIF7</i>				JBTS12				
<i>TCTN1</i>				JBTS13				
<i>TMEM237</i>				JBTS14				
<i>CEP41</i>				JBTS15				
<i>TMEM138</i>				JBTS16				
<i>CPLANE1</i>				JBTS17				OFDSVI
<i>TCTN3</i>				JBTS18				OFDSIV
<i>TMEM231</i>				JBTS20	MKS11			
<i>CSPP1</i>				JBTS21				
<i>PDE6D</i>				JBTS22				
<i>KIAA0586</i>				JBTS23				
<i>TCTN2</i>				JBTS24	MKS8			
<i>CEP104</i>				JBTS25				
<i>KIAA0556</i>				JBTS26				
<i>B9D1</i>				JBTS27	MKS9			
<i>MKS1</i>				JBTS28	MKS1	BBS13		
<i>TMEM107</i>				JBTS29	MKS13			OFDSXVI
<i>ARMC9</i>				JBTS30				
<i>CEP120</i>				JBTS31				
<i>SUFU</i>				JBTS32				
<i>PIBF1</i>				JBTS33				
<i>B9D2</i>				JBTS34	MKS10			
<i>KIF14</i>					MKS12			
<i>BBS1</i>						BBS1		
<i>BBS2</i>						BBS2		
<i>ARL6</i>						BBS3		
<i>BBS4</i>						BBS4		
<i>BBS5</i>						BBS5		
<i>MKKS</i>						BBS6		
<i>BBS7</i>						BBS7		
<i>TTC8</i>						BBS8		
<i>PTHB1</i>						BBS9		

Table 45.2 Genes Mutated in Autosomal Recessive Ciliopathies With Interstitial Nephritis, Cysts, or Both (Cont'd)

Official Gene Name	Disease Designations							
	NPHP	LCA	SLS	JBTS ^a	MKS	BBS	ALMS	OFDS
<i>BBS10</i>						BBS10		
<i>TRIM32</i>						BBS11		
<i>BBS12</i>						BBS12		
<i>WDPCP</i>						BBS15		
<i>LZTFL1</i>						BBS17		
<i>BBIP1</i>						BBS18		
<i>IFT27</i>						BBS19		
<i>IFT74</i>						BBS20		
<i>C8ORF37</i>						BBS21		
<i>ALMS1</i>							ALMS1	

^aIncludes CORS (cerebellooculorenal syndrome) and COACH (cerebellar vermis hypoplasia/aplasia, oligophrenia, ataxia, coloboma, and hepatic fibrosis) syndrome.

^bHypomorphic mutations.

ALMS, Alström syndrome; BBS, Bardet-Biedl syndrome; JBTS, Joubert syndrome; LCA, Leber congenital amaurosis; MKS, Meckel syndrome; NPHP, nephronophthisis; OFDS, oral-facial-digital syndrome; SLS, Senior-Løken syndrome.

Proteomics and assays of ciliogenesis and epithelial morphogenesis have suggested that the NPHP/JBTS/MKS proteins are organized into a network of functionally connected modules.^{512,513} The role of these complexes seems to be to regulate protein trafficking into the cilium. In most cases, mutation leads not to ciliary loss but to an impaired functionality of the cilium. The NPHP1, NPHP4, and RPGRIP1L module localizes to the ciliary transition zone and to cell-cell contacts. It is not essential for ciliogenesis, but it is important for the organization of specialized structures at the apical surface of polarized cells and for epithelial morphogenesis. The IQCB1 and CEP290 module is localized to centrosomes and transition fibers and is indispensable for ciliogenesis in terminal inner medullary collecting duct (IMCD3) cells and for tissue organization. The module consisting of MKS proteins is localized to the transition zone and functionally connected to hedgehog signal transduction. Inversin (INVS), NPHP3, and NEK8 interact in the inversin compartment but their function is unknown. The NPHP-JBTS-MKS network does not overlap with the module integrated by the proteins mutated in BBS (BBSome),⁵¹⁴ despite significant phenotypic overlap, nor with the components of the IFT-A or IFT-B complex proteins despite the fact that defects in IFT may lead to cystic kidney disease and retinal degeneration.

The convergence of proteins mutated in these related ciliopathies at cilia and centrosomes is starting to reveal the pathogenesis of these diseases, which are likely associated with abnormal regulation of protein entry into cilia.⁵¹⁵ Multiple signaling pathways require cilia for proper functioning and have been implicated in ciliopathies, including Wnt and sonic hedgehog signaling.⁵⁰⁸ The pleiotropic features seen in ciliopathies are likely a combination of defects of cilia sensor/mechanosensory functions and cilia-associated signaling that are disrupted by an abnormal cilia-gating process. The observation that proteins that cause NPHP and related ciliopathies when mutated exhibit dual localization at centrosomes and at nuclear foci and play a role in DNA damage response has

led to the hypothesis that defects in DNA damage response participate in the pathogenesis of these diseases.⁵¹⁶

NEPHRONOPHTHISIS

NPHP is an autosomal recessive disorder with an estimated prevalence of 1 in 50,000 live births. It accounts for approximately 5% of ESKD in North American children.⁵¹⁷ It is genetically heterogeneous, with 20 genes identified (see Table 45.2).^{518–524} Homozygous deletions in *NPHP1* account for approximately 21% of all cases of NPHP; the other genes contribute <3% each. No mutations in any of the known genes are found in approximately 60% of patients, indicating that many other genes remain to be discovered.⁵²⁵ Recently, homozygous entire deletion of *NPHP1* has been found in 0.5% of adults with ESKD.⁵²⁶

Mutations in *INVS* and less frequently *NPHP3* lead to renal failure between birth and age 3 years (infantile NPHP). Mutations in the other genes, including *NPHP3*, cause renal failure in the first 3 decades of life (juvenile NPHP).⁵²⁰ Severe retinitis pigmentosa (Senior-Løken syndrome) is frequently associated with *IQCB1*, *CEP290*, and *RPGRIP1L* mutations, but it is less severe and more rarely associated with mutations in other *NPHP* genes. Oculomotor apraxia (Cogan syndrome) is occasionally observed with *NPHP1* and *NPHP4* mutations. *CEP290* and *RPGRIP1L* mutations are frequently associated with cerebellar aplasia or hypoplasia and mental retardation (overlap with JBTS; see later). *INVS* and, more rarely, *NPHP3* mutations can be associated with situs inversus, cardiac ventricular septal defect, hepatic fibrosis, and retinitis pigmentosa. Hypomorphic mutations of *TMEM67* cause NPHP associated with hepatic fibrosis.

The nephrocystins form a multifunctional complex localized in primary cilia, centrosomes, and actin- and microtubule-based structures involved in cell-cell and cell-matrix adhesion signaling as well as in cell division.^{519,521,522,524,527–529} In the kidney, these functional roles could be particularly important

for establishing and maintaining the differentiated state of tubular epithelial cells. Localization of nephrocystins to the cilia explains the association with renal cystic disease and hepatic fibrosis (primary cilia in tubular epithelial cells and cholangiocytes; see earlier discussions of ADPKD and ARPKD), retinitis pigmentosa (rods and cones are modified cilia), and situs inversus and ventricular septal defect (primary cilia in the embryonic node are essential for left–right axis determination).

In the juvenile form, the kidneys are small with a granular capsular surface. On cut sections, the cortex and medulla are thinned, and the corticomedullary margin is indistinct with a variable number of small, thin-walled cysts of distal convoluted and collecting tubule origin. Similar cysts may also be present in the medulla. Grossly visible cysts are often absent. The tubular basement membranes are thickened, even fairly early in the course of the disease, and tubule segments of a single nephron may be encompassed by very dense sclerotic interstitium with sparse chronic inflammatory cell infiltrates. In the infantile form, the kidneys are often enlarged and cystic, and thickening of the tubular basement membranes is less prominent.

Excretory urography and ultrasonography frequently fail to detect cysts because they are small.⁵³⁰ Excretory urography may show inhomogeneous streaking in the medulla caused by accumulation of contrast material in the collecting ducts. Contrast-enhanced CT and MRI are more sensitive to detect small corticomedullary and medullary cysts, but failure to detect cysts does not exclude the diagnosis.

The onset of the disease is insidious. Polyuria and polydipsia are the presenting symptoms. Hypertension is common in the infantile form, but whether absent or present, it is not a prominent feature in the juvenile form. Sodium wasting is common. The urine sediment is characteristically benign. Proteinuria is absent or low grade and there is no microhematuria. Progression to ESKD occurs within the first 3 years in the infantile form and within the first 2 to 3 decades in the juvenile form.

The treatment of NPHP is supportive. Because of the tendency for sodium wasting, volume contraction, and renal azotemia, unnecessary sodium restriction or use of diuretics should be avoided. If kidneys from siblings are considered for transplant surgery, precautions should be taken to obtain them only from unaffected, older relatives, who should be subjected to meticulous diagnostic evaluation.⁵³¹

JOUBERT SYNDROME

JBTS is an autosomal recessive neurologic disease characterized by cerebellar vermis aplasia or hypoplasia with abnormal superior cerebellar peduncles (the “molar tooth sign”), mental retardation, hypotonia, an irregular breathing pattern, and eye movement abnormalities.⁵³² In addition to these core features, patients may exhibit retinal defects (ranging in severity from Leber congenital amaurosis to slowly progressive retinopathies with partially preserved vision), renal defects (NPHP or cystic dysplastic kidneys), and CHF. Rarer features include chorioretinal or optic nerve colobomas, congenital heart malformations, situs inversus, severe scoliosis, skeletal dysplasia, Hirschsprung disease, and midline oral and facial defects, such as cleft lip, cleft palate, or both, notched upper lip, lobulated tongue with multiple frenula, and lingual or

oral soft tumors. The association of JBTS with polydactyly and midline orofacial defects defines the so-called oro-facial-digital type VI syndrome. A distinct subgroup of JBTS defined by cerebellar vermis hypoplasia/aplasia, oligophrenia, congenital ataxia, coloboma, and hepatic fibrosis is known by the acronym COACH, which is usually associated with *TMEM67*, *CC2D2A*, or *RPGRIPL* mutations. Gentile syndrome applies to patients who present with JBTS and CHF in the absence of other clinical features.

Like NPHP and MKS, JBTS exhibits marked genetic heterogeneity. The genetic bases of JBTS are extremely complex and only partly understood, despite the tremendous acceleration in gene discovery enabled by NGS techniques. So far, 34 causative genes have been identified, with autosomal or X-linked recessive inheritance (see Table 45.2).⁵³²

MECKEL SYNDROME

MKS is characterized by bilateral renal cystic dysplasia or PKD, CNS defects (typically occipital encephalocele, but can include the Dandy–Walker malformation or hydrocephalus), postaxial polydactyly, and biliary dysgenesis/CHF. It is a lethal disorder, and most affected infants are stillborn or die within few hours or days after birth.

MKS is genetically heterogeneous with 13 genes so far identified (see Table 45.2).^{533–537} The MKS proteins have been localized to the centrosome, the pericentriolar region, or the cilium itself, and their function is likely involved in forming the barrier between the cell and the cilium at the transition zone.⁵¹⁵

BARDET-BIEDL SYNDROME

BBS is characterized by retinitis pigmentosa, obesity, postaxial polydactyly, learning disabilities, hypogenitalism, and renal abnormalities. Four of these six cardinal features are required for the diagnosis. Other manifestations are diabetes, hypertension, congenital heart disease, ataxia, spasticity, deafness, hepatic fibrosis, and Hirschsprung disease. The manifestations may appear after several years of development.⁵³⁸ Although rare (1 in 120,000 live births), its prevalence in certain geographically isolated communities, such as the Canadian province of Newfoundland and in Kuwait, is more common (1 in 13,500 to 1 in 17,500).³⁹⁰

BBS is genetically heterogeneous, with 21 known BBS genes accounting for approximately 80% of patients clinically diagnosed with the syndrome.^{538–541} The majority of pathogenic mutations are found in *BBS1* and *BBS10*, accounting for 23.2% and 20%, respectively. The disease is primarily inherited in an autosomal recessive manner but with some evidence of a more complex, oligogenic form of inheritance (triallelism and digenic).^{542,543} The localization of the BBS proteins at a subcellular level points to a role in the function of the cilium–centrosome axis. BBS4, BBS6, and BBS8 interact with the pericentriolar material protein 1, a protein important for centriolar duplication. BBS6, BBS10, and BBS12 are chaperonelike proteins. BBS proteins 1, 2, 4, 5, 7, 8, and 9 constitute the BBSome, implicated in vesicular transport toward the cilium. BBS7 and BBS8 play a role in IFT and BBS11 is an ubiquitin ligase.^{514,544}

The diagnosis of BBS is often missed in childhood and made only later in life. Targeted fetal sonography in the

second trimester of pregnancy to detect digital and renal abnormalities has been proposed for the prenatal diagnosis of the disease. The prenatal appearance of enlarged hyperechoic kidneys without corticomedullary differentiation should prompt the diagnosis in a family with BBS, especially when polydactyly is present.^{545,546} In nonaffected families, BBS should be included in the differential diagnosis whenever such an appearance is discovered in utero. The postnatal evolution of the renal ultrasonographic findings is variable, and normalization generally occurs by the age of 2 years.

Renal abnormalities are very common in BBS.⁵⁴⁷ Calyceal clubbing, diverticula, or cysts can be detected in up to 96% of cases. The most common and earliest functional abnormality is a reduced ability to concentrate the urine, resulting in polyuria and polydipsia. Hypertension develops in approximately 50% of patients, and chronic renal insufficiency in 25% to 50%. Despite mental retardation, obesity, and severe visual problems, patients tolerate hemodialysis well. Renal transplantation can be performed, but special attention must be given to controlling hyperphagia and obesity.

ALSTRÖM SYNDROME

ALMS is an autosomal recessive disease that has some phenotypic overlap with BBS. It is characterized by obesity, type 2 diabetes mellitus, retinitis pigmentosa, nerve deafness, CHF, cardiomyopathy, chronic respiratory tract infections, and, frequently, slowly progressive chronic tubulointerstitial nephropathy.^{548,549} The ALMS protein (ALMS1) is of unknown function, is widely expressed in human and mouse tissues, and localizes to centrosomes and the base of cilia.

NEPHRONOPHTHISIS VARIANTS ASSOCIATED WITH SKELETAL DEFECTS (SKELETAL CILIOPATHIES)

Short-rib thoracic dysplasia (SRTD) with or without polydactyly refers to a group of autosomal recessive skeletal ciliopathies that are characterized by a constricted thoracic cage, short ribs, shortened tubular bones, and a “trident” appearance of the acetabular roof. SRTD encompasses Ellis-van Creveld syndrome (EVC) and the disorders previously designated as Jeune syndrome or asphyxiating thoracic dystrophy, short-rib-polydactyly syndrome, and Mainzer-Saldino syndrome. The disease is genetically heterogeneous, with 20 causative genes identified.^{550,551} Some patients have an NPHP-like renal disease, and there is genic overlap between SRTD and JBTS and NPHP (such as *WDR19*, *CEP120*, and *KIAA0586*). Cranioectodermal dysplasia, also known as Sensenbrenner syndrome, is characterized by skeletal abnormalities (e.g., craniosynostosis, narrow rib cage, short limbs, brachydactyly), ectodermal defects, NPHP leading to progressive renal failure, hepatic fibrosis, heart defects, and retinitis pigmentosa. Four genes have been identified with overlap with SRTD and NPHP. Other syndromic forms of NPHP with skeletal disorders include EVC (short stature, short ribs, postaxial polydactyly, nail dystrophy, oral and cardiac defects), with *EVC* and *EVC2* as causative genes, and RHYS syndrome (retinitis pigmentosa, hypopituitarism, nephronophthisis, and skeletal dysplasia).

Whereas proteins associated with NPHP, JBTS, and MKS mainly function at the ciliary transition zone, most proteins

associated with skeletal ciliopathies have been shown to participate in IFT. Ciliary proteins found to be defective in skeletal disorders currently encompass the following four main subgroups: 1. IFT-A subunits and its motor protein, *DYNC2H1*, whose defects disrupt retrograde transport and cause IFT protein accumulation at the ciliary tip; 2. IFT80 and IFT172, 2 of 14 subunits of IFT-B; 3. *NEK1*, a serine-threonine kinase involved in cell-cycle control and ciliogenesis; and 4. *EVC*- and *EVC2*-positive regulators of sonic hedgehog signaling located at the basal body.⁵⁵²⁻⁵⁶²

ORO-FACIAL-DIGITAL SYNDROME TYPE 1

Oro-facial-digital syndrome type 1 is a rare X-linked dominant disorder with prenatal lethality in boys.⁵⁶³ Affected girls may have kidneys indistinguishable from those in ADPKD. The diagnosis is suggested by the extrarenal manifestations, which may include oral (hyperplastic frenula, cleft tongue, cleft palate or lip, malposed teeth), facial (broad nasal root with hypoplasia of nasal alae and malar bone), and digital (brachydactyly, syndactyly, clinodactyly, camptodactyly, polydactyly) anomalies. Some patients have structural CNS anomalies such as agenesis of the corpus callosum, cerebellar agenesis, or the Dandy-Walker malformation. Renal cystic disease is present in 60% of cases after 18 years of age.⁵⁶⁴ These individuals may also have liver and pancreatic cysts.⁵⁶⁵ Mental retardation and tremor can be present in up to 50% of patients. The *OFD1* gene is located in chromosome Xp22.⁵⁶⁶ A combination of Sanger sequencing and gene dosage methods, to detect deletions, yield an 85% mutation detection rate, although inclusion in an NGS panel of ciliopathy genes is now generally used for diagnostics.⁵⁶⁷ Most mutations are private. The 120-kDa *OFD1* protein contains an N-terminal LisH motif, which is important in microtubule dynamics. This protein is a core component of the human centrosome throughout the cell cycle.⁵⁶⁸ Most reported *OFD1* mutations are predicted to cause protein truncation with loss of coiled-coil domains necessary for centrosomal localization. A novel X-linked mental retardation syndrome associated with recurrent respiratory tract infections and macrocephaly caused by ciliary dyskinesia, but no renal phenotype, has been associated with an *OFD1* mutation.⁵⁶⁹

RENAL CYSTIC DYSPLASIA

Renal cystic dysplasia results from an interference with a normal ampullary activity that leads to abnormal metanephric differentiation. When the inhibition of the ampullary activity occurs very early, few collecting ducts are formed and few nephrons develop. The kidney becomes a cluster of cysts with little or no residual parenchyma, and the ureter is absent or atretic. These kidneys may be normal sized, larger than normal [multicystic dysplastic kidney (MCDK)], or markedly shrunken (hypodysplastic kidney). These variations probably represent different stages of the same pathologic process because renal cysts can involute and disappear completely during intrauterine life. When the interference with the ampullary activity occurs later (e.g., as the result of urethral or ureteral obstruction), there may be mild irregularities in branching with a mild generalized dilation of the collecting tubules in the medulla. Most nephrons, however, except the

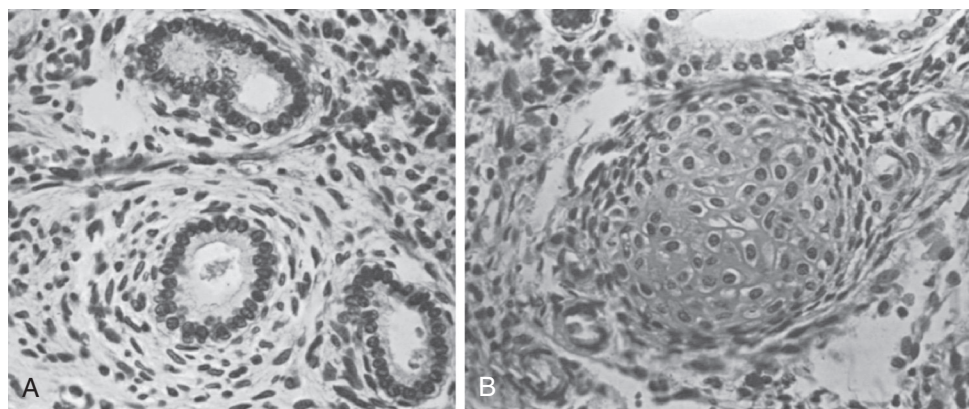


Fig. 45.21 Renal dysplasia. The diagnostic microscopic features include (A) primitive ducts and (B) metaplastic cartilage.

last to be formed, are normal. The cysts are found under the capsule and generally derive from Bowman's spaces (glomerular cysts), the loop of Henle, or terminal ends of collecting tubules. A variety of renal abnormalities in the contralateral kidney can be found in association with cystic dysplastic kidneys. These include renal agenesis, ectopy or fusion, and ureteral duplication or obstruction that may result from injury to the ureteric bud during various stages of development. When the injury to the ureteric bud occurs before a communication with the metanephric blastema has been established, secondary atrophy of the metanephric blastema and renal agenesis ensue. By contrast, if the injury of the bud or ureteral obstruction occurs after renal development is completed, dysplasia does not occur. Thus a spectrum of renal abnormalities ranging from agenesis and severe dysplasia to mild cystic dysplasia with glomerular cysts and a variety of related renal and ureteral abnormalities may result from interferences with normal ampullary activity and metanephric differentiation.

Renal cystic dysplasia may be the consequence of an intrinsic (malformation) or extrinsic (disruption) defect in organogenesis. An intrinsic defect may be caused by a single gene mutation, a chromosomal aberration, or a combination of genetic and environmental factors (multifactorial determination). Extrinsic causes include teratogenic chemicals, metabolic abnormalities, and infections. Evidence for intrinsic or extrinsic defects should be sought by careful review of the pregnancy, family history, and physical examination (pattern of associated abnormalities) as well as by the study of the karyotype. Renal cystic dysplasia frequently occurs as a sporadic event, but they can also occur in the context of many multiorgan malformation syndromes, several of which have defined genetic bases, with these disorders now grouped under the category of congenital anomalies of the kidneys and urinary tract.^{570,571}

Although most dysplastic kidneys are grossly deformed in a fairly characteristic way, most writers accept only two absolute criteria for dysplasia, both of which require histologic confirmation (Fig. 45.21). Of greater importance is the finding of primitive ducts encompassed by mantles of variably differentiated mesenchyma and lined by cuboidal to columnar, sometimes ciliated, epithelium unlike that in any normally developing or mature ducts. Somewhat less important, because of its variable presence, is the finding of metaplastic cartilage (see Fig. 45.21). Cysts of glomerular,

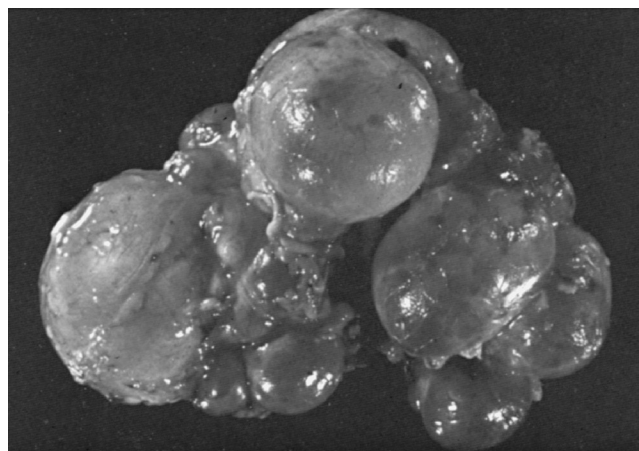


Fig. 45.22 Severe renal cystic dysplasia (multicystic kidney). The renal architecture is markedly distorted.

tubule, and ductal origin may also be present, but because they might represent either a maldevelopment or a histologically similar degenerative change in previously normal but immature structures, they do not provide absolute evidence of parenchymal maldevelopment.

MULTICYSTIC DYSPLASTIC KIDNEY

MCDK is the most common cause of an abdominal mass in infancy and the most common type of bilateral cystic disease in newborns (Fig. 45.22). With the widespread use of fetal sonography, MCDK is now most often diagnosed in utero, usually during the third trimester. MCDK is more often unilateral than bilateral, and boys are more frequently affected than girls. Because affected kidneys tend to involute over weeks or months prenatally and postnatally, the prevalence of unilateral MCDK is higher on fetal screening (1 in 2000) than on neonatal screening (1 in 4000).

Differentiation of MCDK from hydronephrosis in fetuses and newborns is essential because the therapeutic approaches to these conditions differ. The most useful sonographic criteria for identifying an MCDK include the presence of interphases between cysts, a nonmedial location of the larger cysts, absence of an identifiable renal sinus in 100% of the cases, and the absence of parenchymal tissue. The diagnosis can be confirmed by retrograde pyelography, which shows an absence

or atresia of proximal ureter, and by angiography, which demonstrates absence or hypoplasia of the renal artery. Cyst walls often calcify in older patients and may appear as ringlike densities in the region of the kidney.

The manifestations of MCDK depend on whether it is bilateral or unilateral. Bilateral MCDK results in oligohydramnios and the Potter sequence and is incompatible with life. Unilateral MCDK may be diagnosed in a newborn during an evaluation of a renal mass or may go unnoticed until later in life, during evaluation for abdominal or flank discomfort caused by the mass effect of the lesion.⁵⁷² Serial ultrasonography shows that 33% of the MCDKs have completely involuted at 2 years of age, 47% at 5 years, and 59% at 10 years. The development of hypertension or malignant degeneration is very rare.⁵⁷³ Because of its low risk and tendency to involute, MCDK in children is usually managed conservatively.⁵⁷⁴ When indicated, laparoscopic nephrectomy is preferable to open nephrectomy. Attention should be paid to an increased risk for associated urinary tract malformations of the contralateral kidney (e.g., pelviureteric junction obstruction and vesico-ureteric reflux), but voiding cystography is indicated only when the ultrasonographic findings in the contralateral kidney or ureter are abnormal.⁵⁷⁵

OTHER CYSTIC KIDNEY DISORDERS

SIMPLE CYSTS

PREVALENCE

Simple cysts are the most common cystic abnormality encountered in human kidneys.⁵⁷⁶ They may be solitary or multiple and are filled with a fluid that is chemically similar to an ultrafiltrate of plasma. The cysts are very rare in children, but the frequency increases with age.⁵⁷⁷ In autopsy studies and as incidental CT findings, they are found in approximately 25% and 50% in patients 40 and 50 years of age, respectively. A more sensitive CT angiographic study of 1948 potential kidney donors (42% men; mean age, 43 years) showed that 39%, 22%, 7.9%, and 1.6% of 19- to 49-year-old individuals and 63%, 43%, 22%, and 7.8% of 50- to 75-year-old individuals had at least one cortical or medullary cyst at least 2, 5, 10, and 20 mm in diameter, respectively. The 97.5th percentile for the number of cortical and medullary cysts ≥ 5 mm increased with age (1 cyst in men and 1 in women 18–29 years old, 2 in men and 2 in women 30–39 years old, 3 in men and 2 in women 40–49 years old, 5 in men and 3 in women 50–59 years old, and 10 in men and 4 in women 60–69 years old).⁵⁷⁸

PATHOGENESIS

Simple renal cysts are acquired, although there is now evidence that genic and allelic forms of ADPKD associated with mild disease can be mistaken for simple cysts.^{249,579} Several hypotheses have been proposed to explain their pathogenesis. Tubular obstruction and ischemia might play a role. Microdissection studies revealed that diverticula of distal convoluted and collecting tubules are common after age 20 years and increase in number with age. Cysts are thought to derive from progressive dilation and detachment of these diverticula. The cyst walls also appear to be relatively impermeable to low-molecular-weight solutes and to antibiotics. Nonetheless,

the turnover of cyst fluid may be as great as 20 times per day, as measured by tritiated water diffusion. Another study has shown a positive association between the plasma levels of copeptin, a surrogate for vasopressin, and the presence of renal cysts, suggesting the possibility that vasopressin may favor the development of simple renal cysts.⁵⁸⁰

PATHOLOGY

Simple renal cysts are usually lined by a single layer of epithelial cells and filled with a clear, serous fluid. They grow slowly, but huge cysts, up to 30 cm in diameter, have been described. The inner surface of these cysts is glistening and usually smooth, but some cysts may be trabeculated by partial septa that divide the cavity into broadly interconnecting locules. These septated simple cysts should not be confused with multilocular cysts. The cysts are often cortical and distort the renal contour, but they may be deep cortical or apparently medullary in origin. They do not communicate with the renal pelvis. The walls typically are thin and transparent but may become thickened, fibrotic, and even calcified, possibly from earlier hemorrhage or infection.

DIAGNOSIS

Most simple cysts are found on routine imaging studies (Fig. 45.23). Differentiation of simple cysts from RCC is a common problem. Because the appearance of a renal mass on the excretory urogram alone never excludes a malignancy, ultrasonography, CT, or MRI is commonly required to characterize the lesion. Acceptance of definite criteria for the diagnosis of a simple cyst by these imaging techniques has eliminated the use of renal angiography and percutaneous cyst aspiration to characterize renal masses. Calcium deposits are found in 2% of simple cysts and 10% of RCCs. Whereas calcifications appear to be peripheral in simple cysts, they are more central in tumors. Improvements in imaging techniques have also reduced the indications for surgery in the management of patients with benign simple cysts. When the cysts are numerous and bilateral, differentiation from ADPKD may be difficult if liver cysts are not also found. Because of the obvious implications, it is important to avoid a diagnosis of ADPKD in questionable cases unless a familial history consistent with autosomal dominant transmission can be documented or the diagnosis can be confirmed by genetic testing.

MANIFESTATIONS

The cysts are usually asymptomatic, being discovered at the time of a nephro-urologic evaluation for some unrelated problem. They should not distract from the diagnosis of more important intrarenal or extrarenal lesions. Large renal cysts may cause abdominal or flank discomfort, often described as a sensation of weight or a dull ache. Frequently, however, this pain can be explained by a coexisting abnormality such as nephrolithiasis. Rare cases of gross hematuria due to vascular erosion by an enlarging cyst have been documented. However, hematuria is usually attributable to another cause. When the simple cysts lie at or near the hilus, a urographic pattern of calyceal obstruction or hydronephrosis is frequently found. In most but not all cases, these apparent obstructive changes are of no functional significance. A dynamic hippuran/diethylenetriaminepentaacetic acid radioactive renal scan before and after administration of

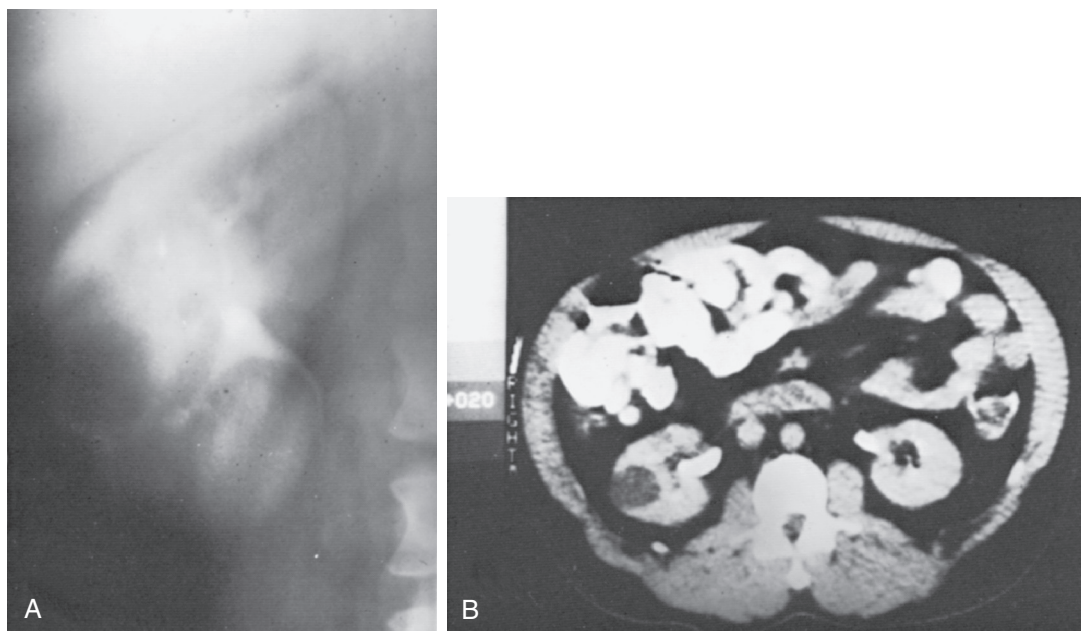


Fig. 45.23 Simple renal cysts. (A) Solitary cortical cyst of the right kidney seen on intravenous urography. (B) Solitary cyst of right renal cortex seen on computed tomography with intravenous contrast enhancement. Oral contrast material was given to highlight the intestine.

furosemide can help assess the degree of obstruction. Rare cases of renin-dependent hypertension caused by solitary intrarenal simple cysts have been described. The proposed mechanism is arterial compression by the cyst that causes segmental renal ischemia. Infection is a rare but dramatic complication of a renal cyst. Simple cysts are not thought to impair renal function, but the presence of simple cysts has been associated with reduced renal function in hospitalized patients younger than 60 years of age.⁵⁸¹ A CT angiographic study of 1948 potential kidney donors showed an association of cortical and medullary cysts 5 mm or larger with higher 24-hour urine albumin excretion as well as with increased body surface area, hypertension, and higher GFR in some analyses, after adjustment for patient age and sex.⁵⁷⁸ Renal cysts are common, particularly in older men, and may be a marker of early kidney injury because they associate with albuminuria, hypertension, and hyperfiltration. Simple cysts infrequently become infected; affected patients present with high fever, flank pain and tenderness, and, frequently, a sympathetic pleural effusion. Most patients are women, and the most common pathogen is *Escherichia coli*. Urine culture results can be negative. Carcinomas do not arise from benign simple cysts. For asymptomatic patients with unequivocal simple cysts, periodic follow-up with ultrasonography is reasonable.

TREATMENT

Treatment of simple renal cyst is indicated only if it is symptomatic or causing obstruction. Intermediate-sized cysts can be aspirated percutaneously, and a sclerosing agent can be instilled into the cavity in an attempt to prevent recurrence. Cysts more than 500 mL in volume are usually drained surgically. Laparoscopic methods are now used routinely. Hypertension has sometimes disappeared after successful aspiration of the cyst fluid or surgical removal of the cyst. Renal vein plasma renin activity is usually elevated in such cases, and the mechanism is thought to be compression of adjacent

vessels by cysts with selective renal ischemia and increased renin production. A surgical approach is usually taken to infected renal cysts, but percutaneous aspiration and drainage of infected cysts have also been used.

LOCALIZED OR UNILATERAL RENAL CYSTIC DISEASE

Localized or unilateral renal cystic disease is a rare condition that involves part or, more rarely, the whole of one kidney with cysts that are indistinguishable from those in ADPKD.⁵⁸² The absence of a family history and the fact that the remaining renal tissue and the liver appear intact help differentiate this condition from asymmetric forms of ADPKD, although mosaicism of an ADPKD mutation is a possible cause. Its etiology and pathogenesis are not understood. The clinical presentation includes a palpable mass, flank pain, gross or microscopic hematuria, and hypertension with well-preserved renal function.

MEDULLARY SPONGE KIDNEY

EPIDEMIOLOGY

MSK, or precalyceal canalicular ectasia, is a common disorder characterized by tubular dilation of the collecting ducts and cyst formation strictly confined to the medullary pyramids, especially to their inner, papillary portions.⁵⁸³ In studies using strict criteria for the quality of acceptable intravenous urograms, the incidence of MSK has been about 13% in patients with calcium urolithiasis but only about 2% in otherwise normal patients. In particular, MSK is associated with a 60% lifetime risk for renal stones, and the prevalence of MSK in patients with renal stones is significantly higher (8.5%; $P < .01$) than in the control population⁵⁸⁴ (1.5%). Among all patients with calcium stones, women have a greater incidence of MSK than men.

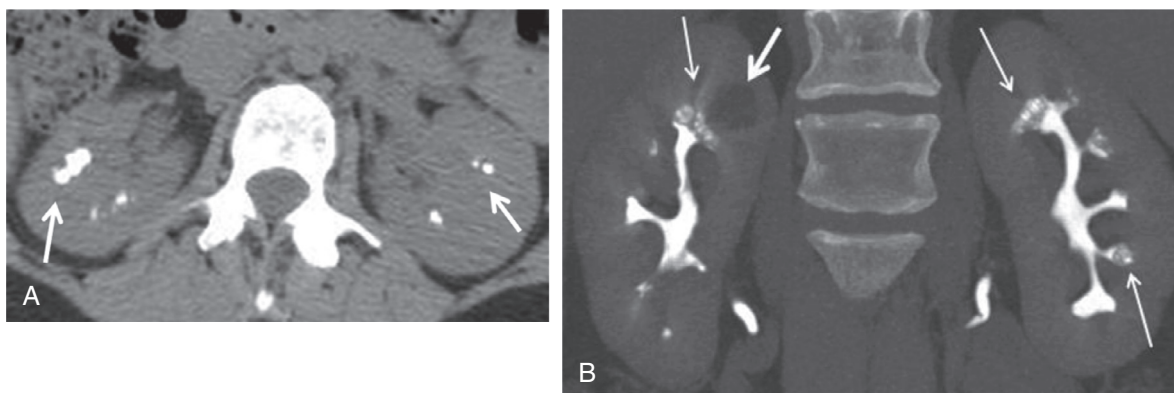


Fig. 45.24 Medullary sponge kidney diagnosed by computed tomography (CT) scan. (A) Noncontrast axial image at the level of the kidneys demonstrates medullary calcinosis (arrows). (B) Coronal maximum intensity projection during the excretory phase of contrast-enhanced CT urography shows a medullary cyst (thick arrow) and dilated collecting tubules showing a papillary brush pattern in both kidneys (thin arrows). (Reprinted with permission from Koraihy FM, Ngo TT, Israel GM, Dahl NK. CT urography for the diagnosis of medullary sponge kidney. *Am J Nephrol*. 2014;39:165–170.)

PATHOGENESIS

MSK has been usually regarded as a nonhereditary disease, but autosomal dominant inheritance has been suggested in several families. In five MSK families, *GDNF* (glial cell–derived neurotrophic factor) gene variants were found to cosegregate with the disease.⁵⁸⁵ A study of family members of 50 patients with MSK identified 59 first- and second-degree relatives of 27 probands in all generations who also had MSK.⁵⁸⁶ There were progressively lower values for urine volume, pH, and excretion of sodium and calcium, and progressively higher levels of serum phosphate noted in probands compared with relatives with bilateral MSK, those with unilateral MSK, and those unaffected by MSK. The investigators interpreted these observations as indicative of a milder form of MSK in the affected relatives. These findings suggest that familial clustering of MSK is common and that the disease has an autosomal dominant inheritance, reduced penetrance, and variable expressivity.

There have been several reports of MSK in patients with Ehlers–Danlos syndrome and in patients with hemihypertrophy. Precalyceal canaliculal ectasia can be observed frequently in patients with ADPKD. MSK has been associated with primary hyperparathyroidism. The rarity of reported cases of this disorder among children favors the interpretation that this is an acquired rather than a congenital disease. Progression of the tubular ectasia and development of tubule dilation and medullary cysts have been documented in some patients.

PATHOLOGY

Despite the name of this disorder, the affected kidney does not closely resemble a sponge. It is usually normal in size or slightly enlarged. The precalyceal canaliculal ectasia may involve one or more renal papillae in one or both kidneys, and the lesions are bilateral in 70% of cases. The dilated ducts communicate proximally with collecting tubules of normal size and often show a relative constriction to approximately normal diameter at the point of their communication with the calyx. Their diameter is often 1 to 3 mm, occasionally 5 mm, and rarely up to 7.5 mm. They often contain small calculi and may be surrounded by normal-looking medullary

interstitium or, in cases of more prominent cystic disease, inflammatory cell infiltration or interstitial fibrosis.

DIAGNOSIS

A definitive diagnosis of MSK can be made by excretory urography when the dilated collecting ducts are visualized on early and later radiographs without the use of compression and in the absence of ureteral obstruction (Fig. 45.24). Deposition of calcium salts within these dilated tubules occurs as renal calculi or nephrocalcinosis. The distribution of the renal calculi in MSK patients is characteristically found in clusters fanning away from the calyx. Because conventional CT has almost completely replaced excretory urography, the diagnosis of MSK may now be made less often. The finding of medullary nephrocalcinosis on CT or medullary hyperchogenicity on ultrasonography may be suggestive, but is not diagnostic, of MSK. Diagnosis of MSK by CT requires multidetector-row CT using high-resolution three-dimensional displays and late urographic images.

MANIFESTATIONS

MSK is usually a benign disorder that may remain asymptomatic and undetected for life. The disease is associated with gross and microscopic hematuria that may be recurrent and with urinary tract infections that often are the first signs of an underlying abnormality. Renal stones consisting of calcium oxalate, calcium phosphate, and other types of calcium salts commonly form in the ectatic collecting ducts and are the most common presentation of this disease.

Impairment of tubular functions, such as a mild concentration defect, a reduced capacity to lower the urine pH after administration of ammonium chloride in comparison with controls, and, possibly, a low maximal excretion of potassium after short-term intravenous potassium chloride loading, may be documented in patients with MSK. Incomplete distal renal tubular acidosis may be found in as many as 30% to 40% of patients.

Whether patients with MSK exhibit specific metabolic abnormalities that predispose them to stone formation different from abnormalities in other patients with stone-forming disorders has been a controversial issue.⁵⁸⁷ Hypercalciuria

and hypocitraturia, a marker of renal tubular acidosis, have been the metabolic risk factors for stone formation more frequently identified in patients with ADPKD. Many, but not all, studies find hypocitraturia to be more common in patients with MSK with stones than in patients with other stone-forming disorders, and the same can be said for hypercalciuria. Some small studies have shown that the hypercalciuria is due to increased intestinal absorption but others have demonstrated a calcium leak. The calcium leak hypothesis could explain reported associations with parathyroid hyperplasia or adenomas and with osteopenia and osteoporosis. An alternative hypothesis is that hypercalciuria in patients with MSK reflects an abnormally high bone turnover as a result of the incomplete renal tubular acidosis seen in many patients with the disease.

Several studies have emphasized the association of hypocitraturia, distal renal tubular acidosis,^{588,589} and hypercalciuria⁵⁸⁷ with MSK. However, other studies have not found hypocitraturia to occur more frequently in patients with MSK than in other patients with renal stones. It has also been suggested that hypercalciuria from a calcium leak may lead to the development of parathyroid adenomas.⁵⁹⁰ However, a critical examination of calcium excretion in patients with MSK and other stone-forming disorders showed that absorptive hypercalciuria was the most common abnormality in MSK, occurring in 59% of patients, whereas only 18% had hypercalciuria resulting from a renal calcium leak.⁵⁹¹ MSK seldom progresses to ESKD, although reduced GFRs have been observed, and a few patients have a relatively poor prognosis because of recurring urolithiasis, bacteriuria, and pyelonephritis.

TREATMENT

There is no specific treatment for MSK. Most patients discovered incidentally can be advised that the disorder is benign and that they can anticipate no serious morbidity and that it is not life-threatening. The treatment of nephrolithiasis and urinary tract infection, when present, is the same as it would be for any patient with these problems. As a general rule, patients with nephrolithiasis should excrete about 2.5 L of urine each day to reduce the risk of stone formation. Potassium citrate and thiazides have been found to be effective in preventing stones in these patients. Fabris and colleagues recommend using potassium citrate as the first step in patients with MSK and a metabolic risk factor regardless of whether this is hypercalciuria, hypocitraturia, hyperuricosuria, or hyperoxaluria.⁵⁹² They recommend starting with 20 mEq of citrate per day in two divided doses, increasing the administration gradually by 10 mEq at a time, if tolerated, for patients in whom a citraturia level >450 mg/24 hours is not achieved initially, until the desired citrate level is reached, provided that the urine pH in a 24-hour collection is less than 7.5. Careful monitoring of the urine pH is necessary to ensure that pH stays below 7.5 in a 24-hour urine collection so as to prevent further formation of calcium phosphate stones in the ectatic tubules. With this regimen, these investigators have achieved not only an increase in urine citrate but also a significant reduction in urine calcium excretion (presumably as a result of the activation of the epithelial calcium channel TRPV5 in the distal nephron by the higher luminal pH) and, most importantly, a marked reduction in the stone event rate, from 0.58 to 0.10 stones per year per patient. They also

observed an improvement in bone densitometry, with a total vertebral T-score increasing from -2.82 to -1.98 and a total hip T-score increasing from -2.03 to -1.86 after an average follow-up of 6.5 years.⁵⁹³ These investigators reserve the use of thiazides for patients who continue to pass stones or have hypercalciuria despite receiving an optimal dose of citrate.

Patients with MSK appear to be more susceptible to urinary tract infections, and routine preventive measures seem warranted, especially in female patients. Repeated unnecessary investigations for hematuria should be avoided. Relapsing urinary tract infections may be due to infected renal stones and may require long-term antimicrobial suppression when the source of the infections cannot be eliminated.

ACQUIRED CYSTIC KIDNEY DISEASE

EPIDEMIOLOGY

Acquired cystic kidney disease (ACKD) is characterized by small cysts distributed throughout the renal cortex and medulla of patients with ESKD and is unrelated to inherited renal cystic disease. There is no agreement on the extent of cystic change required for the diagnosis, ranging from one to five cysts per kidney in radiologic studies to cystic changes in 25% to 40% of renal volume for tissue-based studies. Its prevalence and severity are higher in men than in women and increase with the duration of azotemia. Acquired cysts are found in 7% to 22% of patients with renal failure and serum creatinine values exceeding 3 mg/dL before dialysis, in 35% who have undergone dialysis for <2 years, in 58% for 2 to 4 years, in 75% for 4 to 8 years, and in 92% for longer than 8 years.⁵⁹⁴ ACKD is unrelated to age, dialysis methods, race, and the cause of renal failure. In one study, no reduction in the frequency or severity of this disease was observed in 43 patients treated with hemodiafiltration in comparison with 43 patients treated with conventional hemodialysis after a mean follow-up of 63 months despite significantly lower levels of serum parathyroid hormone and alkaline phosphatase with hemodiafiltration. Multiple logistic regression analysis indicated that the duration of RRT was the only risk factor for the presence of ACKD.⁵⁹⁵ Cysts can regress after successful renal transplant surgery but conversely can develop in transplanted kidneys affected by long-term rejection. Cyclosporine has been incriminated as predisposing native kidneys to cyst formation.

A very important feature in ACKD is the occurrence of renal tumors. The overall prevalence of RCC in patients undergoing hemodialysis evaluated radiologically or at autopsy is approximately 1% to 4%.⁵⁹⁶ Carcinoma in dialysis recipients is three times more common in the presence than in the absence of acquired renal cysts, and six times more common in large than in small cystic kidneys. Overall, the incidence of renal malignancy in patients undergoing dialysis has been estimated to be 50 to 100 times greater than in the general population. The RCCs associated with ACKD have a lower risk for metastasis and a better prognosis than RCCs not associated with ACKD.

The risk for RCC remains high after renal transplantation in patients with ACKD. A study of 961 patients who received a kidney transplant between 1970 and 1998 included 561 patients who underwent prospective ultrasound screening of the native kidneys between 1997 and 2003.⁵⁹⁷ Approximately 23% of them were found to have ACKD. Including 19 patients

with formerly diagnosed RCC, the study found that the prevalence of RCC was 4.8% among all patients, 19.4% among the patients with ACKD, and 0.5% in those without ACKD. RCC was bilateral in 26% of cases. Tumor histology was clear cell RCC in 58% and papillary RCC in 42% of cases. Only one patient had a lung metastasis, and no patient died. Another study conducted ultrasound examination of the native kidneys every 6 months after renal transplantation between 1991 and 2007.⁵⁹⁸ RCCs were diagnosed in 10 patients after a mean follow-up of 61.8 months. Two lesions were solid and eight were cystic, with the average size 2.1 cm. Four were clear cell type, and six papillary carcinomas. None of the patients had metastatic disease.

PATHOGENESIS

The development of the cysts and tumors seems to be tied to the pronounced epithelial hyperplasia observed microscopically. The hyperplasia, in turn, seems to be a result of the uremic state even though there appears to be no relation between the occurrence of acquired cysts and the efficacy of dialysis. If ACKD is present at the time of successful transplantation, that process seems to regress or at least not to increase in severity. Conceivably, the loss of renal mass causes the production of renotropic factors that stimulate hyperplasia.

PATHOLOGY

ACKD is usually bilateral and equal, with even severely affected kidneys weighing <100 g (30% <50 g), although about 25% weigh >150 g, including a few exceptional specimens of >1000 g (Fig. 45.25). In nephrectomy and autopsy specimens, the cysts vary in number and type from a few subcapsular cysts up to 2 to 3 cm in diameter to numerous smaller cysts that are diffusely distributed. The cysts are generally smaller than those in ADPKD. Microdissection studies have demonstrated the continuity of the cysts with both proximal and distal tubules and have suggested their origin both in the fusiform dilation of tubule segments and in multiple small tubule diverticula. Some, but not all, immunohistochemical studies have shown that the cysts in ACKD are mostly derived from proximal tubules.⁵⁹⁹

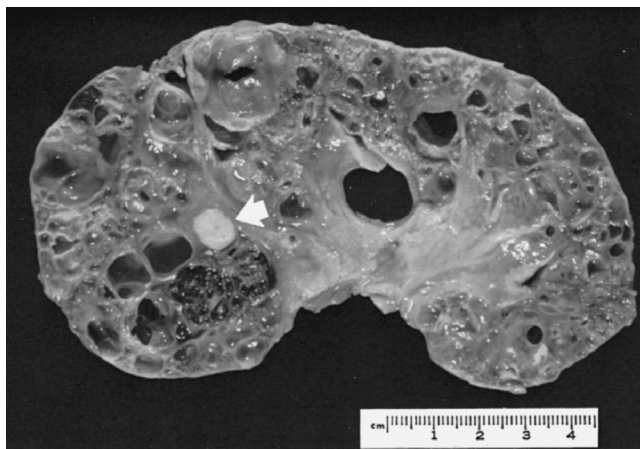


Fig. 45.25 Acquired cystic disease in a 320-g kidney from a patient with a 10-year history of hemodialysis. There were bilateral, multifocal renal cell carcinomas (arrow) with multiple systemic metastases.

In a significant fraction of reported cases, the cysts contain single or, more often, multiple papillary, tubular, or solid neoplasms arising from the cyst lining and consistent with renal cell “adenomas” or adenocarcinomas. The genetic changes underlying the development of most of these tumors are different from those occurring in sporadic clear cell RCCs. Compared with sporadic RCC, ACKD-associated RCC tends to display lower Fuhrman nuclear grade, less proliferative activity, and diploidy in most cases, reflecting less aggressive behavior. The predominant type is clear cell, but papillary RCCs are overrepresented in comparison with the sporadic RCC in the general population.^{596,597} In addition, RCC with distinctive histologic features has been associated with ACKD. These tumors are characterized by abundant eosinophilic cytoplasm; a variably solid, cribriform, tubulocystic, and papillary architecture; and deposits of calcium oxalate crystals.^{600–602} For more discussion of ACKD-associated RCC, please see Chapter 41.

DIAGNOSIS

Ultrasonography is a sensitive method to detect renal cysts and ACKD. However, complex cysts with intracystic septations, intracystic hemorrhage, mural nodules, and peripheral calcifications are sometimes difficult to distinguish from RCC with this method. CT with contrast enhancement is superior to ultrasonography in the evaluation of complex cysts (Fig. 45.25), but contrast enhancement is required to differentiate between benign and potentially malignant cystic lesions, and the intravenous administration of iodinated contrast media carries a risk of worsening the renal function in patients who are not yet undergoing dialysis or who have impaired renal function after transplantation. MRI, by contrast, provides high-resolution images with excellent tissue contrast even without the administration of gadolinium, which is contraindicated in patients with impaired renal function.⁶⁰³ T1- and T2-weighted turbo spin-echo MRI sequences provide very good contrast among different tissues, and modern techniques such as diffusion-weighted sequences help improve diagnostic accuracy.

Because RCC is an important complication of ACKD, screening with ultrasonography has been recommended after 3 years of dialysis, followed by screening for neoplasm at 1- or 2-year intervals thereafter. However, because RCC is a relatively rare cause of death among dialysis recipients, a more aggressive renal imaging program, including annual screening, would be unlikely to significantly reduce mortality and therefore would not be cost-effective.⁶⁰⁴ In the end, the clinical decision must be based on the individual patient, with consideration given to both the known risk factors for carcinoma—including prolonged dialysis, the presence of ACKD, large kidneys, and male sex—and the patient’s age and general fitness. Screening with ultrasonography or MRI at 1- to 2-year intervals may be beneficial in selected populations such as young dialysis recipients or in transplant recipients with ACKD.

MANIFESTATIONS

ACKD develops insidiously. Most patients have no symptoms. When symptoms occur, gross hematuria, flank pain, renal colic, fever, palpable renal mass, and rising hematocrit are most common. Retroperitoneal hemorrhage may manifest as acute pain, hypotension, and shock. Rarely, the presentation

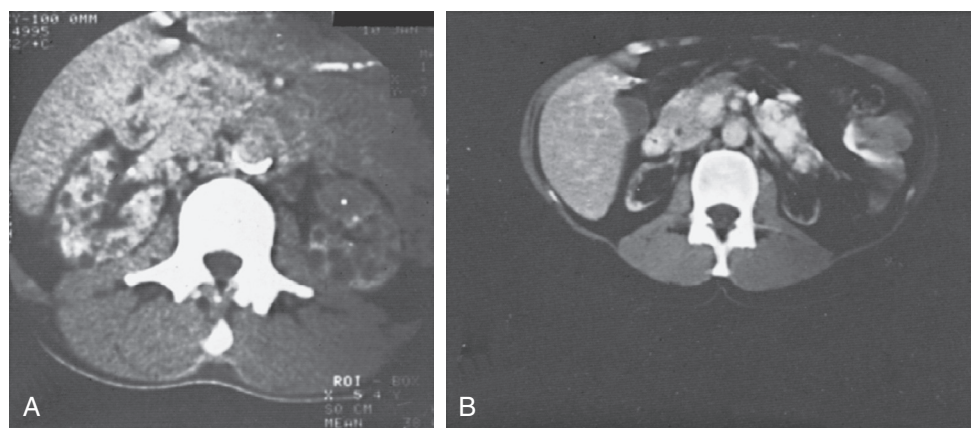


Fig. 45.26 Acquired renal cystic disease. (A) Computed tomography (CT) scan with intravenous contrast. This man had renal failure caused by diabetic nephropathy and had received hemodialysis for 6 years before this examination. There is bilateral renal enlargement with diffuse cysts in the cortex and medulla. A solid tissue tumor (*white dot*) is seen in the anterior part of the left kidney. (B) CT scan of the original kidneys in a patient with a functioning renal allograft. Note the marked atrophy of the renal parenchyma in contrast to the cystic changes seen in (A).

consists of symptoms from metastatic RCC. Approximately 20% of ACKD-associated RCCs metastasize (vs. 50% for sporadic RCCs).

TREATMENT

Bleeding episodes in ACKD, either intrarenal or perirenal, are often treated conservatively with bed rest and analgesics. Persistent hemorrhage, however, may require nephrectomy or therapeutic renal embolization and infarction. Because the risk of undetected RCC is high in patients with retroperitoneal hemorrhage, nephrectomy is recommended in those in whom carcinoma cannot be ruled out. If a few larger cysts are associated with flank pain, percutaneous aspiration (with cytologic examination) is a reasonable temporizing measure. ACKD may regress after successful renal transplantation (Fig. 45.26).

Renal masses larger than 3 cm detected in patients with ACKD are treated by excision. For tumors smaller than 3 cm, the options are nephrectomy for those who can undergo surgery or annual CT follow-up with resection if the lesions enlarge. Although metastases are less likely to occur from small than from large tumors, small tumor size is not a guarantee against metastasis. Resection even of small neoplasms seems prudent in preparation for transplantation. Because carcinoma in the setting of ACKD is often multicentric and bilateral, some writers recommend bilateral nephrectomy in these cases. If this procedure is not performed, frequent monitoring of the contralateral kidney is advised. Laparoscopic bilateral radical nephrectomy in patients with ESKD, ACKD, and suspicious tumors has been proposed as a more desirable alternative to traditional open surgery.⁶⁰⁵

RENAL CYSTIC NEOPLASMS

Renal cystic neoplasms encompass a number of entities that cannot be reliably distinguished from one another on preoperative imaging studies. These entities include cystic RCC, multilocular cystic nephromas, cystic partially differentiated nephroblastomas, and mixed epithelial and stromal tumors.⁶⁰⁶

CYSTIC RENAL CELL CARCINOMA

Multilocular and unilocular RCCs account for about 5% of RCCs and are characterized by their cystic nature—<25% of solid component—and by the absence of necrosis. They are usually clear cell type and of low grade and virtually never metastasize or cause death.⁶⁰⁷ They should be distinguished from RCCs with a large cystic component due to extensive necrosis (pseudocystic necrotic carcinoma), which have an aggressive behavior, often leading to metastasis and death. Surgical excision is usually needed for diagnosis because fine-needle aspiration is not sufficiently accurate.

MULTILOCULAR CYSTIC NEPHROMA

Cystic nephroma is a rare benign cystic neoplasm encountered in children and adults, with a bimodal distribution of age and gender (65% in patients younger than 4 years of age with a male:female ratio of 2:1 and the remainder in patients older than 30 years of age with a male:female ratio of 1:8). Cystic nephroma appears as an encapsulated multilocular mass, the locules of which are not connected to each other or to the pyelocalyceal system. They are lined by a single layer of nondescript, flattened, or cuboidal cells and “hobnail” cells with abundant eosinophilic cytoplasm and large apical nuclei. The septa are composed of connective tissue and may contain scattered atrophic renal tubules. Multilocular cystic nephroma is a benign lesion, but malignant transformation can occur in rare cases.

CYSTIC PARTIALLY DIFFERENTIATED NEPHROBLASTOMA

Cystic partially differentiated nephroblastoma is a rare benign cystic renal neoplasm that is histologically identical to cystic nephroma except for Wilms tumor elements within the septa. It mostly occurs in children younger than 2 years of age, with rare adult occurrences. It is cured by complete excision.

MIXED EPITHELIAL AND STROMAL TUMOR

Mixed epithelial and stromal tumor is a rare type of cystic renal neoplasm, with about 50 cases reported. Contrary to cystic nephroma and cystic partially differentiated nephroblastoma, which are purely cystic and have thin septa, the mixed epithelial and stromal tumor is partly cystic and has thicker wall-forming solid areas. All affected patients but one have been female, with a mean age of 46 years. The role of female hormones in the pathogenesis of this tumor is supported by a female predominance, a history of long-term estrogen treatment in many patients, and the expression of estrogen and progesterone receptors by tumor stromal cells. Mixed epithelial and stromal tumors are benign, and resection is curative.

RENAL CYSTS OF NONTUBULAR ORIGIN

CYSTIC DISEASE OF THE RENAL SINUS

The cystic disorders of the renal sinus are benign conditions that with modern imaging techniques can be clearly distinguished from more serious mass-occupying lesions of the renal pelvis or renal parenchyma. Two types of cystic lesions have been described in this area: hilus cysts and parapelvic cysts.

Hilus cysts, which have been identified only at autopsy, are thought to be caused by regressive changes in the fat tissue of the renal sinus, especially in kidneys with abundant fat in the renal sinus associated with renal atrophy. The cysts result from fluid replacement of adipose tissue that undergoes regressive changes owing to localized vascular disease and atrophy because of recent wasting. A single layer of flattened mesenchymal cells lines the wall of such a cyst, and the cystic fluid is clear and contains abundant lipid droplets.

Parapelvic cysts are of lymphatic origin and are much more common. The walls of the cysts are very thin and are lined by flat endothelial cells. The composition of the cystic fluid resembles that of lymph. The mechanism responsible for the dilation of the lymphatics is not known. Parapelvic cysts may be multiple and bilateral. They are in direct contact with the extrarenal pelvic surface and extend into the renal sinus, distorting the infundibula and calyces. The kidneys may appear slightly enlarged, but the enlargement is exclusively caused by the expansion of the renal sinus, and the area of the renal parenchyma remains normal. Bilateral parapelvic cysts (cystic disease of the renal sinus) can be confused with ADPKD on excretory urography, but the distinction between the two entities is straightforward on CT or MRI.

Parapelvic cysts are most frequently diagnosed after the fourth decade of life. They are usually discovered in the course of evaluations for conditions such as urinary tract infections, nephrolithiasis, hypertension, and prostatism. Despite considerable calyceal distortion, the pressure in these lymphatic cysts is low and not likely to result in significant functional obstruction. Indeed, renal function in patients with bilateral multiple parapelvic cysts is usually normal. Occasionally, parapelvic cysts are the only finding in the course of evaluation for otherwise unexplained lumbar or flank pain. The therapeutic approach to parapelvic cysts should be conservative.

PERIRENAL LYMPHANGIOMAS

Perirenal lymphangiomas are characterized by dilation of the lymphatic channels around the kidneys that leads to the development of unilocular or multilocular cystic masses.⁶⁰⁸ Lymphatic obstruction may play a role in its pathogenesis, and rare familial cases suggest a genetic component. Perirenal lymphangiomas have also been observed in patients with TSC.⁶⁰⁹ Pregnancy is reported to exacerbate the condition, possibly because the renal lymphatics play a role in handling an enhanced interstitial fluid flow during this condition.⁶¹⁰ Mild renal functional impairment and hypertension can occur transiently and revert to normal in the postpartum period.

SUBCAPSULAR AND PERIRENAL URINOMAS (URINIFEROUS PSEUDOCYSTS)

Subcapsular and perirenal urinomas are encapsulated collections of extravasated urine in the subcapsular and perirenal spaces. They are usually secondary to obstructive uropathies, such as posterior urethral valve, pelviureteric junction, or vesicoureteric junction obstruction, ureteric calculus, or trauma. They are caused by pyelosinus backflow, which can occur when the intrapelvic pressure rises to ≥ 35 cm H₂O, leading to rupture of calyceal fornices. Whereas subcapsular urinomas are situated between the renal parenchyma and renal capsule, perirenal urinomas are located between the renal capsule and Gerota fascia. Treatment includes temporary decompression by placement of a pigtail catheter in the most dependent point of the urinoma and correction of the underlying disorder.

PYELOCALYCEAL CYSTS

Also termed “pyelocalyceal diverticula” or “calyceal or pyelorenal cysts” or “diverticula,” pyelocalyceal cysts represent congenital, probably developmental, saccular diverticula from a minor calyx (type I) or from the pelvis or adjacent major calyx (type II). Type I is more common, is usually located in the poles (especially the upper), and tends to be smaller and less often symptomatic than the centrally located type II. Both types are usually <1 cm in diameter but occasionally may be quite large. The cysts are encompassed by a muscularis, are lined by a usually chronically inflamed transitional epithelium, and usually contain urine or cloudy fluid.

Pyelocalyceal cysts occur sporadically, affect all age groups, and usually are unilateral. They may be detected in as many as 0.5% of excretory urograms but normally are asymptomatic unless they are complicated by nephrolithiasis or infection. The frequency of stone formation in calyceal diverticula has been reported to be between 10% and 40%. Transitional cell carcinoma arising in a pyelocalyceal cyst has been seldom reported. Surgical intervention is indicated only when conservative management of this complication fails.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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